# Thinking about the Economy, Deep or Shallow?*

Pierfrancesco Mei[†] Lingxuan Wu[‡]

December 17, 2025

**Abstract**

We propose a theory of *shallow thinking* to capture people's limited understanding of general equilibrium. We represent general equilibrium as a causal chain among multiple variables with feedback loops. We assume that people only think through finite steps, which our customized survey estimates at about 2.6. Longer causal chains are harder to understand, leading to belief over- and underreaction across variables and shocks. In a New Keynesian model with active monetary policy, long-term interest rates underreact to cost-push shocks but overreact to monetary policy shocks. News about future cost-push shocks triggers inflation, and more persistent cost-push shocks cause higher inflation.

*Keywords:* causal reasoning, mental models, general equilibrium, inflation expectations, monetary policy, return predictability, business cycle fluctuations.

*JEL:* D50, D83, D84, E12, E30, E52, E62, E70

*We thank Peter Andre, Francesca Bastianello, Adrien Bilal, John Campbell, Gabriel Chodorow-Reich, Ben Enke, Maryam Farboodi (discussant), Joel Flynn, Xavier Gabaix, Thomas Graeber, Sam Hanson, Zhen Huo, Oleg Itskhoki, Baiyun Jing, Spencer Kwon, David Laibson, Chen Lian, Avi Lipton, Matthew Rabin, Chris Roth, Karthik Sastry, Josh Schwartzstein, Dmitriy Sergeyev, Andrei Shleifer, Stefanie Stantcheva, Jeremy Stein, Ludwig Straub, Adi Sunderam, Alireza Tahbaz-Salehi, Chris Wolf, and seminar and conference participants at BC Carroll, Berkeley, Brown, Chicago Booth Asset Pricing Conference, Columbia Business School, Harvard, HBS, Indiana, IZA Beliefs Workshop, Kellogg, LSE, New York Fed Junior Macro Workshop, NYU Stern, UCLA Anderson, USC Marshall, UT Austin McCombs, UW Foster, Wharton, and Yale for their helpful comments. We thank Sam Cohen for research assistance and Roberto Colarieti for help with the survey design. The study received IRB approval from Harvard (IRB22-1403, IRB24-0959). We acknowledge financial support from the Michael S. Chae Macroeconomic Policy Fund and the Molly and Domenic Ferrante Fund, both awarded through Harvard. Wu is grateful for support from the Alfred P. Sloan Foundation Pre-Doctoral Fellowship in Behavioral Macroeconomics, awarded through the NBER. First version: Nov 2024.

[†]Goldman Sachs, pierfrancesco.mei@gs.com.

[‡]NYU Stern, lingxuanwu@stern.nyu.edu, *Corresponding Author*.

# 1 Introduction

It is well documented that ordinary people underappreciate "equilibrium effects" across micro- and macroeconomic settings (Nagel, 1995; Dal Bó, Dal Bó and Eyster, 2018; Andre et al., 2022). For example, many people fail to appreciate the directional effects of macro shocks on unemployment and inflation, and think of propagation channels that differ from those emphasized by economists. How do we formalize equilibrium effects in macro models? As all macro outcomes adjust in response to any shock, are some effects harder to appreciate than others?

We propose a theory of *shallow thinking* to model people's limited understanding of shock propagation in general equilibrium, measure it, and uncover its implications. We present a systematic way to unpack a general equilibrium model as a causal chain among multiple variables with feedback loops. We assume that, in forming beliefs, people think through only finite steps of shock propagation along the causal chain, and hence, longer chains are less understood.[1] Our customized survey finds that, on average, people understand about 2.6 steps, far below infinity that corresponds to rational expectations.

By truncating the causal chain of reasoning, shallow thinking predicts a rich pattern of belief under- and overreaction across variables and shocks, leading to implications that align with a range of evidence in macro and finance. In a New Keynesian model with an active Taylor rule, shallow thinking predicts that long-term nominal interest rates overreact to monetary policy shocks and underreact to cost-push shocks. It implies that news about future cost-push shocks leads to inflation, as opposed to deflation predicted by rational expectations. It further suggests that more persistent cost-push shocks are more inflationary, rather that less inflationary under rational expectations. In an RBC model, shallow thinking amplifies and prolongs the economy's responses to productivity shocks and leads to a stock market boom and crash.

In Section 2, we introduce shallow thinking by unpacking the textbook New Keynesian model as a system of causal relations. We construct a 2-period setting by considering

[1] By distilling all "equilibrium effects" and suggesting differential difficulty in understanding them, we enrich previous work that focuses only on strategic interactions in general equilibrium (García-Schmidt and Woodford, 2019, Farhi and Werning, 2019, Iovino and Sergeyev, 2023, Bianchi-Vimercati, Eichenbaum and Guerreiro, 2024, Greenwood and Hanson, 2015, Bastianello and Fontanier, 2025*a*,*b*).

transitory news shocks (i.e., period-1 shocks that become known in period 0), before generalizing to persistent shocks in Section 5. Under the standard assumption that agents observe contemporaneous variables, the period-1 equilibrium is a static general equilibrium, with no role for beliefs. However, the period-0 equilibrium depends on agents' beliefs about period-1 outcomes, as households and firms make forward-looking decisions. We introduce shallow thinking of period-1 general equilibrium and explore its implications for macro outcomes and asset prices in period 0.

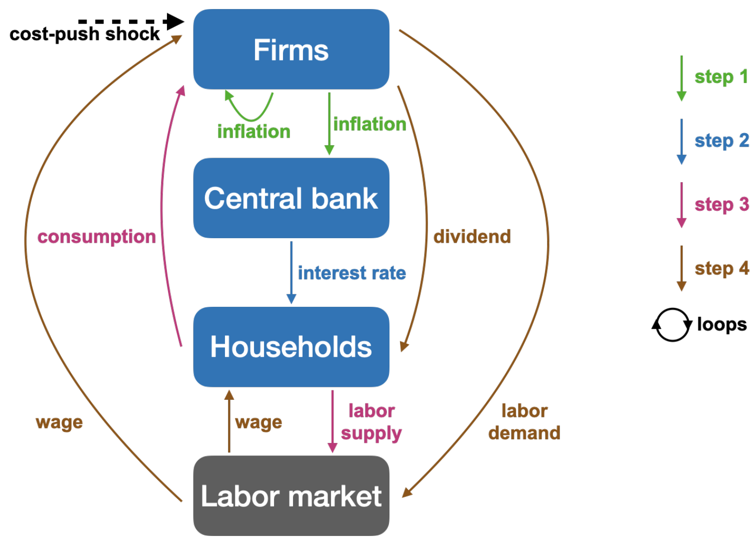



Figure 1: Period-1 New Keynesian economy as directed cyclic graph of causal relations

*Notes:* This figure depicts the causal relations among macroeconomic variables in the period-1 New Keynesian economy as a directed cyclic graph. Each node represents an agent type (firms, the central bank, households) or a competitive market (the labor market). For a competitive market, the price is interpreted as being set by a fictitious Walrasian auctioneer in response to supply and demand shifts, as explained in Section 2.1. We define a *causal relation* as the direct dependence of one variable on another (i.e., a partial derivative), arising from agents' best responses. Dashed arrows indicate shocks affecting some agents (e.g., cost-push shocks that trigger firms to raise prices in the absence of any changes in marginal cost), while solid arrows represent variables—decisions by actual agents (e.g., inflation, interest rate, and consumption) or fictitious auctioneers (e.g., real wage). Each arrow points to the agents (including fictitious ones) responding to it. Variables are color-coded to illustrate the step-by-step propagation of a cost-push shock.

Figure 1 depicts the causal relations among macro variables in the period-1 economy using a directed cyclic graph, which is key to introducing shallow thinking. To draw this graph, we start from the basics of general equilibrium: agents' best responses and price determination in competitive markets. We cast each competitive market as a fictitious auctioneer who sets the price in response to changes in demand and supply, capturing Walras's idea of price adjustment via tâtonnement. With such auctioneers as fictitious agents, we represent general equilibrium as a system of best response functions of all the real and fictitious agents. We define a *causal relation* as the direct dependence of one variable on another (i.e., a partial derivative) driven by these best responses. As reflected in Figure 1, each solid arrow represents a variable chosen by some agents, pointing to the agents it directly influences, and an dashed arrow indicates a shock that directly changes some agents' behavior. The graph is cyclic, representing the equilibrium as a fixed point, appreciated by rational agents who take infinite steps starting from the shock.

Deviating from rational expectations, we hypothesize that individuals foresee only a finite number of steps in shock propagation on the graph, which we refer to as their *depth of thinking* $d$. Notably, we use the step-by-step propagation to model people's thinking process, rather than capturing true dynamics in the economy, since all variables indeed respond to any shock in equilibrium. We color-code Figure 1 to illustrate the propagation of a cost-push shock, i.e., a supply shock that triggers firms to raise prices without a change in marginal cost. A depth-1 agent acknowledges only the most obvious implication—firms will raise their prices, causing inflation—while overlooking changes in all other variables. A depth-2 agent further appreciates that the central bank will raise the interest rate in response to higher inflation according to the Taylor rule, but fails to foresee additional implications. A depth-3 agent understands that a higher interest rate will discourage household consumption and incentivize labor supply. A depth-4 agent recognizes that changes in household behavior will affect the firms and the labor market, and so on. This iterative process captures the idea that more distant causal effects are harder to grasp. It nests rational expectations as infinite depth of thinking, under which agents correctly assess the strength of all causal relations and feedback loops embedded in the graph.

We allow for belief heterogeneity parsimoniously—assuming that, in the population, the depth of thinking $d$ follows a geometric distribution with continuation rate $\lambda \in [0, 1]$. A higher $\lambda$ means that people think more deeply, with $\frac{1}{1-\lambda}$ being the average depth of thinking and $\lambda = 1$ nesting rational expectations (i.e., infinite depth). The average expectations, which drive the equilibrium outcomes in a large class of models including those analyzed in this paper, are tractably parametrized by $\lambda$—they are *as if* generated by a representative agent who knows all causal relations but dampens them by $\lambda$. Hence, $\lambda$ is the only parameter we need to measure in order to apply our theory to macro models.

In Section 3, we develop a survey to test shallow thinking and measure $\lambda$ based on a novel prediction. Our theory predicts that impulse responses of variables that are more distant from shocks are understood by fewer people, and this declining relationship informs $\lambda$. This prediction arises because each shallow agent perceives changes only in a subset of variables that are close to the shocks in the directed graph. It allows us to estimate $\lambda$ from a systematic pattern of belief heterogeneity at the population level.

To test this prediction, we survey US households and present them with hypothetical scenarios involving classic macro shocks, such as oil supply shocks and monetary policy shocks, building on Andre et al. (2022). For each shock, respondents provide directional forecasts for key macro variables, such as inflation, unemployment, and interest rates, in 12 months. We use directional responses from the empirical literature as the benchmark to assess the correctness of respondents' forecasts. We confirm that variable-shock combinations that are further apart on the causal chain are forecasted correctly by fewer people. This finding is robust to including a rich set of fixed effects, such as individual-by-variable and individual-by-shock fixed effects, suggesting that it is not driven by certain individuals understanding certain variables or shocks better than others.

Our estimation strongly rejects the null hypothesis of $\lambda = 1$ and suggests $\lambda \approx 0.61$, implying an average depth of thinking ($\frac{1}{1-\lambda}$) of only about 2.6—far below infinity assumed under rational expectations. Our findings keenly support our parametrization of belief formation in deviation from rational expectations. While we nest rational expectations as a limit case of infinite depth of thinking, an alternative view is that rational agents may have learned the responses of variables to shocks through experience, without necessarily understanding the workings of the economy. Our survey does not require respondents to explain their predictions and finds that many fail to provide correct directional responses, casting doubt on this view. Further, our estimation of $\lambda$ builds upon a systematic belief error: for the same variable, it is less understood when it is more distant from shocks, rather than resting on an arbitrary belief error that may result from noisy measurement.

In Sections 4 and 5, we show that shallow thinking generates belief under- and overreaction across variables and shocks, and leads to several implications in macro and finance that align with empirical evidence. We show that, under shallow thinking, long-term nominal interest rates overreact to monetary policy shocks—as documented by Cochrane and Piazzesi (2002), Gürkaynak, Sack and Swanson (2005), Hanson and Stein (2015), and others—but underreact to supply shocks, consistent with findings by Bauer, Pflueger and Sunderam (2024) and Cieslak (2018). Moreover, macro variables have additional predictive power for bond excess returns beyond current yields, as found by Ludvigson and Ng (2009), Cooper and Priestley (2009), Joslin, Priebsch and Singleton (2014), and Cieslak and Povala (2015). Our model further predicts that news about future cost-push shocks leads to inflation—as found by Känzig (2021)—rather than deflation as predicted by rational expectations. Finally, more persistent cost-push shocks lead to higher inflation, which contrasts with lower inflation predicted by rational expectations.

Shallow thinking reconciles all these findings through dampening perceived causal relations and generating variable-by-shock belief misreaction, which has two layers of meaning. First, beliefs of the same variable misreact differently to different shocks. In the 2-period setting, the period-0 long-term interest rate *underreacts* to cost-push news shocks, because agents underperceive the Taylor rule—that the central bank will raise the short-term interest rate in response to inflation in period 1. In contrast, the period-0 long-term interest rate *overreacts* to monetary policy news shocks, as agents underappreciate an offsetting loop: a positive monetary policy shock will lower inflation, prompting the central bank to slightly lower the short-term rate per the Taylor rule. Consequently, in the presence of multiple shocks, current yields (driven by interest rate expectations) mainly reflect monetary policy shocks, while macro variables such as inflation expectations load more on other shocks, providing additional predictive information on bond excess returns.

Second, in response to the same shock, beliefs of different variables misreact differently, which collectively modulate the equilibrium outcomes. In particular, cost-push news shocks lead to overreaction of inflation expectations and underreaction of expectations of other variables to various degrees, jointly resulting in current inflation, instead of deflation as predicted by rational expectations. The overreaction of inflation expectations arises from misperceiving the relative strength of two counteracting feedback loops. Inflation amplifies itself via a self-loop of *pricing complementarity*, as firms raise prices in response to higher aggregate inflation. It is offset by a longer loop via *monetary policy response*, whereby the interest rate hike in response to inflation lowers the real wage, exerting downward pressure on inflation. Shallow agents better understand the short amplification loop and thus perceive net amplification, even though the long offset loop objectively dominates. Due to the overestimation of future inflation and the underappreciation of the future interest rate hike and associated economic downturn, firms set higher prices and households consume more in period 0, leading to inflation. In contrast, rational agents fully anticipate the future interest rate hike and the resulting economic downturn, and their responses result in current deflation.

We further extend shallow thinking to an infinite-horizon setting and suggest that more persistent cost-push shocks cause higher inflation, in contrast to lower inflation predicted by rational expectations. This difference in comparative statics arises because a more persistent shock strengthens all general equilibrium feedback loops, particularly boosting the long offset loop of inflation via *monetary policy response* discussed previously. A rational agent recognizes that a more persistent cost-push shock is offset more, resulting in lower inflation. In contrast, shallow agents anticipate higher future inflation through the short amplification loop of *pricing complementarity*, and their responses bring that about. Taking these results together, shallow thinking suggests that supply shocks are harder to combat with monetary policy, and more so with more persistent shocks.

In extending shallow thinking to infinite-horizon models with persistent shocks, we define cross-variable causal relations as the *sequence-space Jacobians* à la Auclert et al. (2021), generalizing the partial derivatives in the case of static general equilibrium. This technique is applicable to other models that can be represented in the sequence space. As another example, we consider an RBC economy in Appendix C and illustrate that shallow thinking amplifies and prolongs the economy's responses to productivity shocks and results in a stock market boom and crash. Hence, shallow thinking brings to the RBC model amplification and persistence which are realistic features of the data (King and Rebelo, 1999), in a way consistent with the classic Kindleberger (1978) narrative of crises, where innovations are followed by asset market booms and crashes.

**Connection with level-*k* thinking.** We contribute new insights to the literature on level-*k* thinking in macro and finance (García-Schmidt and Woodford, 2019, Farhi and Werning, 2019, Iovino and Sergeyev, 2023, Bianchi-Vimercati, Eichenbaum and Guerreiro, 2024, Greenwood and Hanson, 2015, Bastianello and Fontanier, 2025*a*,*b*). While level-*k* thinking originated in experimental and game-theoretical literature to model *limited strategic sophistication*, we propose that general equilibrium (GE) encompasses more than strategic interactions. Accordingly, we model the underappreciation of GE as a form of *limited causal reasoning*. Conceptually, cognitive psychology has established that human reasoning struggles with complex causal systems, even absent strategic interaction.[2] Pragmatically, we argue that preserving the full complexity of GE has new and realistic implications.

First, in the stylized 2-period setting we analyze, period-1 GE does not involve beliefs. Therefore, it is free of strategic interactions, and level-*k* thinking coincides with rational

[2]For example, people's knowledge of causally complex systems—whether mechanical devices or natural phenomena—is sparse and shallow (Rozenblit and Keil, 2002; Wilson and Keil, 1998). They pay special attention to earlier nodes in causal chains (Ahn et al., 2000) and their predictions based on causal networks are insufficiently sensitive to the strengths of causal links (see Rottman and Hastie, 2014 for a review). While our primary contribution is in macro and finance, our study also advances the causal reasoning literature (Waldmann, 2017), which typically feature simple examples in short experiments, by examining a complex real-world domain involving many variables and long-term data accumulation.

expectations. More broadly, by focusing on strategic interactions, previous work collapses workhorse macro models into simpler representations. For example, García-Schmidt and Woodford (2019) consider a 3-equation representation of the New Keynesian model. Farhi and Werning (2019) study the Keynesian cross whereby consumption amplifies itself. With only one feedback loop, level-$k$ thinking predicts a uniform under- or overreaction of beliefs (depending on the sign of feedback), and shallow thinking would predict similarly.[3]

We argue, however, that workhorse macro models are better represented as a long causal chain with multiple variables and feedback loops, supported by new survey evidence and implications that align with existing findings. For example, we account for the underappreciation of monetary policy, a finding by Bianchi, Lettau and Ludvigson (2022), Coibion et al. (2023*b*), Wu (2023), and Bauer, Pflueger and Sunderam (2024) across various types of agents. As a result, interest rate expectations may underreact to a supply shock—as found by Bauer, Pflueger and Sunderam (2024)—even if inflation expectations overreact. Moreover, with multiple feedback loops, shallow thinking dampens them differentially by length, potentially reversing the sign of the perceived net effect and the equilibrium responses relative to rational expectations. In summary, we suggest if agents only partially understand the economy, modelers should not simplify the underlying economic structure for them when characterizing belief formation.

**Further literature review.** Our theory captures bounded rationality absent information frictions, and hence complements work on information frictions (Lucas, 1972; Gabaix and Laibson, 2001, Mankiw and Reis, 2002; Woodford, 2003*a*; Nimark, 2008; Angeletos and La'O, 2009; Angeletos and Lian, 2018; Angeletos and Huo, 2021; Angeletos and Sastry, 2021) and inattention (Sims, 2003; Maćkowiak and Wiederholt, 2009; Gabaix, 2020). We find that respondents make directional forecast errors even with full information about the shocks, in support of our theory.[4] Relative to papers on learning under misspecified models (Evans and Honkapohja, 2001; De Grauwe (2011); Eusepi and Preston, 2018; Farmer, Nakamura and Steinsson, 2024; Molavi, Tahbaz-Salehi and Vedolin, 2024; Molavi, 2025) or model uncertainty (Hansen and Sargent, 2007), we characterize misspecified mental models based on the true model analytically, measure it, and uncover implications

[3]Indeed, Angeletos and Lian (2023*a*) show that this property is shared across several prominent models of bounded rationality and incomplete information.

[4]Moreover, prior work generates *horizon-dependent dampening* due to bounded rationality (Gabaix, 2020; Farhi and Werning, 2019) or information frictions (Angeletos and Lian, 2018; Angeletos and Huo, 2021). We focus on *cross-variable dampening* as a complementary aspect, and discuss the connection in Section 5.1.

that align with evidence. We also offer a tractable implementation in workhorse macro models using the sequence-space Jacobian (Auclert et al., 2021).

Our theory contributes to the belief overreaction literature, by generating under- and overreaction in a way that reconciles several bond market puzzles as previously discussed. Barberis (2018) reviews earlier papers in this literature; Bordalo et al. (2020), Afrouzi et al. (2023), and Azeredo da Silveira, Sung and Woodford (2024) present recent developments. In contrast to papers that generate under- or overreaction hinging on the sequence of shocks (Barberis, Shleifer and Vishny, 1998; Daniel, Hirshleifer and Subrahmanyam, 1998; Rabin, 2002; Wachter and Kahana, 2024; Kwon and Tang, 2024; Bianchi, Ludvigson and Ma, 2025), our predictions only depend on the propagation mechanisms of shocks.

We provide a theory of heterogeneous mental models, contributing to the literature that measures mental models in specific scenarios (Stantcheva 2021, 2023*b*; Andre et al. 2022, 2025; Andre, Schirmer and Wohlfart 2024; Bastianello, Décaire and Guenzel, 2025). Andre et al. (2022) show that people's macro forecasts in response to hypothetical shocks are highly heterogeneous. Our theory predicts belief heterogeneity in a way confirmed in our survey which builds on theirs. Prior to this paper, Wu (2023) measures people's imperfect mental models using existing forecasts of professional forecasters and policymakers. This paper tailors a survey for households and derives implications for macro and finance.

Our use of directed *cyclic* graphs connects to work using directed *acyclic* graphs (DAGs) for different purposes. Building on Auclert et al. (2021), who use DAGs to solve macro models and break cycles for technical convenience, we draw a cyclic graph to capture equilibrium feedback loops and represent mental models.[5] We complement Spiegler (2016, 2020), who examines agents fully utilizing a perceived DAG with causal links that differ from the true DAG (e.g., reverse causality, ignoring confounders). We focus on the consequences of not thinking through all causal links, predicting and confirming that agents consider different links depending on the shock.[6] Further afield, Pearl (2009) and others in causal inference use DAGs for identification.[7] Decades of macro research have been devoted to identifying the true model, which we take as given.

[5] Dogra (2024) elucidates the paradoxes of rational expectations and propose "process models" that correspond to DAGs. We distill causal relations and loops from equilibrium models for shallow thinking.

[6] Andre et al. (2025) suggest that people give narratives about the macroeconomy that can be represented by DAGs, though our general equilibrium models feature cyclic dependencies. Our theory suggests that people may think with DAGs *because* they do not take enough steps to appreciate feedback loops.

[7] For reviews, see Heckman and Pinto (2015) and Imbens (2020). Our treatment of equilibrium prices as determined by demand and supply shifts aligns with Imbens (2020)'s suggested representation in his Figure 12B, drawing on seminal work on simultaneous equations (Tinbergen, 1930; Haavelmo, 1943).

# 2 Shallow Thinking in a New Keynesian Economy

We set up the textbook New Keynesian model in Section 2.1 with transitory news shocks, i.e. shocks that realize in period 1 and become known in period 0. In Section 2.2, we represent the period-1 general equilibrium (GE) as a system of causal relations in a directed cyclic graph. This causal representation of GE is our broader theoretical contribution, encompassing the New Keynesian model and other applications. In Section 2.3, we introduce and discuss shallow thinking based on our representation of GE.

## 2.1 The New Keynesian Economy

We consider the New Keynesian model à la Woodford (2003*b*) and Galí (2015). The economy consists of three types of agents (firms, households, and a central bank) and a competitive labor market. We take a log-linear approximation around the steady state for simplicity and use lower-case letters for log-linear deviations.

We study period-1 shocks that become known in period 0. Since the New Keynesian model is purely forward-looking, the economy returns to its steady state from period 2 onwards, which we assume the agents understand. Appendix D.1 provides the full details of the infinite-horizon model, useful later for studying persistent shocks. Here, we focus on period 1, obtained by setting all log deviations from period 2 onwards to zero.

We represent the period-1 GE as a system of causal relations as follows: (i) we interpret price determination in a competitive market (i.e., the labor market) as a fictitious Walrasian auctioneer setting a price in response to supply and demand shifts; (ii) we maintain the structural form of *best responses* of all agents (including Walrasian auctioneers), expressing their decisions as functions of decision-relevant variables. The guiding principle is that in each equation, the variable on the left-hand side is what an agent chooses and the variables on the right-hand side are external to their control but influence their decisions.

**Firms.** There is a continuum of firms indexed by $j \in [0, 1]$ that produce using labor to satisfy demand and set prices subject to Calvo rigidity. In period 1, firms choose labor demand, pay dividends, and reset prices if possible, taking as given the aggregate inflation rate $\pi_1$, real wage $w_1$, and aggregate demand $c_1$. Each firm produces a differentiated good, which collectively forms a constant-elasticity bundle that households consume, and charges a markup $\mu$ in the steady state.

All firms produce to satisfy demand using the same linear technology in labor, giving rise to the aggregate dividend and labor demand

$$
\begin{aligned}
div_1 &= c_1 - \frac{1}{\mu - 1} w_1 \\
n_1^d &= c_1
\end{aligned}
\tag{1}
$$

To anticipate our analysis of the labor market, we interpret labor demand $n_1^d$ as a demand curve $n_1^d = \hat{n}_1^d + e^d w_1$, which shifts by

$$
\hat{n}_1^d = c_1 \tag{2}
$$

and has an elasticity $e^d = 0$ in this model, as firms only use labor in production.

A $(1-\theta)$ share of firms can reset their prices in period 1 to maximize dividends, and each chooses

$$
p_{j1}^* = p_0 + (1 - \beta\theta)\left[w_1 + \sum_{k=0}^{\infty} (\beta\theta)^k \pi_1\right]
$$

where $\beta$ is the household time discount rate, the inverse of which equals the steady-state interest rate. Period-0 aggregate inflation results from the pricing behavior of the $(1-\theta)$ share of resetting firms as $\pi_1 = (1-\theta)\left(p_{j1}^* - p_0\right)$. Following the tradition at least since Clarida, Galí and Gertler (1999), we consider a cost-push shock $\epsilon_1^\pi$, which triggers firms to raise prices without a change in marginal cost. Thus inflation is determined by the Phillips curve

$$
\pi_1 = \theta\kappa w_1 + (1-\theta)\pi_1 + \epsilon_1^\pi \tag{3}
$$

with $\kappa \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}$ capturing the slope with respect to the real wage $w_1$. Importantly, we do not move $\pi_1$ on the right-hand side to the left. We intentionally preserve the dependence of $\pi_1$ on itself, which encapsulates the within-period complementarity in individual price-setting, as each firm takes aggregate inflation as given.

**Households.** There is a continuum of households who live infinitely and maximize their lifetime utility, discounted by $\beta$, which is separable in consumption and labor supply. In period 1, households choose consumption and labor supply, taking as given the nominal interest rate $i_1$, the real wage $w_1$, and dividend $div_1$. Their optimal consumption and labor supply decisions are given by

$$c_1 = -\sigma^{-1}\beta i_1 + \frac{(1-\beta)(\mu-1)\nu}{\sigma+\mu\nu} div_1 + \frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu} w_1 \tag{4}$$

$$n_1^s = \nu^{-1}\beta i_1 - \frac{(1-\beta)(\mu-1)\sigma}{\sigma+\mu\nu} div_1 + \nu^{-1}\left[1-\sigma\frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu}\right] w_1$$

where $\sigma^{-1}$ is the elasticity of intertemporal substitution, and $\nu^{-1}$ is the the Frisch elasticity of labor supply. In writing down these best responses, we have already solved the household maximization problem so that the left-hand-side variables are optimal decisions and the right-hand-side variables are taken as given by households.

Similar to labor demand, we interpret labor supply $n_1^s$ as a supply curve $n_1^s = \hat{n}_1^s + e^s w_1$, which shifts by

$$\hat{n}_1^s = \nu^{-1}\beta i_1 - \frac{(1-\beta)(\mu-1)\sigma}{\sigma+\mu\nu} div_1 \tag{5}$$

due to changes in non-wage variables, and has an elasticity $e^s = \nu^{-1}\left[1-\sigma\frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu}\right]$ as households work more when the wage is higher.

**Central bank.** The central bank follows a Taylor rule with a monetary policy shock $\epsilon_1^i$,

$$i_1 = \phi\pi_1 + \epsilon_1^i \tag{6}$$

**Labor market.** Finally, to close the model, the wage is determined by equilibrating labor supply and demand $n_1^s = n_1^d$. We introduce a fictitious labor market auctioneer who sets the wage as the intersection of supply and demand curves,

$$\hat{n}_1^s + e^s w_1 = \hat{n}_1^d + e^d w_1$$

$$w_1 = \left(e^s - e^d\right)^{-1}\left(\hat{n}_1^d - \hat{n}_1^s\right) \tag{7}$$

This captures Walras's idea of tâtonnement, where the auctioneer raises the wage if there is excess demand for labor and reduces it otherwise, as illustrated in Figure 2.

**Period-1 equilibrium.** In period 1, agents observe all variables, best respond, and the labor market clears. The equilibrium is characterized by (1-7). This period-1 equilibrium is a static general equilibrium, where beliefs play no role.

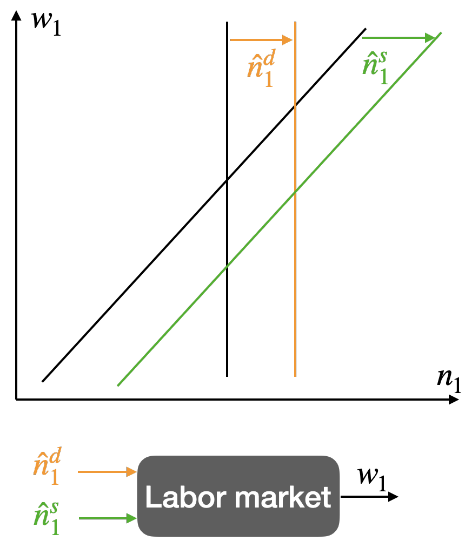



Figure 2: Wage determination in the competitive labor market

*Notes:* This figure depicts the labor market which gives rise to the real wage $w_1$ that balances supply shift $\hat{n}_1^s$ and demand shift $\hat{n}_1^d$. In this New Keynesian model, the labor demand curve is inelastic since firms only use labor in production.

## 2.2 Static General Equilibrium as a System of Causal Relations

We represent the period-1 general equilibrium in the New Keynesian economy as a system of causal relations, and draw it as a directed graph.

The period-1 equilibrium in response to the cost-push shock $\epsilon_1^\pi$ and the monetary policy shock $\epsilon_1^i$ is fully characterized by (1-7). We collect all macro variables in a vector $V_1 \equiv \left(i_1, \pi_1, div_1, \hat{n}_1^d, c_1, \hat{n}_1^s, w_1\right)'$ and the two shocks correspondingly in $S_1 \equiv \left(\epsilon_1^i, \epsilon_1^\pi, 0, 0, 0, 0, 0\right)'$.[8]

To capture the causal relations in the economy, we define $M$ as a matrix of all partial derivatives among the macro variables in (1-7), detailed in (D17). Each element of $M$ is a *partial derivative* that describes how one variable responds to another variable in period 1, which we define as a *causal relation*. A causal relation arises from either agents' best responses to decision-relevant variables (e.g., consumption $c_1$ responding to the interest rate $i_1$) or the determination of prices from supply and demand (i.e., the Walrasian auctioneer's response). We characterize the period-1 equilibrium as follows.

**Proposition 1.** (GE as a system of causal relations) *The period-1 New Keynesian economy is characterized by the fixed point to the system of causal relations among all* agents' actions and

[8]While we focus on these two shocks in this paper, it is straightforward to incorporate additional shocks.

competitive prices, $V_1 \equiv \left(i_1, \pi_1, div_1, \hat{n}_1^d, c_1, \hat{n}_1^s, w_1\right)'$, *as*

$$\underbrace{V_1}_{variables} = \underbrace{M}_{causal\ relations} V_1 + \underbrace{S_1}_{shocks} \tag{8}$$

*The equilibrium can be solved to yield*

$$V_1 = (I - M)^{-1} S_1 = \sum_{n=1}^{\infty} M^{n-1} S_1 \tag{9}$$

*which is a sum of all equilibrium effects of varying distance, where each* $M^{n-1}S_1$ *term is an n-step effect of a shock on a variable via* $n-1$ *intermediate variables.*

Each equation in system (8) describes how an outcome on the left depends on a set of causes on the right, where $S_1$ represents the direct (or partial equilibrium) effects of shocks and $MV_1$ captures the indirect (or general equilibrium) effects.[9] Formula (9) expresses the solution to (8) as the sum of all equilibrium effects of varying distance. The 1-step effect $S_1$ is the direct (or partial equilibrium) effect of shocks, while all subsequent terms represent indirect (or general equilibrium) effects of varying distance.

**Mapping the system of causal relations to graph.** We visualize this causal system (8) as a directed graph in Figure 1, formally supporting the intuition outlined in the introduction. Each node represents a type of agent (including the fictitious labor market auctioneer), and each arrow indicates a macro variable—either an agent's decision or the real wage determined in the competitive labor market.

## 2.3 Shallow Thinking of Static General Equilibrium

Motivated by evidence from behavioral economics and psychology on limitations of causal reasoning, we assume that agents foresee only a finite number of steps in shock propagation in the directed graph. We outline two key assumptions that shape agents' heterogeneous beliefs and lead to a parsimonious characterization of the average beliefs.

Specifically, in period 0, we assume that agents form expectations about period-1 outcomes only based on the news shocks $S_1$ as follows.

[9] To incorporate shocks that directly affect multiple variables (e.g., a household preference shock impacting both consumption and labor supply), we can extend (8) as $V_1 = MV_1 + J^{direct} S_1$ in which $J^{direct}$ captures the direct effects of each shock on a subset of variables and $S_1$ may differ in dimensionality from $V_1$.

**Assumption 1.** Individuals each have a finite *depth of thinking* $d \in \mathbb{N}^+$, with expectations

$$\mathbb{E}_0^d [V_1] \equiv \sum_{n=1}^{d} M^{n-1} S_1 \tag{10}$$

which implies an iterative formula for $d > 1$, starting from $\mathbb{E}_0^0 [V_1] = 0$,

$$\mathbb{E}_0^d [V_1] = M\mathbb{E}_0^{d-1} [V_1] + S_1 \tag{11}$$

Definition (10) formalizes the idea that a depth-$d$ agent only understands the effects of shocks that take no more than $d$ steps. Equivalently, the iterative formula (11) suggests that a depth-$d$ agent can think one step further compared to a depth-$(d-1)$ agent. For example, in response to a cost-push shock $\epsilon_1^{\pi}$ (Figure 1), a depth-1 agent acknowledges only the most obvious implication: firms will raise their prices (causing inflation $\pi_1$). A depth-2 agent can further appreciate that the central bank will raise the interest rate $i_1$ in response to higher inflation. A depth-3 agent understands that a higher interest rate $i_1$ will discourage household consumption and incentivize labor supply. A depth-4 agent recognizes that changes in household behavior will affect the firms and the labor market.

This iteration continues infinitely, and only a depth-$\infty$ (i.e., rational) agent correctly assesses the strength of all loops and accurately forecasts the period-1 equilibrium, i.e., $\mathbb{E}_0^{rational} [V_1] = V_1$. In this sense, the rational expectations hypothesis is nested by letting the agents take infinite steps in the causal graph to converge to the fixed point. Furthermore, under the standard assumption that agents observe all contemporaneous variables in their best responses in period 1, they do not have to understand others' behavior. As a result, in modeling period-0 beliefs, level-$k$ thinking coincides with rational expectations.

**Principles of representing GE.** By definition, shallow agents do not appreciate the fixed point, and their beliefs depend on the causal representation of GE. We emphasize a specific representation of GE underlying shallow thinking—once we cast competitive markets as fictitious Walrasian auctioneers that determine prices, there is one systematic representation of GE that consists of the responses of all agents to decision-relevant variables, as captured by (8) in Proposition 1. There are alternative representations that lead to the same fixed point as (8). However, all these alternatives, including the classic 3-variable representation in terms of $(i_1, \pi_1, c_1)$ which we discuss in Appendix D.2, implicitly mix in relations that *only hold in equilibrium* and consequently fail to reflect best responses.

Hence, our later test of shallow thinking is a joint test of our assumption of iterative belief formation (11) and the specific representation $M$ for iteration.

We have also made an implicit choice to represent GE in terms of aggregate variables, so that causal relations are about how one aggregate variable affects another and are thus comparable. It can be promising for future work to consider more disaggregated models and allow for differential difficulty in understanding different relations.[10]

**Two types of belief distortions.** A finite-depth agent suffers from two types of belief distortions. First, if the agent only thinks through a small number of steps, they zero out changes in variables that are further away from the shock. For those variables, the agent would not even understand their correct directional responses, which will be the basis of our later measurement of shallow thinking. Second, even if such an agent takes enough steps to perceive change in a variable, they generally misforecast the its response in level, due to underappreciation of feedback loops.

**Belief heterogeneity and average beliefs.** We allow for rich belief heterogeneity with a parsimonious parametrization, which leads to an analytical characterization of the average beliefs that are comparable to the true equilibrium outcomes (8, 9).

**Assumption 2.** In the population, individual depth of thinking $d$ follows a geometric distribution over $\mathbb{N}^+$ with continuation rate $\lambda \in [0, 1]$, i.e.,

$$\mathbb{P}(d \geq n) = \lambda^{n-1}, \ \forall n \in \mathbb{N}^+ \tag{12}$$

We assume that everyone can take at least one step, meaning that they understand the direct effects of shocks. A $\lambda$ share of them take at least two, a $\lambda^2$ share take at least three, and so on. A higher shallow thinking parameter $\lambda$ means that individuals are deeper on average, with $\frac{1}{1-\lambda}$ being the average depth of thinking,[11] and $\lambda = 1$ nesting the rational expectations hypothesis (i.e., people having infinite depth of thinking). With this parametric assumption, we could aggregate heterogenous beliefs into average beliefs, which will drive the period-0 equilibrium as we analyze in Section 4.

[10]For example, it might be easier to understand that the oil price affects the gas price than that the general price level affects the interest rate.

[11]The expectation of a geometric distribution is $\mathbb{E}[d] \equiv \sum_{n=1}^{\infty} \mathbb{P}(d = n) \cdot n = \sum_{n=1}^{\infty} \mathbb{P}(d \geq n) = \frac{1}{1-\lambda}$.

**Proposition 2.** (Average beliefs) *The average beliefs* $\overline{\mathbb{E}}_0[V_1] \equiv \sum_{n=1}^{\infty} \mathbb{P}(d=n) \cdot \mathbb{E}_0^n[V_1]$ *are sums of all effects of shocks of varying distance*

$$\overline{\mathbb{E}}_0[V_1] = \sum_{n=1}^{\infty} \lambda^{n-1} M^{n-1} S_1 \tag{13}$$

*Further, the average beliefs satisfy a fixed point*

$$\overline{\mathbb{E}}_0[V_1] = \underbrace{\lambda M}_{\text{average perceived causal relations}} \overline{\mathbb{E}}_0[V_1] + S_1 \tag{14}$$

Equation (13) is comparable to (9) that expresses the equilibrium as a sum of all effects of varying distance, but with more distant effects dampened more, since fewer people appreciate them. Equation (14) further suggests that, the average beliefs are formed *as if* as a fixed point by a representative agent who knows all causal relations in $M$ but underappreciates them by a factor $\lambda$, parallel to (8) that characterizes the equilibrium as a fixed point. The proof is simple, by summing all $n$-step effects with decaying weights.[12]

In summary, the backbone of shallow thinking is that people understand only a limited number of steps in shock propagation, as captured by Assumption 1. Assumption 2 serves as a convenient aggregator to generate average beliefs. Its nature is parametric rather than conceptual, akin to how Calvo pricing is a useful parametrization of nominal rigidity but not essential. With these two assumptions, one can generate heterogeneous and average beliefs in a macro model with a single additional parameter, $\lambda$. If the model also satisfies an additional assumption introduced next, we can estimate $\lambda$ with a panel regression using a survey that elicits directional forecasts of shock propagation.

# 3 Measuring Shallow Thinking

In this section, we test shallow thinking and estimate $\lambda$ using a survey. First we provide a brief overview of our survey design, which examines the general public's understanding of shock propagation, with details in Appendix A.1. Then we formalize the theoretical

[12]Moreover, formula (14) is exactly the formula of imperfect mental models studied in Wu (2023). That paper extracts an empirical moment using comovements in multivariate forecasts data and rejects the null of $\lambda = 1$ (rational expectations). That exercise is *quantitative*, comparing level forecasts to the true conditional expectations. Measurement in this paper only uses qualitative forecasts (i.e., beliefs being directionally correct) elicited in a customized survey. The estimates of $\lambda$ from two papers are remarkably close.

prediction that changes in variables more distant from shocks are perceived by fewer people and use that to measure $\lambda$.

**Overview of survey design.** We assess the US general public's understanding of shock propagation by asking them to forecast the directional responses of key macro variables to hypothetical shocks in 12 months. We conducted an online survey of 1,000 respondents in June and October 2024 through LUCID Marketplace, with demographics quotas following the census distribution.[13]

We study six classic macro shocks in three groups: oil supply shocks (oil) and monetary policy shocks (MP) as group 1, government spending shocks (G) and personal income tax shocks (PIT) as group 2, and corporate income tax shocks (CIT) and transfer payment shocks (TP) as group 3. Half of the respondents are randomly assigned to group 1, which includes the two shocks analyzed in the model, while a quarter are assigned to each of groups 2 and 3 for additional evidence. For each shock, respondents provide directional forecasts of a set of macro variables, such as inflation and the interest rate. For each variable, respondents select "up," "down," "unchanged," or "I don't know" to indicate the expected change in response to the shock over the next 12 months.

Table 1 presents the baseline specification with eleven macro variables and their true directional responses to shocks according to the empirical literature reviewed in Table B2. In addition, it lists the *causal distance* $D_{vs}$, which is a model-implied measure of the distance between a variable $v$ and a shock $s$ on the directed graph that we define and discuss shortly. Based on the true directional responses, we determine the correctness of each respondent's forecasts accordingly. This baseline specification comprises variable-shock combinations with statistically significant directional responses. The baseline causal distance $D_{vs}$ is derived from our New Keynesian model, enriched with decreasing-returns production and a Taylor rule dependent on both inflation and unemployment. We consider various robustness checks for the selection of variable-shock combinations $(v, s)$ and the model-implied causal distance $D_{vs}$ in Tables B3 and B4.

[13] Due to resource constraints, we survey US households and apply the calibrated $\lambda$ to both households and firms in our model. Notably, firm managers are largely uninformed about recent aggregate inflation or monetary policy and have inflation expectations that are far from anchored, similar to households (Candia, Coibion and Gorodnichenko, 2024; Coibion, Gorodnichenko and Kumar, 2018). Furthermore, professional forecasters also underappreciate the causal relations in the economy (Wu, 2023), including the response of monetary policy to economic conditions (Cieslak, 2018; Bauer, Pflueger and Sunderam, 2024).

Table 1: Baseline version of causal distance and correct directions

| | Group 1 (50% of sample) | | Group 2 (25%) | | Group 3 (25%) | |
|---|---|---|---|---|---|---|
| | Oil ↑ | MP ↑ | G ↑ | PIT ↑ | CIT ↑ | TP ↑ |
| Output | 3↓ | 2↓ | 1↑ | 2↓ | 3↓ | 2↑ |
| Interest rate | 2↑ | 1↑ | | | | |
| Price | 1↑ | 3↓ | 2↑ | 3↓ | | |
| Unemployment | 3↑ | 2↑ | 2↓ | 2↑ | 3↑ | 3↓ |
| Labor hours | 3↓ | 2↓ | 2↑ | 2↓ | 3↓ | 3↑ |
| Durable consumption | 3↓ | 2↓ | 2↓ | 2↓ | 3↓ | 2↑ |
| Non-durable consumption | 3↓ | 2↓ | 2↓ | 2↓ | 3↓ | 2↑ |
| Dividend | | | | | 2↓ | 2↑ |
| Personal income tax | | | 1↑ | 1↑ | | |
| Corporate income tax | | | 1↑ | | 1↑ | |
| Government borrowing | | | 1↑ | 1↓ | 1↓ | |

*Notes:* Six shocks are oil supply shock (oil), monetary policy shock (MP), government spending shock (G), personal income tax shock (PIT), corporate income tax shock (CIT), and transfer payment shock (TP). The latter four all concern the federal government. Each cell indicates the causal distance $D_{vs}$, which is formally defined shortly, and the directional response (up or down) from the empirical literature reviewed in Table B2. Robustness versions of variable-shock selection and causal distance are in Tables B3 and B4.

## 3.1 Empirical Implications of Shallow Thinking

We formally establish the theoretical prediction that changes in more distant variables are understood by fewer people, under an additional assumption.

To set the stage, we index respondents by $n$, variables by $v$, and shocks by $s$ in our survey. We capture the correctness of respondents' directional forecasts, based on the true directional changes from the empirical literature, as an indicator $1_{nvs}$. Among the four possible options ("up," "down," "unchanged," or "I don't know"), only the correct direction ("up" or "down") counts as correct directional belief, since we already restrict to variable-shock combinations with statistically significant impulse responses.

**Definition 1.** We define *correct directional belief* $1_{nvs}$ as 1 if respondent $n$ correctly forecasts the directional response of variable $v$ to shock $s$, and 0 otherwise.

As discussed earlier, a shallow agent perceives changes only in variables that are close to shocks. Across the population, since people are of heterogeneous depths of thinking, changes in variables further removed from shocks are understood by fewer people. This is a population-level prediction about the correct directional belief $1_{nvs}$ for our empirical test, without the need to determine the depth of thinking $d$ for each individual.

To formally define the distance of a variable relative to a shock for our test, we introduce some additional notation. Notice that beliefs $\mathbb{E}^d[V_1]$ are linear in the shocks $S_1$, as determined in (10). With only a slight abuse of notation, we let $v \in V_1$ be a variable and $s \in S_1$ be a shock. Thus, $\frac{\partial v}{\partial s}$ is the true sensitivity of variable $v$ to shock $s$, whereas $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ is the perceived sensitivity by a depth-$d$ individual. And their signs indicate the true and perceived directional responses of variables to shocks, respectively.

**Definition 2.** We define the *causal distance* $D_{vs}$ as the minimum $d$ such that $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ has the same sign as $\frac{\partial v}{\partial s}$, for each variable $v \in V_1$ and shock $s \in S_1$.

That is, causal distance $D_{vs}$ corresponds to the depth of the shallowest individual who can correctly perceive the directional response of $v$ to $s$. In our example with transitory cost-push and monetary policy shocks, $D_{vs}$ equals the depth of the shallowest agent who perceives any change of $v$ in response to $s$. That is, $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ is 0 for all $d < D_{vs}$. Nonetheless, Definition 2 is more general when applied to persistent shocks and other models.

**Assumption 3.** Model parameters $M$ satisfy that $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ has the same sign as $\frac{\partial v}{\partial s}$ for all $d \geq D_{vs}$ across all $(v, s)$.

This assumption holds true when more distant causal relations either amplify or offset the impact of the shock, once the correct direction is established, but do no overturn it. This assumption is true in the New Keynesian model we study under a standard calibration and is only useful to establish a reduced-form estimation of $\lambda$ next. For instance, in response to the cost-push shock $\epsilon_1^\pi$, the central bank will raise the interest rate $i_1$ to offset the shock, but does not lead to deflation. That is, the perceived inflation response, $\frac{\partial \mathbb{E}^d[\pi_1]}{\partial \epsilon_1^\pi}$, is positive for all $d \geq 1$. And since $\mathbb{E}^d[i_1] = \phi \mathbb{E}^{d-1}[\pi_1]$ from (11), the perceived interest rate response, $\frac{\partial \mathbb{E}^d[i_1]}{\partial \epsilon_1^\pi}$, is positive for all $d \geq 2$.

Under Assumption 3, any agent with a depth of thinking $d$ greater than or equal to the causal distance $D_{vs}$ will correctly forecast the directional change of variable $v$ in response to shock $s$, while any agent with $d < D_{vs}$ will not. Thus, the expectation of $1_{nvs}$ conditional on causal distance $D_{vs}$ is the share of respondents with depth $d \geq D_{vs}$, which is $\lambda^{D_{vs}-1}$ given Assumption 2, suggesting a reduced-form estimation of $\lambda$ as follows.

**Proposition 3.** (Share of correct directional beliefs) *The share of correct directional belief $1_{nvs}$ among all respondents, conditional on causal distance $D_{vs}$, satisfies*

$$\mathbb{E}^{population}[1_{nvs} | D_{vs} = D] = \lambda^{D-1}, \ \forall D \in \mathbb{N}^+ \tag{15}$$

*where $\mathbb{E}^{population}$ denotes the expectation in the population of survey respondents. Consequently,*

1. *an ordinary least squares regression $1_{nvs} = \gamma D_{vs} + \alpha + \epsilon_{nvs}$ yields a negative slope $\gamma$;*
2. *a nonlinear least squares regression $1_{nvs} = b_1 \cdot b_2^{D_{vs}-1} + b_0 + \epsilon_{nvs}$ identifies $\lambda$ with $b_2$.*

We consider both the linear and the nonlinear specifications. The null of $\gamma = 0$ and $b_2 = 1$ includes rational expectations and any other theories of beliefs that do not correlate with causal distance $D_{vs}$. Our estimation results in Section 3.2 will show that $\gamma$ is negative and $b_2$ is lower than 1, both with high levels of statistical significance.

The nonlinear specification offers a direct estimation of $\lambda$ from a panel regression, following directly from (15).[14] A linear specification is valuable for two reasons. First, it allows us to empirically control for fixed effects to purge confounding sources of belief heterogeneity, and we will show that our slope estimate $\gamma$ is indeed robust to such controls. Second, it does not hinge on the parametric Assumption 2, since a negative $\gamma$ by itself indicates that some agents only perceive changes in variables closer to shocks.

## 3.2 Correctness of Belief Declines in Causal Distance

We examine the predictability of correct directional belief $1_{nvs}$ by causal distance $D_{vs}$ as prescribed by Proposition 3.

Figure 3 shows the expectation of correct directional belief $1_{nvs}$, conditional on causal distance $D_{vs}$, together with the 99.9% confidence interval, produced following Cattaneo et al. (2024). The blue dot indicates the conditional expectation and the red diamond purges individual-by-variable and individual-by-shock fixed effects $\delta_{nv}, \delta_{ns}$, the purpose of which we discuss soon. The conditional expectation of $1_{nvs}$ declines drastically in causal distance $D_{vs}$ in both cases. Most respondents correctly forecasts step-1 variables (those that are directly affect by shocks). Only half of them understand variables that are 2 steps away. For variables that are 3 steps away, the correct share is around one quarter, which is as good as randomly selecting an answer from the four options.

Table 2 presents various specifications of the ordinary least squares (OLS) regression

$$1_{nvs} = \gamma D_{vs} + \alpha + \delta_{nv} + \delta_{ns} + \epsilon_{nvs} \tag{16}$$

[14] While we impose Assumption 3 on the model, it is much less restrictive than it may appear. In models where Assumption 3 does not hold for all $(v, s)$, Proposition 3 can still be applied to the subset of $(v, s)$ where it holds. If even that is not possible, one can estimate $\lambda$ by minimizing distance between the distribution of measured $1_{nvs}$ and the corresponding theory-implied distribution parametrized by $\lambda$.

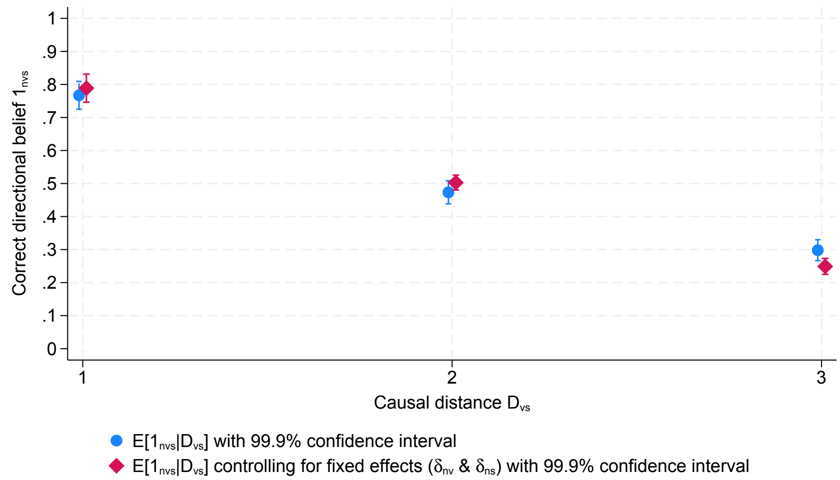



Figure 3: Expectation of correct directional belief $1_{nvs}$ conditional on causal distance $D_{vs}$

*Notes:* This figure shows the expectation of correct directional belief $1_{nvs}$, conditional on causal distance $D_{vs}$, together with the 99.9% confidence interval. The blue dot indicates the conditional expectation. The red diamond represents the conditional expectation, controlling for individual-by-variable and individual-by-shock fixed effects $\delta_{nv}, \delta_{ns}$. Correct directional belief $1_{nvs}$ equals 1 if respondent $n$ correctly forecasts the directional response of variable $v$ to shock $s$, and 0 otherwise. Causal distance $D_{vs}$ is derived from the New Keynesian model. Table 1 lists the correct directional responses and causal distance.

and the nonlinear least squares (NLS) regression

$$1_{nvs} = b_1 \cdot b_2^{D_{vs}-1} + b_0 + \epsilon_{nvs} \tag{17}$$

of correct directional belief $1_{nvs}$ on causal distance $D_{vs}$ as in Proposition 3. Standard errors are clustered at the individual level, since they may correlate across all answers submitted by an individual.

Column (1) uses causal distance $D_{vs}$ as the only predictor in (16) and finds a negative coefficient $\gamma$ that is statistically significant and an $R^2$ of 10%. The fact that the coefficient is significant attests to the validity of our parametrization. We interpret rational expectations as fully thinking through shock propagation. An alternative interpretation is that people may have learned how variables respond to shocks through experience, without understanding the underlying mechanisms. Even though we do not require survey participants to explain their forecasts, they appear like they forecast based on partial understanding of the workings of the economy. Such a negative coefficient is also immune to the concern that respondents given random answers whose correctness will not correlate with $D_{vs}$.

Column (2) shows that individual fixed effects matter too, increasing the $R^2$ to 23%.

Table 2: Regression of correct directional belief $1_{nvs}$ against causal distance $D_{vs}$

| | OLS | | | | | NLS |
|---|---|---|---|---|---|---|
| Correct directional belief $1_{nvs}$ | (1) | (2) | (3) | (4) | (5) | (6) |
| Causal distance $D_{vs}$ | -0.22*** | -0.24*** | -0.24*** | -0.27*** | | |
| | (0.01) | (0.01) | (0.01) | (0.01) | | |
| $1_{D_{vs}=2}$ | | | | | -0.29*** | |
| | | | | | (0.02) | |
| $1_{D_{vs}\geq 3}$ | | | | | -0.54*** | |
| | | | | | (0.02) | |
| $b_2 - 1$ | | | | | | -0.39*** |
| | | | | | | (0.05) |
| Observations | 10763 | 10763 | 10763 | 10763 | 10763 | 10763 |
| $R^2$ | 0.10 | 0.23 | 0.30 | 0.63 | 0.63 | 0.11 |
| Individual FE | | Yes | Yes | Absorbed | Absorbed | |
| Variable FE | | | Yes | Absorbed | Absorbed | |
| Shock FE | | | Yes | Absorbed | Absorbed | |
| Individual-variable FE | | | | Yes | Yes | |
| Individual-shock FE | | | | Yes | Yes | |

Standard errors in parentheses

Standard errors clustered at individual level

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

*Notes*: This table shows the regression results of correct directional belief $1_{nvs}$ against causal distance $D_{vs}$, using the ordinary least squares (OLS) specification (16) and the nonlinear least squares (NLS) specification (17). Correct directional belief $1_{nvs}$ equals 1 if respondent $n$ correctly forecasts the directional response of variable $v$ to shock $s$ and 0 otherwise. Causal distance $D_{vs}$ is derived from a New Keynesian model. The OLS specification tests the null hypothesis that the slope is 0, controlling for individual-by-variable and individual-by-shock fixed effects $\delta_{nv}, \delta_{ns}$. The NLS specification identifies $\lambda$ with $b_2$, with a null hypothesis of $b_2 = 1$. Hence, we show the estimate $b_2 - 1$ and the associated $p$-value.

That means some people are more likely to be correct than others, consistent with our assumption of heterogeneous depths of thinking.

Column (3) shows that the slope estimate and its statistical significance are robust to the inclusion of variable and shock fixed effects. Controlling for these additional fixed effects addresses two important concerns. First, people may understand some variables or some shocks better in ways that happen to correlate with distance. For example, theories of optimal inattention (Sims, 2003; Gabaix, 2014) would imply that agents optimize their perceptions of variables or shocks that are more relevant for their utility.[15] Our findings

[15] In Table B5, we run the same regression on subsamples of decision-relevant and decision-irrelevant variables separately. We find that, while the intercept on the decision-relevant subsample is indeed higher, indicating a higher unconditional probability of correct directional forecast, the slope is also higher, which suggests a faster decline in causal distance. Further, once we include individual-variable and individual-

show that, for the same variable, its response is understood more poorly when it is further away from a shock. A second concern is that some shocks (like monetary policy shocks) may take longer to transmit into the economy or some variables may respond more slowly, which may lead people to predict no changes over a fixed horizon. Again, these possibilities are absorbed by shock and variable fixed effects.[16]

Column (4) further controls for individual-by-variable or individual-by-shock fixed effects. They absorb potentially confounding sources of individual heterogeneity. Such individual heterogeneity may stem from individual-specific cognitive capacity under optimal inattention, or individual-specific experience (Malmendier and Nagel, 2016). These fixed effects also absorb heterogeneity in mental models that do not hinge on distance. For example, if a person believes in a post-pandemic quantity-constrained model of the economy, they will predict that prices respond to all shocks but quantities are fixed.

Column (5) demonstrates that, relative to step-1 variables (that are directly shocked), step-2 variables are understood by fewer people, and step-3-and-above variables by even fewer, confirming what we observe from Figure 3. This is a direct confirmation that among all equilibrium effects, some are better understood than others, a key thesis of our theory. Without a formal theory that distinguishes among equilibrium effects, one may intuitively expect a difference in understanding step-1 variables (that are directly affected) and others. But our empirics say more than that—there is a significant decline in the share of correct directional belief $1_{nvs}$ from step 2 to step 3 and beyond, substantiating our theory.

Last, column (6) shows the nonlinear estimation (17) and strongly rejects the null of $b_2 = 1$. The estimation suggests that $\hat{\lambda} \approx 0.61$. That means, people on average think about only 2.6 steps, even under our parametric assumption that there is a tail distribution of people who are deeper than 3. We take this value as our calibration of $\lambda$.[17]

The steep declining pattern suggests that our focus on limited depth of thinking is an important one, compared to potential differences in perceived causal models. When applying shallow thinking to macro models, we assume for simplicity that agents know the

shock fixed effects, the intercepts and slopes on the two subsamples become statistically indistinguishable.

[16] We also note that we have already purposefully selected variable-shock combinations that have statistically significant impulse responses at the 12-month horizon for our specification.

[17] Alternatively, one could infer $\lambda$ from the average probability of correctly forecasting step-2 variables, $\mathbb{E}^{population}[1_{nvs}|D_{vs} = 2]$, based on (15). That is approximately 0.5, as shown in Figure 3. The difference relates to a slight discrepancy between Assumption 2—which assumes that everyone correctly understands the directional responses of step-1 variables—and the findings in Figure 3 that they mostly, but not always, do. It may arise from respondents' occasional misunderstanding of or inattention to certain survey questions, as any noise only lowers the indicator $1_{nvs}$. We proceed with the higher value of $\lambda$ to be conservative.

true causal links and strength. Alternatively, if people could iterate their models infinitely many times but believed in different causal models or strength of causal relations, they have to be wrong in a way that correlates with causal distance in the New Keynesian model to explain our findings, which is far from obvious. Moreover, our estimation of the average depth of thinking is less than 3, suggesting that perceived strength of causal relations more distant than that does not matter much. Later, we present formulas suggesting that the average perceived strength of GE feedback loops declines exponentially with their length under shallow thinking, while other model parameters (such as the Taylor rule coefficient) and any misperceptions thereof only affect it proportionally.

**Robustness checks and additional results.** We conduct various robustness checks in Appendix B.2. A reasonable concern is that since the causal distance between variables and shocks is model-implied, altering the model changes the causal distance and may affect the results. First, we note that with variable and shock fixed effects, the variation behind our estimation is from the relative distance between variables rather than their absolute distance. We also consider two robustness versions of the model-implied distance and show that our finding is robust.

One may also worry that some of the directional responses may be less well established empirically than others. We show that our findings hold when we only consider oil supply and monetary policy shocks and when we exclude potentially controversial variable-shock combinations (such as the inflation response to monetary shocks—the classic price puzzle).

Further, in Appendix B.3, we suggest that the limited depth of thinking vis-à-vis causal relations in the economy is an individual characteristic and reflects limited knowledge.

# 4 Implications for Inflation and Long-Term Rates

We show that shallow thinking leads to belief under- and overreaction across variables and shocks and makes predictions regarding inflation and long-term rates that align with empirical evidence. To this end, we study transitory news shocks introduced in Section 2. We present analytical results and quantify them based on a standard quarterly calibration of the New Keynesian model, with all parameters listed in Table 3.

We assume that the yield of a 2-period bond $y_0^{(2)}$, i.e., the long-term yield, is determined by the expectations hypothesis with average shallow thinking beliefs as[18]

$$y_0^{(2)} = \frac{i_0 + \overline{\mathbb{E}}_0 [i_1]}{2} \tag{18}$$

We note that our analysis of transitory news shock is general in this 2-period setting: while we compare period-0 beliefs and equilibrium outcomes under shallow thinking against rational expectations regarding news about period-1 shocks, the same comparison holds for persistent shocks that last for 2 periods. That is because in the log-linearized economy, a 2-period persistent shock is the sum of a period-0 shock and a period-1 shock that becomes known in period 0. As the economy's response to a period-0 shock is independent of agents' belief formation, the comparison across theories of expectations regarding any 2-period persistent shock is solely driven by its news shock component.

Table 3: Quarterly calibration of the New Keynesian economy

| Parameter | Description | Value | Estimate/Target |
|---|---|---|---|
| ***Beliefs*** | | | |
| $\lambda$ | Continuation rate of depth of thinking | 0.61 | Our survey evidence |
| ***Firms*** | | | |
| $\theta$ | Price stickiness | 0.75 | Average price duration of 1 year |
| $\kappa$ | Phillips curve slope | 0.086 | $\kappa = \theta^{-1}(1-\theta)(1-\beta\theta)$ |
| $\mu$ | Steady state markup | 1.1 | |
| ***Households*** | | | |
| $\beta$ | Discount factor | 0.99 | Steady state annual $\bar{r} = 4\%$ |
| $\sigma^{-1}$ | Elasticity of intertemporal substitution (EIS) | 1 | |
| $\nu^{-1}$ | Frisch elasticity of labor supply | 0.5 | |
| ***Central bank*** | | | |
| $\phi$ | Taylor rule coefficient | 1.5 | |

## 4.1 Inflation Expectations Overreact to Cost-Push Shocks

Here we show that inflation expectations overreact to cost-push shocks $\epsilon_1^{\pi}$. We characterize the period-1 equilibrium and period-0 expectations in Proposition 4.

**Proposition 4.** (Period-1 cost-push shock) *The period-1 equilibrium in response to a cost-push*

[18] Our model focuses on beliefs while abstracting away from risk. A microfoundation of this formula is that that an intermediary prices the 2-period bond on behalf of all households by averaging their beliefs. We also assume that the 2-period bond is in zero supply, so that its price does not affect the macro outcomes.

*shock $\epsilon_1^\pi$ features*

$$\pi_1 = \frac{1}{1 - \underbrace{(1-\theta)}_{\text{pricing complementarity}} + \underbrace{\phi\theta\kappa}_{\text{monetary policy loop}} \cdot \underbrace{\sigma^{-1}(\nu+\sigma)}_{\text{Keynesian cross}}}\epsilon_1^\pi \tag{19}$$

$$i_1 = \phi\pi_1 \tag{20}$$

*whereas the period-0 average expectations upon observing the news about $\epsilon_1^\pi$ are*

$$\overline{\mathbb{E}}_0[\pi_1] = \frac{1}{1-\lambda(1-\theta)+\lambda^4\cdot\phi\theta\kappa\cdot K_1(\lambda)}\epsilon_1^\pi \tag{21}$$

$$\overline{\mathbb{E}}_0[i_1] = \lambda\phi\overline{\mathbb{E}}_0[\pi_1] \tag{22}$$

*with $K_1(\lambda)$ increasing in $\lambda$ under our calibration and $K_1(1) = \sigma^{-1}(\nu+\sigma)$.*

To understand these results, we start with equilibrium inflation (19) and the average inflation expectation (21). The period-1 equilibrium is independent of agents' belief formation, as (19) is independent of $\lambda$. It coincides with rational expectations, i.e., (21) under $\lambda = 1$. Recall that both the equilibrium and the average expectations are the sum of all $n$-step effects, as established in (9) and (13), with more distant effects dampened more for expectations. We organize all these effects into three groups that involve different loops, color-coded in Figure 4, and inflation $\pi_1$ is directly involved in two of them.

The first loop is a self-loop of *pricing complementarity*, shown in purple. A higher inflation $\pi_1$ incentivizes all firms to price higher, thus amplifying itself. This effect has a strength of $(1-\theta)$. As we sum up the infinite geometric series going through this loop, its strength appears in the denominator of $\pi_1$ in (19). It is dampened by $\lambda$ in $\overline{\mathbb{E}}_0[\pi_1]$ in (21) since only a $\lambda$ share of people understand each loop.

The second one is a 4-step loop involving *monetary policy*, shown in orange. As inflation $\pi_1$ rises, the central bank raises the interest rate $i_1$, which encourages labor supply $\hat{n}_1^s$, lowering the real wage $w_1$. As a lower wage prompts firms to reduce prices, this offsets the inflation response. This loop with a strength of $\phi\theta\kappa$ takes 4 steps to close, meaning that whenever it loops once, only $\lambda^4$ share of the people perceive the next loop, resulting in a dampening of $\lambda^4$ in $\overline{\mathbb{E}}_0[\pi_1]$ in (21) relative to $\pi_1$ in (19).

The final loop concerns the *Keynesian cross*, shown in gray. As the central bank raises the interest rate $i_1$, it discourages household consumption $c_1$, leading firms to lower dividends

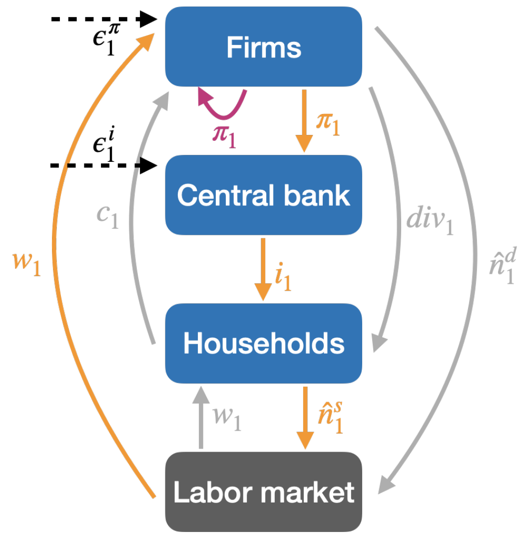



Figure 4: Causal relations and loops of period-1 economy

*Notes:* This figure illustrates the three loops of causal relations in the period-1 economy in different colors, to accompany the discussion of Propositions 4 and 5. Three loops are the pricing complementarity self-loop (in purple), the monetary policy loop (in orange) and the Keynesian cross (in gray).

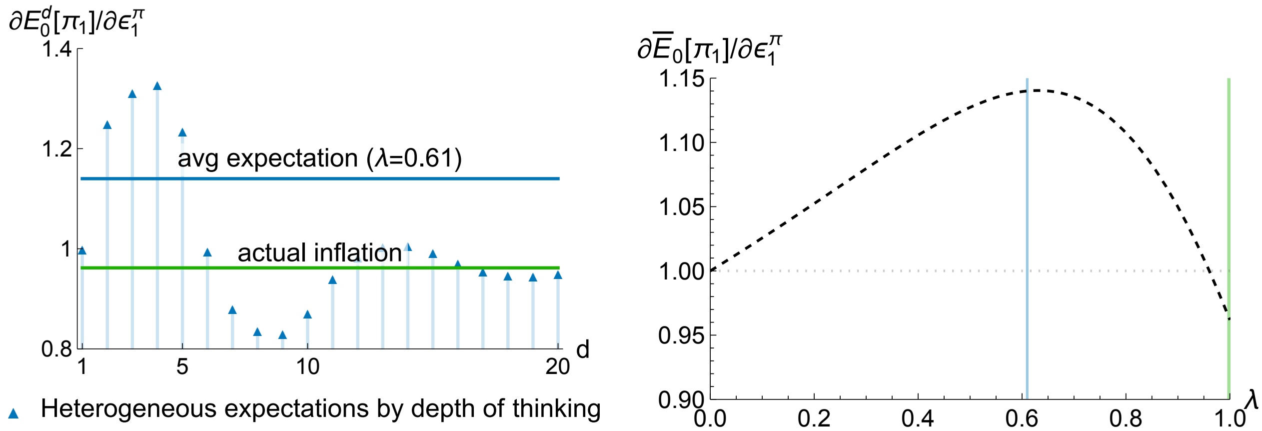
$div_1$ and reduce labor demand $\hat{n}^d$, resulting in a lower wage $w_1$. Consequently, households want to consume even less, triggering additional adjustments by firms. This Keynesian cross strengthens any effect that impacts households, thus compounding the monetary policy loop. Once again, summing the infinite geometric series leads the strength of this loop $\sigma^{-1}(\nu + \sigma)$ to appear in the denominator of $\pi_1$ in (19), with its dampening for expectations $\overline{\mathbb{E}}_0[\pi_1]$ in (21) captured by a polynomial $K_1(\lambda)$.

Overall, the average inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ in (21) is modified relative to the true inflation $\pi_1$ in (19), with different loops exponentially dampened by lengths. In contrast, other model parameters (such as the Taylor rule coefficient $\phi$) and any misperceptions thereof only change the perceived strength proportionally. This distinction highlights the quantitative importance of shallow thinking as we have alluded to.

Once we establish the inflation response, the equilibrium interest rate response $i_1$ in (20) follows directly as the Taylor rule coefficient $\phi$ times inflation. The average interest expectation $\overline{\mathbb{E}}_0[i_1]$ in (22) is $\lambda$ times $\phi$ times the inflation expectation, as it takes one more step for agents to appreciate the response of interest rate to inflation.

In the limit of $\lambda = 0$, all agents take only one step. They do not perceive any feedback on inflation, i.e., $\overline{\mathbb{E}}_0[\pi_1] = \epsilon_1^\pi$, and overlook changes in all other variables, e.g., $\overline{\mathbb{E}}_0[i_1] = 0$.

Figure 5 plots underlying heterogenous inflation expectations $\overline{\mathbb{E}}_0^d[\pi_1]$ in panel (a), and the average inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ as a function of $\lambda$ in panel (b), in response to cost-push shocks $\epsilon_1^\pi$. In panel (b), the blue vertical line indicates our calibrated $\lambda$, whereas the green vertical link corresponds to rational expectations, which coincides with the true equilibrium response $\pi_1$.




(a) Heterogeneous inflation expectations $\overline{\mathbb{E}}_0^d[\pi_1]$

(b) Average inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$

Figure 5: Beliefs in response to cost-push news shocks $\epsilon_1^\pi$

*Notes:* Panel (a) plots the heterogenous inflation expectations $\overline{\mathbb{E}}_0^d[\pi_1]$ (relative to the size of the shock) across individuals, in response to news about a cost-push shock $\epsilon_1^\pi$. Panel (b) plots the average inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ as a function of the shallow thinking parameter $\lambda$. The blue line indicates our calibration of $\lambda$, while the green line represents rational expectations ($\lambda = 1$).

As shown in Panel 5a, a depth-1 agent anticipates a one-to-one response of inflation to cost-push shock. As an agent thinks more deeply, they acknowledge the amplification of inflation through pricing complementarity and hence forecast higher inflation response. However, once they reason 5 steps, they start to recognize the monetary-policy loop which offsets inflation response. As agents take more steps in reasoning, their inflation expectations fluctuate non-monotonically and converge towards the rational expectation. Under our calibration of $\lambda = 0.61$, people on average understand only about 2.6 steps, leading to an average inflation expectation that exceeds the actual inflation response.

Panel 5b varies $\lambda$ and shows that the inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ is non-monotonic. Our calibration implies that the inflation expectation exceeds the size of the direct effect of one, whereas the true inflation response $\pi_1$ is below one. That is, shallow agents think that the cost-push shock will be amplified, though actually it will be offset. The mechanism is that, in determining inflation (19), there is a shorter amplification loop of pricing complementarity and a much longer offsetting loop via monetary policy response.

Shallow agents understand the shorter loop relatively better than the longer loop and perceive net amplification. This occurs even though the longer offset loop is actually stronger than the shorter amplification loop, meaning that the actual net effect is offsetting.

**Strength and length of GE feedback loops jointly determine belief misreaction.** We note that the predicted relationship between the inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ and the deviation from rational expectations under shallow thinking differs qualitatively from theories of sticky information (Mankiw and Reis, 2002) and cognitive discounting (Gabaix, 2020). They predict a monotonic increase in the inflation expectation from 0 to the true equilibrium value $\pi_1$, as one moves from full stickiness/myopia to full information rational expectations, and predict underreaction only.

At a high level, these theories predict a monotonic relationship because their agents understand the relative importance of multiple GE feedback loops and only dampen the net effect in their perception. In contrast, shallow thinking suggests that longer GE feedback loops are harder to understand. Under our calibration, shallow agents on average perceive amplification of inflation response rather than offset. Appendix D.3 further examines this mechanism by varying the strength of GE feedback.

## 4.2 Over- and Underreaction of Long-Term Rate

Here we study the response of interest rate expectations to a monetary policy shock $\epsilon_1^i$ and compare with the cost-push shock $\epsilon_1^\pi$ studied before. We characterize the period-1 equilibrium and period-0 expectations in Proposition 5.

**Proposition 5.** (Period-1 monetary policy shock) *The period-1 equilibrium in response to a monetary policy shock $\epsilon_1^i$ features*

$$i_1 = \left[1 + \frac{\phi\theta\kappa\sigma^{-1}(\nu+\sigma)}{1-(1-\theta)}\right]^{-1}\epsilon_1^i \tag{23}$$

$$\pi_1 = -\frac{\theta\kappa\sigma^{-1}(\nu+\sigma)}{1-(1-\theta)} i_1 \tag{24}$$

*whereas the period-0 average expectations upon observing the news about $\epsilon_1^i$ are*

$$\overline{\mathbb{E}}_0[i_1] = \left[1 + \frac{\lambda^4\phi\theta\kappa K_1(\lambda)}{1-\lambda(1-\theta)}\right]^{-1}\epsilon_1^i \tag{25}$$

$$\overline{\mathbb{E}}_0\left[\pi_1\right] = -\frac{\lambda^3\theta\kappa K_1\left(\lambda\right)}{1-\lambda\left(1-\theta\right)}\overline{\mathbb{E}}_0\left[i_1\right] \tag{26}$$

*with* $K_1(\lambda)$ *increasing in* $\lambda$ *under our calibration and* $K_1(1) = \sigma^{-1}(\nu+\sigma)$.

These results relate to those regarding the cost-push shocks in Proposition 4, but with a subtle and consequential difference. In this case, *all* general equilibrium effects offset the interest response to a monetary policy shock, whereas the inflation response to a cost-push shock involves both amplifying and offsetting feedback loops.

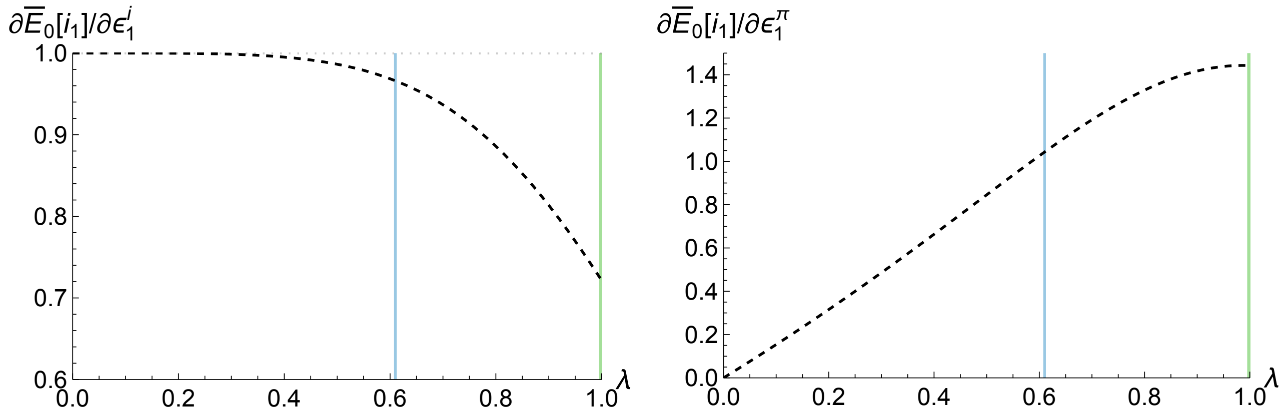
To appreciate that, we analyze the interest rate in (23), which is the sum of all $n$-step effects, as in (9). These effects belong to three different loops—pricing complementarity, the monetary policy loop, and the Keynesian cross—as previously established and displayed in Figure 4. Among the three loops, the interest rate response is directly involved in *only one*: the monetary policy loop, which offsets the interest rate response to a monetary policy shock in 4 steps—a higher interest rate $i_1$ encourages labor supply $\hat{n}_1^s$, which then lowers the real wage $w_1$, leading to lower inflation $\pi_1$ through firms' pricing decisions, ultimately prompting the central bank to lower the interest rate $i_1$ according to the Taylor rule. This 4-step monetary policy offset loop, with a strength $\phi\theta\kappa$, compounds with the other two loops—pricing complementarity and the Keynesian cross—which correspond to the $\frac{1}{1-(1-\theta)}$ and $\sigma^{-1}(\nu+\sigma)$ terms in (23). That occurs because pricing complementarity strengthens any effect on inflation, while the Keynesian cross reinforces any effect impacting households.

For the interest rate expectation $\overline{\mathbb{E}}_0[i_1]$ in (25), the 4-step monetary policy loop is dampened by $\lambda^4$, the pricing complementarity self-loop is dampened by $\lambda$, and the Keynesian cross is also dampened, captured by $K_1(\lambda)$, as in the prior case with cost-push shocks.

The equilibrium inflation response $\pi_1$ in (24) depends on the interest rate response $i_1$. It is compounded by the pricing complementarity loop $\frac{1}{1-(1-\theta)}$ and the Keynesian cross $\sigma^{-1}(\nu+\sigma)$, as any effect of the interest rate on inflation involves responses of both firms and households. The inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ derives accordingly from the interest rate expectation $\overline{\mathbb{E}}_0[i_1]$ but is additionally dampened by $\lambda^3$, since it takes 3 steps for the interest rate to affect inflation.

In the limit of $\lambda = 0$, shallow agents perceive no general equilibrium effects, and thus $\overline{\mathbb{E}}_0[i_1] = \epsilon_1^i, \overline{\mathbb{E}}_0[\pi_1] = 0$.

Figure 6 plots the average interest rate expectation $\overline{\mathbb{E}}_0[i_1]$ in response to monetary policy and cost-push news shocks respectively, both as functions of $\lambda$ in dashed black lines. The blue vertical line indicates our calibrated $\lambda$, whereas the green vertical link corresponds to rational expectations and the true equilibrium response $i_1$.




(a) Monetary policy news shock $\epsilon_1^i$

(b) Cost-push news shock $\epsilon_1^\pi$

Figure 6: Average interest rate expectations in response to two shocks

*Notes:* Panel (a) plots the average interest expectation $\overline{\mathbb{E}}_0[i_1]$ (relative to the size of the shock) as a function of the shallow thinking parameter $\lambda$, in response to news about a monetary policy shock $\epsilon_1^i$, and panel (b) plots its response to news about a cost-push shock $\epsilon_1^\pi$. The blue line indicates our calibration of $\lambda$, while the green line represents rational expectations ($\lambda = 1$).

Panel 6a indicates that the average interest rate expectation $\overline{\mathbb{E}}_0[i_1]$ overreacts to the monetary policy shock compared to the true response, thus suggesting that the period-0 long-term yield $y_0^{(2)}$ overreacts too according to (18). This aligns with a large body of literature on the excess sensitivity of long-term interest rates to monetary policy shocks, including Cochrane and Piazzesi (2002), Gürkaynak, Sack and Swanson (2005), and Hanson and Stein (2015), among others. The underlying mechanism is that, all general equilibrium feedback loops offset the interest rate response to a monetary policy shock, and as they are dampened to varying degrees, shallow agents perceive less offset than the true overall effect and thus overestimate the future interest rate.

In contrast, panel 6b suggests that the average interest rate expectation $\overline{\mathbb{E}}_0[i_1]$ underreacts to the cost-push shock compared to the true response, leading to underreaction of the period-0 long-term yield $y_0^{(2)}$. This occurs because shallow thinking agents underappreciate the Taylor rule, i.e., $\overline{\mathbb{E}}_0[i_1] = \lambda\overline{\mathbb{E}}_0[\pi_i]$, despite that the inflation expectations overreact. This aligns with findings that long-term interest rates responded too little to inflation surprises before the March 2022 interest rate hike (Bauer, Pflueger and Sunderam, 2024) and that even professional forecasters systematically underappreciate the extent of monetary easing during recessions (Cieslak, 2018; Wu, 2023).

As we over- and underreaction of long-term interest rates depending on the shocks, unconditionally, whether the long-term rate is excessively sensitive compared to the short rate depends on the relative importance of shocks. Indeed, Hanson, Lucca and Wright (2021) show that, after 2000, long-term interest rates are overly sensitive to changes in the short-term rates and predictably revert, suggesting overreaction. However, this pattern does not hold prior to 2000. Through the lens of our theory, one possibility is that there were more supply shocks (such as oil supply shocks) before 2000, contributing to underreaction in the mix and obscuring overreaction to monetary policy shocks.[19]

**Belief over- and underreaction across variables and shocks.** We have established a rich pattern of belief misreaction to news shocks, summarized in Table 4, which varies across variables and shocks. It contrasts with existing theories that predict underreaction to all shocks, such as sticky information (Mankiw and Reis, 2002) and cognitive discounting (Gabaix, 2020), or those that give rise to overreaction via extrapolation or diagnostic expectations (Barberis, 2018; Bordalo, Gennaioli and Shleifer, 2018). Shallow thinking predicts that the average expectations of inflation and interest rate overreact to shocks that directly affect themselves and underreact to shocks that directly affect other variables.

Table 4: Belief over- and underreaction in the New Keynesian economy

| News shock | Inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ | Interest rate expectation $\overline{\mathbb{E}}_0[i_1]$ |
|---|---|---|
| Cost-push $\epsilon_1^\pi$ | Overreaction | Underreaction |
| Monetary policy $\epsilon_1^i$ | Underreaction | Overreaction |

*Notes:* This table summarizes belief misreaction to different news shocks. The average expectations of variable $v$ overreact (compared to the true equilibrium) to shock $s$ if $\left|\frac{\partial \overline{\mathbb{E}}_0[v_1]}{\partial \epsilon_1^s}\right| > \left|\frac{\partial v_1}{\partial \epsilon_1^s}\right|$, and underreact otherwise.

## 4.3 Predictability of Bond Excess Returns

Building on previous results, here we consider the bond excess returns in an economy impacted by both the cost-push shock $\epsilon_1^\pi$ and the monetary policy shock $\epsilon_1^i$. We show shallow thinking implies that bond excess returns can be predicted by macro variables, controlling for current yields, as established empirically by Ludvigson and Ng (2009),

[19]We show the comparative statics of this univariate coefficient of long rate on short rate with respect to the relative importance of monetary policy shocks in Appendix D.4, under persistent shocks.

Cooper and Priestley (2009), Joslin, Priebsch and Singleton (2014), Cieslak and Povala (2015), and Cieslak (2018), among others.

We consider the excess return of holding a 2-period (i.e., long-term) bond from period 0 to period 1, relative to holding a 1-period (i.e., short-term) bond,[20]

$$xr^{(2)}_{0\to 1} \equiv \underbrace{-i_1 + 2y^{(2)}_0}_{\text{return of long-term bond}} - \underbrace{i_0}_{\text{return of short-term bond}} = \overline{\mathbb{E}}_0[i_1] - i_1 \tag{27}$$

where the second equality follows from (18). When the period-0 expectation of period-1 interest rate $\overline{\mathbb{E}}_0[i_1]$ exceeds its actual value $i_1$, the long-term bond is undervalued in period 0 and will appreciate in period 1, leading to a positive excess return, and vice versa.

In particular, we study the predictability of bond excess return $xr^{(2)}_{0\to 1}$ by the average inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$, as noted by Joslin, Priebsch and Singleton (2014) and Cieslak (2018), controlling for the forward rate $\overline{\mathbb{E}}_0[i_1]$,[21]

$$\underbrace{xr^{(2)}_{0\to 1}}_{\text{bond excess return}} = \beta_\pi \underbrace{\overline{\mathbb{E}}_0[\pi_1]}_{\text{inflation expectation}} + \beta_i \underbrace{\overline{\mathbb{E}}_0[i_1]}_{\text{forward rate}} + \alpha + \epsilon_{0\to 1} \tag{28}$$

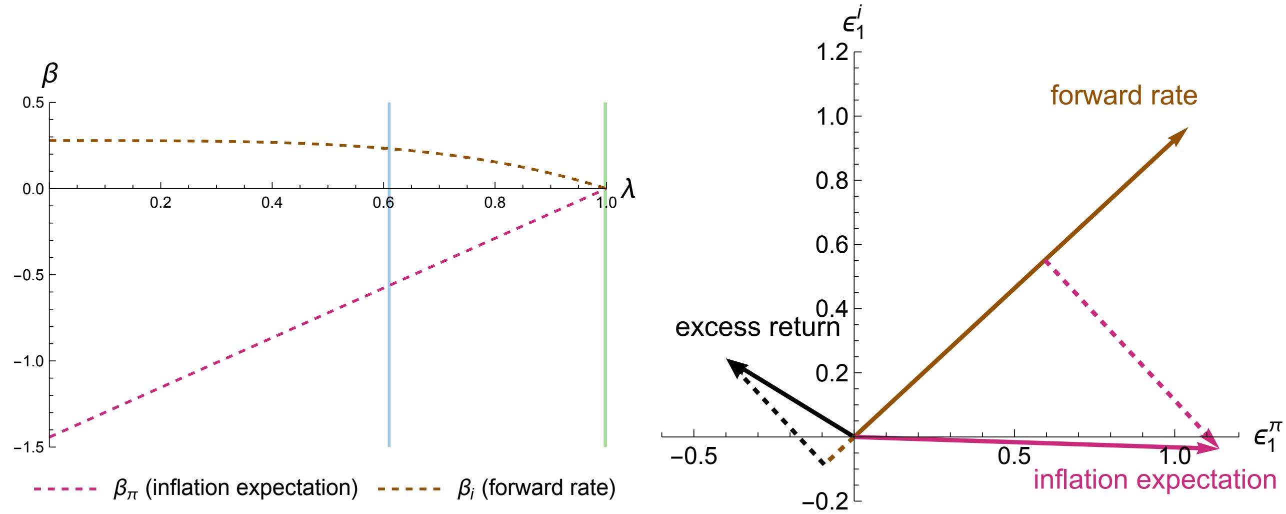
Figure 7a illustrates the theory-implied coefficients $\beta_\pi$ and $\beta_i$ as functions of $\lambda$, in brown and purple respectively. As long as $\lambda < 1$, our theory predicts a negative $\beta_\pi$, which is what Joslin, Priebsch and Singleton (2014) and Cieslak (2018) find.

To understand the mechanism, we start by examining the limits of $\lambda = 0$ and 1 and build intuition with shock loadings illustrated in Figure 7b. In the limit of $\lambda = 0$, we have $\overline{\mathbb{E}}_0[\pi_1] = \epsilon^\pi_1, \overline{\mathbb{E}}_0[i_1] = \epsilon^i_1$ from Propositions 4 and 5. As the interest rate expectation underreacts to the cost-push shock $\epsilon^\pi_1$ and overreacts to the monetary policy shock $\epsilon^i_1$, the excess return $xr^{(2)}_{0\to 1}$ loads negatively on the former and positively on the latter, as determined by (27), i.e., $\beta_\pi < 0, \beta_i > 0$, further illustrated in Figure D2a. In the limit of rational expectations ($\lambda = 1$), and only in this case, the excess return $xr^{(2)}_{0\to 1}$ is 0, as there is no expectational error. Thus, both coefficients $\beta_\pi$ and $\beta_i$ are 0, illustrated in Figure D2b.

We establish that $\beta_\pi < 0$ for all $\lambda \in (0, 1)$. To see that, as shown in Figure 7b, as long as $\lambda > 0$ (i.e., some understanding of shock propagation), the inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$

[20] The return of the 2-period bond is the difference between its period-1 log price deviation, $-i_1$, and its period-0 log price derivation, $-2y^{(2)}_0$.

[21] This regression, with expectations as predictors, is equivalent to a bivariate Coibion and Gorodnichenko (2015) regression. The coefficients reflect the misappreciation of causal relations between variables, which Wu (2023) systematically analyzes across many variable pairs.




(a) Coefficients $\beta_\pi$ and $\beta_i$

(b) Loadings on news shocks (calibrated $\lambda$)

Figure 7: Predictability of bond excess returns, with news shocks

*Notes:* Panel (a) plots the theory-implied coefficients $\beta_\pi$ and $\beta_i$ from the predictive regression of bond excess returns $xr^{(2)}_{0\to1}$ in (28) on the inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ and forward rate $\overline{\mathbb{E}}_0[i_1]$, as functions of the shallow thinking parameter $\lambda$. The period-0 economy is impacted by news about period-1 cost-push shocks $\epsilon_1^\pi$ and monetary policy shocks $\epsilon_1^i$. The blue vertical line corresponds to our calibration of $\lambda$, while the green line represents rational expectations ($\lambda = 1$).
Panel (b) illustrates the loadings of bond excess return $xr^{(2)}_{0\to1}$, inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ and forward rate $\overline{\mathbb{E}}_0[i_1]$ on the cost-push news shock $\epsilon_1^\pi$ (x-axis) and the monetary policy news shock $\epsilon_1^i$ (y-axis), as solid vectors. The vectors are determined under our calibration of $\lambda$, but their quadrant placements are generically true when $\lambda \in (0,1)$. Figures D2a and D2b show the cases with $\lambda = 0, 1$. The two dashed vectors show the residuals of bond excess return $xr^{(2)}_{0\to1}$ and inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ projected onto the forward rate $\overline{\mathbb{E}}_0[i_1]$. Their opposite directions imply a negative $\beta_\pi$, following the Frisch-Waugh-Lovell theorem.

lies in the fourth quadrant as it loads positively on the cost-push shock and negatively on the monetary policy shock, and the forward rate $\overline{\mathbb{E}}_0[i_1]$ is in the first quadrant as it loads positively as both shocks. If $\lambda < 1$ (i.e., not fully understanding shock propagation), the excess return $xr^{(2)}_{0\to1}$ is placed in the second quadrant as the interest rate expectation overreacts to the monetary policy shock and underreacts to the cost-push shock. This figure also shows the residuals of $xr^{(2)}_{0\to1}$ and $\overline{\mathbb{E}}[\pi_1]$ projected onto $\overline{\mathbb{E}}_0[i_1]$, denoted $xr^{(2)}_{0\to1}|\overline{\mathbb{E}}_0[i_1]$ and $\overline{\mathbb{E}}_0[\pi_1]|\overline{\mathbb{E}}_0[i_1]$ respectively, as dashed vectors. These two residuals point to opposite directions due to the quadrant placements of the three vectors involved. According to the Frisch-Waugh-Lovell theorem, the coefficient $\beta_\pi$ from the bivariate regression (28) equals the univariate regression coefficient of the residual $xr^{(2)}_{0\to1}|\overline{\mathbb{E}}_0[i_1]$ on the residual $\overline{\mathbb{E}}_0[\pi_1]|\overline{\mathbb{E}}_0[i_1]$. Opposite directions of these residuals hence imply a negative $\beta_\pi$.

In this subsection, we show that that inflation expectations negatively predict bond excess returns, in an economy with two shocks. In reality, the economy is impacted by multiple shocks. Mathematically, in a linear model with $N$ shocks, $N$ independent predictors from the yield curve would span all shocks. However, in practice, the entire yield curve is well captured by the first three principal components.[22] Our theory suggests that other macro variables may contain additional information about non-monetary-policy shocks, and can therefore predict bond excess returns, as established empirically.

## 4.4 Cost-Push News Shocks Cause Inflation

So far we have focused on belief misreaction and asset prices. Based on these results, here we establish that news about period-1 cost-push shocks is inflationary, rather than deflationary as predicted by rational expectations, and less contractionary.

**Period-0 equilibrium.** In period 0, the shocks have not materialized, but firms and households are forward-looking. The period-0 equilibrium arises from firms' and households' optimal behavior, given their perfect observation of all period-0 variables and their beliefs about period-1 outcomes, along with the central bank's Taylor rule and the labor market clearing condition.

The period-0 equilibrium consists of seven variables $\left\{i_0, \pi_0, div_0, n_0^d, c_0, n_0^s, w_0\right\}$, similar to the period-1 equilibrium. Among these seven variables, three of them (the interest rate $i_0$, dividend $div_0$ and labor demand $n_0^d$) depend only on the contemporaneous values of the other variables. These contemporaneous causal relations are the same as their period-1 counterparts in Section 2.1. Three variables (inflation $\pi_0$, consumption $c_0$ and labor supply $n_0^s$) depend on agents' beliefs about period-1 outcomes, since firms's pricing decisions and households' consumption and labor supply decisions are forward-looking, detailed next. Last, the wage $w_0$ arises from the labor market clearing condition $n_0^s = n_0^d$.

The period-0 inflation $\pi_0$ satisfies

$$\pi_0 = \theta\kappa\left(w_0 + \beta\theta\overline{\mathbb{E}}_0\left[w_1\right]\right) + (1-\theta)\left(\pi_0 + \beta\theta\overline{\mathbb{E}}_0\left[\pi_1\right]\right) \tag{29}$$

which increases in expectations of both future real wage $w_1$ and inflation $\pi_1$, as firms want to front run a higher future marginal cost.

[22]Indeed, in our model, in the presence of multiple shocks with the same persistence, all forward rates are collinear with each other, but not with other macro variables or expectations thereof.

The period-0 consumption $c_0$ and labor supply $n_0^s$ follow

$$c_0 = -\sigma^{-1}\beta\left(i_0 - \overline{\mathbb{E}}_0[\pi_1] + \beta\overline{\mathbb{E}}_0[i_1]\right) + \frac{(1-\beta)(\mu-1)\nu}{\sigma+\mu\nu}\left(div_0 + \overline{\mathbb{E}}_0[div_1]\right) + \frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu}\left(w_0 + \overline{\mathbb{E}}_0[w_1]\right) \tag{30}$$

$$n_0^s = \nu^{-1}\beta\left(i_0 - \overline{\mathbb{E}}_0[\pi_1] + \beta\overline{\mathbb{E}}_0[i_1]\right) - \frac{(1-\beta)(\mu-1)\sigma}{\sigma+\mu\nu}\left(div_0 + \overline{\mathbb{E}}_0[div_1]\right) + \nu^{-1}\left[1 - \sigma\frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu}\right]w_0 - \frac{(1-\beta)(1+\nu)\nu^{-1}\sigma}{\sigma+\mu\nu}\overline{\mathbb{E}}_0[w_1] \tag{31}$$

Households react to expectations of the future interest rate, dividend, and wage, as well as future inflation, as a higher inflation lowers the real interest rate from period 0 to 1.

In determining the period-0 equilibrium, the only decision-relevant beliefs are about inflation $\pi_1$, wage $w_1$, interest rate $i_1$ and dividend $div_1$. Figure 8a illustrates the determination of period-0 equilibrium, where these beliefs act like shocks to firms and households.[23]

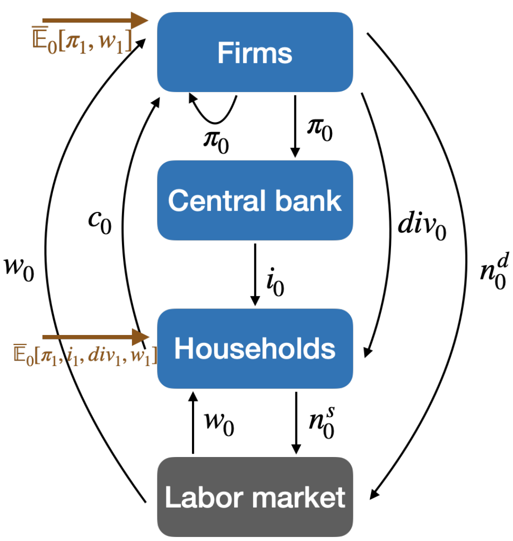
In particular, the inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ serves two roles here: it acts like a cost-push shock for firms as they want to front run future inflation, and it functions as a demand shock for households since a higher inflation lowers the real interest rate. This observation will prove useful when we discuss the effects of cost-push shocks next.

**Cost-push news shocks cause inflation.** Figure 8b depicts the period-0 inflation $\pi_0$ and output $y_0$ (which is simply the consumption $c_0$), in response to news about a period-1 cost-push shock $\epsilon_1^\pi$. Unlike the period-1 equilibrium, which is independent of $\lambda$, the period-0 equilibrium does depend on $\lambda$. The blue line indicates the equilibrium under our calibration of $\lambda$, whereas the green line stands for the rational expectations equilibrium.

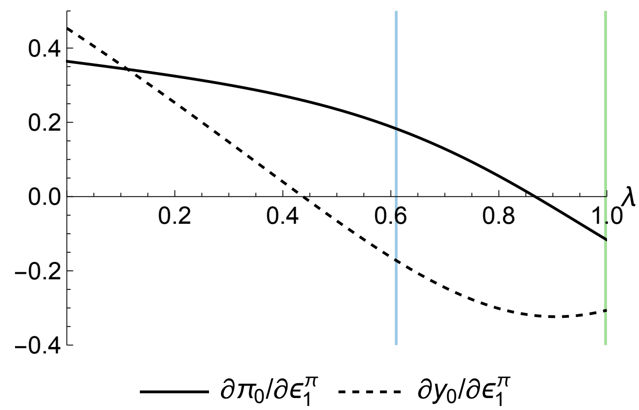
A cost-push news shock is *inflationary* in period 0 under the calibrated shallow thinking parameter $\lambda$, consistent with empirical evidence (Känzig, 2021), but *deflationary* under rational expectations. The latter occurs because rational agents understand that the central bank will raise the interest rate $i_1$ in response to higher inflation $\pi_1$, causing a contraction of the period-1 economy. Anticipating that, households cut back on their consumption in period 0, resulting in a period-0 output contraction as well.

In contrast, in the limit of extremely shallow agents ($\lambda = 0$), they only perceive a change

[23] In Figure 8a that determines the equilibrium given beliefs, we equalize the actual labor demand and supply $n_0^d, n_0^s$ as the market clearing condition. We only introduce fictitious Walrasian auctioneers for the purpose of characterizing beliefs. In the equilibrium given any beliefs, markets always clear.




(a) Period-0 equilibrium




(b) Period-0 inflation $\pi_0$ and output $y_0$, in response to cost-push news shocks $\epsilon_1^\pi$

Figure 8: Period-0 equilibrium in response to news shocks

*Notes:* Panel (a) demonstrates the period-0 equilibrium in response to transitory news shocks, which are period-1 shocks observed in period 0, driven by firms' and households' beliefs about period-1 outcomes. Panel (b) plots the period-0 inflation $\pi_0$ and output $y_0$ (which equals consumption $c_0$) in response to news about a cost-push shock $\epsilon_1^\pi$, normalized relative to the size of the shock, as functions of the shallow thinking parameter $\lambda$. The solid line stands for inflation $\pi_0$ and the dashed line represents output $y_0$. The blue vertical line indicates the equilibrium under our calibration of $\lambda$, while the green vertical line represents the rational expectations equilibrium with $\lambda = 1$.

in future inflation, $\overline{\mathbb{E}}_0[\pi_1] = \epsilon_1^\pi$, but not in any other variable. As we have established with (29-31), inflation expectations act like a cost-push shock for firms and a demand shock for households. As a result, firms set higher prices and households consume more and work less, leading to period-0 inflation and output expansion.

With our calibrated $\lambda$, agents partially expect an economic downturn in period 1. In this case, news about period-1 cost-push shocks causes inflation instead of deflation and leads to a smaller output contraction compared to the rational expectations equilibrium.

In terms of the inflation response to cost-push news shocks, shallow thinking reverses the sign relative to the rational expectations equilibrium to align with empirical evidence. It happens as beliefs of different variables misreact differently to the same shock, and they have counteracting effects on behavior. In contrast, theories of sticky information (Mankiw and Reis, 2002) and cognitive discounting (Gabaix, 2020) preserve the sign of the rational expectations equilibrium but modify only the magnitude of period-0 deflation.

**Transitory news shocks inform persistent shocks.** So far, we have focused on transitory news shocks, but this analysis is more informative than it may seem. As we have noted at the beginning of this section, comparing theories of beliefs under transitory news shocks is equivalent to comparing them under 2-period shocks, since the period-0 equilibrium response to contemporaneous shocks is independent of belief formation.

Furthermore, analyzing transitory news shocks informs persistent shocks. Under a first-order approximation, a persistent shock is the sum of a current shock and a series of news shocks. Shallow thinking implies that any news cost-push shock is inflationary in period 0, whereas rational expectations predict otherwise. First, it follows that a persistent cost-push shock is more inflationary under shallow thinking. Second, as a *more* persistent cost-push shock is the sum of more news shocks, shallow thinking predicts *higher* current inflation, contrary to lower current inflation as predicted by rational expectations.

Next, we will generalize shallow thinking to persistent shocks to substantiate these two claims. The generalization will address a missing piece from the intuitive analysis above: in responding to a persistent shock, agents must think about how variables in future period $t_1$ depend on shocks in further future periods $t_2$.

# 5 Persistent Shocks and Dynamic General Equilibrium

We generalize to persistent shocks in the New Keynesian economy and show that the insights gained from transitory news shocks still hold, while new lessons emerge. In particular, more persistent cost-push shocks lead to higher inflation under shallow thinking, in contrast to lower inflation predicted under rational expectations.

## 5.1 Causal Representation, Shallow Thinking Beliefs, and Equilibrium

We introduce shallow thinking of dynamic general equilibrium, by focusing on the cross-variable causal relations and abstracting away from the cross-horizon dimension. We represent the infinite-horizon New Keynesian model as a system of causal relations in the sequence space, generalizing Section 2. This representation of dynamic general equilibrium also applies to the RBC model studied in Appendices C and F.

We consider persistent shocks $\{\epsilon_t\}_{t\geq 0}$ that are observed at time $0^-$, and assume that agents form expectations $\mathbb{E}^d[\cdot]$ once and for all at time $0^-$, i.e., $\mathbb{E}_t^d[v_s] \equiv \mathbb{E}^d[v_s]$ for $s > t$. This assumption simplifies the analysis, as it nests the full information rational expectations benchmark and is a reasonable starting point.[24]

The infinite-horizon New Keynesian model is characterized by sequences of seven variables $\left\{i_t, \pi_t, div_t, n_t^d, c_t, n_t^s, w_t\right\}_{t\geq 0}$. We will refer to this collection of variables as $\mathcal{V}$. The first six variables are agents' actions, which we collect as $\mathcal{V}^{action}$, whereas the last is a price formed in the competitive labor market. Note that we start with the labor supply and demand $n_t^s, n_t^d$ in order to reinterpret them as supply and demand curves shortly. In that reinterpretation, with only slight abuse of notation, we will also use $\mathcal{V}$ and $\mathcal{V}^{action}$ to denote variables of interests with $n^s, n^d$ replaced by $\hat{n}^s, \hat{n}^d$.

Among these, three variables $i_t, div_t, n_t^d$ only depend on contemporaneous values of other variables, laid out in (D1, D2, D6). Three variables $\pi_t, c_t, n_t^s$ are forward-looking, capturing best responses of forward-looking firms and households—inflation $\pi_t$ is linear in $w_t, \pi_t$ and $\left\{\overline{\mathbb{E}}_t\left[w_{t+k}\right], \overline{\mathbb{E}}_t\left[\pi_{t+k}\right]\right\}_{k\geq 1}$, and consumption and labor supply $c_t, n_t^s$ are linear in $i_t, div_t, w_t$ and $\left\{\overline{\mathbb{E}}_t\left[i_{t+k}\right], \overline{\mathbb{E}}_t\left[\pi_{t+k}\right], \overline{\mathbb{E}}_t\left[div_{t+k}\right], \overline{\mathbb{E}}_t\left[w_{t+k}\right]\right\}_{k\geq 1}$, detailed in (D3-D5). The last variable $w_t$ arises from equilibrating labor supply $n_t^s$ and demand $n_t^d$. These equations completely characterize the equilibrium, given beliefs.

**REE as causal relations in sequence space.** To determine shallow thinking beliefs, we first characterize the rational expectations equilibrium (REE) in terms of causal relations. For REE, by replacing each expectation $\overline{\mathbb{E}}_t\left[v_\tau\right]$ with the true outcome $v_\tau$ under rational expectations, the six variables in $\mathcal{V}^{action}$ that agents choose can be represented in the sequence space, following Auclert, Rognlie and Straub (2024) and Auclert et al. (2021), as

$$\mathbf{v}^{REE} = \sum_{\mathbf{u}\in\mathcal{V}} \mathbf{J}_{\mathbf{vu}}^{REE}\mathbf{u}^{REE} + \epsilon^v, \ \forall \mathbf{v} \in \mathcal{V}^{action} \tag{32}$$

where

$$\left(\mathbf{J}_{\mathbf{vu}}\right)_{ts} \equiv \begin{cases} \frac{\partial v_t}{\partial u_s} & s \leq t \\ \frac{\partial v_t}{\partial \overline{\mathbb{E}}[u_s]} & s > t \end{cases} \tag{33}$$

[24]We believe this is reasonable beyond convenience for two reasons. First, an important reason for expectation updates over time is that agents gradually learn about shocks. We focus on the rationality of beliefs in the absence of any information frictions, and our survey design mimics this environment. Second, expectations can update as agents learn how aggregate variables respond to shocks *through repeated experiences*, similar to how economists study shock propagation using time series data. We leave this for future research, but we note that in our survey we consider conventional shocks, such as oil supply shocks and monetary policy shocks. These are age-old shocks, so our estimation already incorporates knowledge that people have accumulated over time. This contrasts with unconventional policy shocks (such as forward guidance), to which people had no prior exposure until recently.

is the Jacobian of the sequence of one variable $\mathbf{v} \equiv (\{v_t\}_{t\geq 0})'$ with respect to the sequence of another variable $\mathbf{u} \equiv (\{u_t\}_{t\geq 0})'$, and $\boldsymbol{\epsilon}^v \equiv \left(\left\{\epsilon_t^v\right\}_{t\geq 0}\right)'$ denotes the sequence of a structural shock. The Jacobians $\mathbf{J}_{\mathbf{vu}}$ are upper triangular matrices in the New Keynesian model, which is purely forward-looking.[25]

For labor supply and demand $\mathbf{n}^s, \mathbf{n}^d$, by separating their dependence on the wage $\mathbf{w}$ from the rest, we interpret them as supply and demand curves in the sequence space

$$\mathbf{v}^{REE} = \mathbf{J}_{\mathbf{vw}}\mathbf{w}^{REE} + \hat{\mathbf{v}}^{REE},\ \mathbf{v} \in \left\{\mathbf{n}^s, \mathbf{n}^d\right\}$$

with elasticities $\mathbf{J}_{\mathbf{n}^s\mathbf{w}}, \mathbf{J}_{\mathbf{n}^d\mathbf{w}}$ and shifts $\hat{\mathbf{n}}^{s,REE}, \hat{\mathbf{n}}^{d,REE}$ defined as

$$\hat{\mathbf{v}}^{REE} = \sum_{\mathbf{u}\in\mathcal{V}\backslash\{\mathbf{w}\}} \mathbf{J}_{\mathbf{vu}}\mathbf{u}^{REE} + \boldsymbol{\epsilon}^v,\ \mathbf{v} \in \left\{\mathbf{n}^s, \mathbf{n}^d\right\} \tag{34}$$

In the New Keynesian model, demand elasticity $\mathbf{J}_{\mathbf{n}^d\mathbf{w}}$ is a matrix of zeros in this model since firms only use labor inputs and supply elasticity $\mathbf{J}_{\mathbf{n}^s\mathbf{w}}$ is an upper triangular matrix, as households' labor supply responds to current and future wages.

By equalizing supply and demand $\mathbf{n}^{s,REE} = \mathbf{n}^{d,REE}$, we can interpret the wage $\mathbf{w}^{REE}$ as resulting from supply and demand shifts $\hat{\mathbf{n}}^{s,REE}, \hat{\mathbf{n}}^{d,REE}$

$$\mathbf{w}^{REE} = (\mathbf{J}_{\mathbf{n}^s\mathbf{w}} - \mathbf{J}_{\mathbf{n}^d\mathbf{w}})^{-1}\left(\hat{\mathbf{n}}^{d,REE} - \hat{\mathbf{n}}^{s,REE}\right) \tag{35}$$

This rule of price determination generalizes its counterpart in the period-1 economy (7).

Taking stock of the rational expectations equilibrium, formula (32) describes all agents' actions $\mathcal{V}^{action}$, where we use $\hat{\mathbf{n}}^{s,REE}, \hat{\mathbf{n}}^{d,REE}$ as in (34) instead of $\mathbf{n}^{s,REE}, \mathbf{n}^{d,REE}$, and equation (35) characterizes the wage from the competitive labor market. We stack the sequences of outcomes to form a long vector $\mathbf{V} \equiv (\{\mathbf{v}'\}_{v\in\mathcal{V}})'$, sequences of shocks as $\mathbf{S} \equiv \left(\left\{\boldsymbol{\epsilon}^{v'}\right\}_{v\in\mathcal{V}}\right)'$, and correspondingly stack the Jacobians as a giant matrix $\mathbf{M} \equiv \left(\{\mathbf{J}_{\mathbf{vu}}\}_{v,u\in\mathcal{V}}\right)$. $\mathbf{M}$ captures all *causal relations* in dynamic general equilibrium, which are now defined as the *sequence-space Jacobians*, instead of partial derivatives in the period-1 economy.

**Proposition 6.** (Dynamic GE as a system of causal relations in the sequence space) *The rational expectations equilibrium in the New Keynesian economy is characterized by the fixed point*

[25](32) can accommodate optimal choices that depend on past changes, such as investment in the RBC model. In that case, we interpret $v_t$ as the time-$0^-$ planned optimal action at time $t$ as a function of the perceived sequence of decision-relevant variables $\left\{\overline{\mathbb{E}}\left[u_s\right]\right\}_{s\geq 0}$ and define the Jacobian accordingly.

*to the system of causal relations in the sequence space among all* agents' actions and competitive prices, $\mathbf{V} \equiv \left(\mathbf{i}', \boldsymbol{\pi}', \mathbf{div}', \hat{\mathbf{n}}^{d'}, \mathbf{c}', \hat{\mathbf{n}}^{s'}, \mathbf{w}'\right)'$, *as*

$$\underbrace{\mathbf{V}^{REE}}_{\text{sequence of variables}} = \underbrace{\mathbf{M}}_{\text{causal relations in sequence space}} \mathbf{V}^{REE} + \underbrace{\mathbf{S}}_{\text{sequence of shocks}} \tag{36}$$

*The rational expectations equilibrium can be solved to yield*

$$\mathbf{V}^{REE} = (\mathbf{I} - \mathbf{M})^{-1} \mathbf{S} = \sum_{n=1}^{\infty} \mathbf{M}^{n-1} \mathbf{S} \tag{37}$$

*which is a sum of all equilibrium effects of varying distance, where each* $\mathbf{M}^{n-1}\mathbf{S}$ *term is an n-step effect of the sequence of a shock on the sequence of a variable via sequences of* $n-1$ *intermediate variables.*

This proposition generalizes Proposition 1 in the case of period-1 general equilibrium to dynamic general equilibrium with persistent shocks.

**Shallow thinking.** With Proposition 6, we generalize Assumption 1 to capture shallow thinking about dynamic general equilibrium, by replacing $(V, M, S)$ with the long vector and stacked Jacobian $(\mathbf{V}, \mathbf{M}, \mathbf{S})$. Consequently, the average expectations characterized by Proposition 2 is generalized, also by replacing $(V, M, S)$ with $(\mathbf{V}, \mathbf{M}, \mathbf{S})$.

As alluded to before, we focus on the causal relations across variables and abstract away from the horizon-dependent dampening studied by Angeletos and Lian (2018), Farhi and Werning (2019) and Gabaix (2020). We assume that if agents understand how consumption $c_t$ depends on the contemporaneous interest rate $i_t$, they also understand how $c_t$ depends on future $i_s$. This is reflected in that the dependence of $\mathbb{E}^d[c_t]$ on $\mathbb{E}^{d-1}[\mathbf{i}]$ is mediated via the causal relations $\mathbf{M}$, which collects all the Jacobians $\mathbf{J_{vu}}$ in (33). One could further generalize our theory to introduce additional dampening across periods, by modifying $\mathbf{M}$.[26]

**Equilibrium under shallow thinking.** We study a *temporary equilibrium* in which agents maximize their utilities, taking as given the average expectations $\overline{\mathbb{E}}_t[\cdot]$, and markets clear.

[26]For example, by replacing $(\mathbf{J_{vu}})_{ts}$ with $m^{s-t}(\mathbf{J_{vu}})_{ts}$ for $s > t$ with a factor $m < 1$ and adjusting $\mathbf{M}$ accordingly, we can allow for the possibility that agents underappreciate the dependence of one variable on future variables relative to its dependence on contemporaneous variables.

Specifically, given the average beliefs $\overline{\mathbb{E}}[\cdot]$, the equilibrium $\left\{i_t, \pi_t, div_t, n_t^d, c_t, n_t^s, w_t\right\}_{t\geq 0}$ is characterized period by period by firms' and households' optimal behavior given their perfect observation of time-$t$ variables and expectations about future variables, as in (D1-D5), along with the Taylor rule (D6) and the labor market clearing condition (D7).[27]

## 5.2 More Persistent Cost-Push Shocks Cause Higher Inflation

We consider the responses of shallow thinking beliefs and equilibrium to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$. In particular, we highlight that a more persistent (higher $\rho$) cost-push shock leads to higher inflation under shallow thinking, contrary to the prediction of rational expectations.

**REE and shallow thinking beliefs.** In response to an exponentially decaying shock, both beliefs and equilibrium outcomes decay exponentially at the same rate. Proposition 7 establishes the rational expectations equilibrium (REE) and shallow thinking beliefs, characterizing their time-$t$ values relative to the time-$t$ size of the shock $\epsilon_t^\pi$.

**Proposition 7.** (Persistent cost-push shock) *The rational-expectations equilibrium (REE) response to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$ features*

$$\pi_t^{REE} = \left[1 - \frac{(1-\theta) + \left(\rho - \phi\right)\theta\kappa \frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}}{1-\beta\theta\rho}\right]^{-1} \epsilon_t^\pi \tag{38}$$

$$i_t^{REE} = \phi \pi_t^{REE} \tag{39}$$

*whereas the average expectations under shallow thinking are*

$$\overline{\mathbb{E}}[\pi_t] = \left[1 - \frac{\lambda(1-\theta) + \left(\lambda^3\rho - \lambda^4\phi\right)\theta\kappa K(\lambda,\rho)}{1-\beta\theta\rho}\right]^{-1} \epsilon_t^\pi \tag{40}$$

$$\overline{\mathbb{E}}[i_t] = \lambda\phi\overline{\mathbb{E}}[\pi_t] \tag{41}$$

[27] Notably, if we consider the set of all possible temporary equilibrium outcomes (supported by arbitrary beliefs), shallow thinking expectations do not fall into that set. That is because shallow thinking agents underappreciate contemporaneous causal relations that hold in any equilibrium (e.g., agents' best responses to contemporaneous conditions, Taylor rule, and the labor market clearing condition). In contrast, level-$k$ thinking beliefs fall into the set of temporary equilibrium outcomes, since level-$k$ thinking agents only make assumptions about other agents' beliefs but perfectly understand the contemporaneous causal relations.

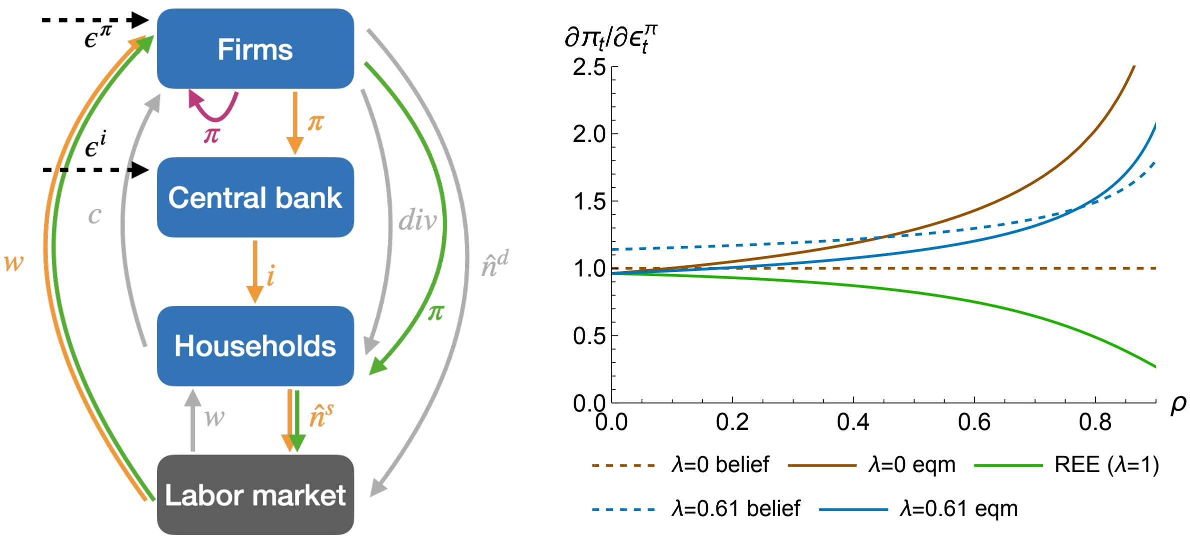
*with $K(\lambda, \rho)$ increasing in $\lambda$ and $\rho$ under our calibration and $K(1, \rho) = \frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}$.*

This proposition shows how exactly the persistence of shock $\rho$ matters, nesting Proposition 4 with $\rho = 0$. A positive $\rho$ gives rise to new terms and modifies existing terms, which we dissect in order by analyzing REE inflation (38) and shallow expectations (40).

Regarding new terms, four general equilibrium loops across sequences of variables are involved, instead of three, displayed in Figure 9a. Relative to the case with period-1 shocks (Figure 4), persistent shocks give rise to new causal relations—the responses of household consumption and labor supply to inflation $\pi$, plotted in green. This results in the fourth loop, the *inflation-on-households loop* (in green), in addition to pricing complementarity (in purple), the monetary policy loop (in orange), and the Keynesian cross (in gray). This inflation-on-household loop takes 3 steps to close and amplifies the inflation response to cost-push shocks, since a higher inflation $\pi$ simultaneously encourages consumption $c$ and discourages labor supply $\hat{n}^s$, which leads to a higher wage $w$, feeding into inflation $\pi$. As a result, this loop is dampened by $\lambda^3$ in expectations (40). Further, the strength of this loop is proportional to $\rho$, meaning that it only exits when $\rho > 0$ and is stronger when $\rho$ is higher. It is because only future inflation impacts household behavior by changing the real interest rate, and the expected inflation decays with rate $\rho$.

Concerning the strength of these terms, a positive $\rho$ does two things, in addition to activating the inflation-on-households loop. First, it strengthens the general equilibrium feedback loops by $\frac{1}{1-\beta\theta\rho}$, because firms' pricing decisions are forward-looking and respond to a sum of future changes discounted by $\beta\theta$. Second, it boosts the Keynesian cross via $K(\lambda, \rho)$. This additional boost occurs because households' decisions are also forward-looking. Since the Keynesian cross reflects the feedback between firms and households, the effect of persistence is compounded. As $\rho$ increases, all general equilibrium feedback loops become stronger, and the strengthening of inflation-on-households and monetary policy loops compounds with that of the Keynesian cross.

**More persistent cost-push shocks cause higher inflation.** Figure 9b plots the average inflation expectation $\overline{\mathbb{E}}[\pi_t]$ and the equilibrium inflation $\pi_t$, both relative to the time-$t$ size of the shock, as functions of persistence $\rho$. The dashed lines indicate expectations, and the solid lines represent the equilibrium. Different colors correspond to different values of $\lambda$, with $\lambda = 0$ in brown, our calibrated $\lambda$ in blue, and $\lambda = 1$ (rational expectations) in green. In the case with rational expectations, the beliefs coincide with the equilibrium.




(a) Causal relations in the sequence space

(b) Inflation $\pi_t$ in response to cost-push shock $\epsilon_t^\pi$

Figure 9: New Keynesian economy with persistent shocks

*Notes:* Panel (a) illustrates the four loops of causal relations across sequences of variables in the New Keynesian economy with persistent shocks in different colors, to accompany the discussion of Proposition 7. Four loops are the pricing complementarity self-loop (in purple), the inflation-on-households loop (in green), the monetary policy loop (in orange) and the Keynesian cross (in gray).
Panel (b) plots the equilibrium inflation $\pi_t$ and the average expectation $\overline{\mathbb{E}}[\pi_t]$ (relative to the size of the shock) in response to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$, as functions of the persistence $\rho$, under different values of $\lambda$. The dashed lines indicate expectations, and the solid lines represent the equilibrium. Different colors stand for different values of $\lambda$, with $\lambda = 0$ in brown, our calibration of $\lambda$ in blue, and $\lambda = 1$ (rational expectations) in green. Under $\lambda = 1$, the beliefs coincide with the rational expectations equilibrium (REE).

We note from Figure 9b that two insights obtained with transitory news shocks extend here, and a new lesson emerges regarding persistence. First, the average inflation expectation exceeds the equilibrium inflation when shocks are not too persistent, suggesting overreaction, as the dashed blue line lies above the solid blue line.[28] Second, cost-push shocks are more inflationary under shallow thinking than under rational expectations, since the solid blue line lies above the solid green line. Last, a more persistent cost-push shock leads to higher inflation under shallow thinking, but lower inflation under rational expectations, since the solid blue line increases in $\rho$ while the solid green line decreases.

To understand this new lesson regarding persistence, we start with the limiting cases of $\lambda = 0$ and $\lambda = 1$ (rational expectations). In the limit of $\lambda = 0$, inflation expectations always have the same size as the shock, with the dashed brown line being flat, as agents

[28] We also find that the average interest rate expectation underreacts to cost-push shocks but overreacts to monetary policy shocks (Figure D5), extending the case with transitory news shocks.

perceive no general equilibrium effects. As inflation expectations affect agents' behavior by acting like a cost-push shock for firms and a demand shock for households, a more persistent shock leads to a larger cumulative effects of future disturbance, resulting in higher inflation through households' and firms' responses.

In contrast, in the limit of $\lambda = 1$ (rational expectations), as analyzed based on Proposition 7, a higher $\rho$ strengthens all general equilibrium effects and, in particular, boosts the inflation-on-households and monetary policy loops relative to the pricing complementarity self-loop. The monetary policy loop offsets the inflation response, whereas the other two amplify it. As the monetary policy loop is the strongest among them and rational agents appreciate that, a higher $\rho$ strengthens the offset and leads to a lower inflation.

Under our calibrated $\lambda$, the average inflation expectation increases in $\rho$. The mechanism is that, while the monetary policy offset loop is objectively the strongest, it is also the longest and gets dampened the most in expectations. Thus, as shallow agents better understand the shorter amplification loops than the monetary policy offset loop, they believe that a more persistent shock leads to more amplification. Through their responses to expectations, the equilibrium inflation is higher when $\rho$ is higher. That increasing relationship is less drastic than in the limit case of $\lambda = 0$, as shallow agents partially understand the monetary policy reaction and its effects on the economy.

# 6 Conclusion

This paper develops a theory of *shallow thinking* as the structure of belief formation, validates its empirical content using a customized qualitative survey, and illustrates its implications for macro and finance. We develop a systematic representation of general equilibrium as a system of causal relations. We introduce shallow thinking by assuming that people only take finite steps on the causal chain starting from shocks. The key implication of our theory is that more distant causal relations and longer feedback loops have less influence on beliefs, which leads to differential belief misreactions across variables and shocks. We find that that, on average, people think about only 2.6 steps of shock propagation—far below infinity assumed by rational expectations.

Our theory predicts that, in a New Keynesian model with an active Taylor rule, long-term nominal rates underreact to cost-push shocks but overreact to monetary policy shocks. It occurs as agents underappreciate shock propagation and offsetting feedback loops and reconciles multiple bond market puzzles. Moreover, news about future cost-push shocks leads to inflation—consistent with empirical findings—whereas rational expectations predict deflation. Additionally, more persistent cost-push shocks result in higher inflation, again contrary to rational expectations. These differences arise because shallow agents overweigh short feedback loops that amplify the effect of cost-push shocks on inflation, relative to a long offset loop via monetary policy response. In an RBC model, shallow thinking amplifies and prolongs output fluctuations in response to productivity shocks and leads to a stock market boom and crash.

At a high level, we acknowledge the immense complexity of the economy. If it has taken decades for our best economists to understand how it works—or if they are still figuring it out—we must carefully consider how much actual economic agents understand and study its consequences. In addition to its empirical relevance, our model of iterative determination of expectations can be applied to more complicated models with better computational tractability than rational expectations (Moll, 2025). It may also be fruitful to explore how people form their mental models through learning and how their understanding varies across aggregation levels, such as in the context of production networks and spatial economics.

# Bibliography


**Afrouzi, Hassan, Spencer Y. Kwon, Augustin Landier, Yueran Ma, and David Thesmar.** 2023. "Overreaction in Expectations: Evidence and Theory." *The Quarterly Journal of Economics*, 138(3): 1713–1764.

**Ahn, Woo-kyoung, Nancy S. Kim, Mary E. Lassaline, and Martin J. Dennis.** 2000. "Causal Status as a Determinant of Feature Centrality." *Cognitive Psychology*, 41(4): 361–416.

**Andre, Peter, Carlo Pizzinelli, Christopher Roth, and Johannes Wohlfart.** 2022. "Subjective Models of the Macroeconomy: Evidence From Experts and Representative Samples." *The Review of Economic Studies*, 89(6): 2958–2991.

**Andre, Peter, Ingar Haaland, Christopher Roth, Mirko Wiederholt, and Johannes Wohlfart.** 2025. "Narratives about the Macroeconomy." *The Review of Economic Studies*, Forthcoming.

**Andre, Peter, Philipp Schirmer, and Johannes Wohlfart.** 2024. "Mental Models of the Stock Market." *Working Paper*.

**Angeletos, George-Marios, and Chen Lian.** 2018. "Forward Guidance without Common Knowledge." *American Economic Review*, 108(9): 2477–2512.

**Angeletos, George-Marios, and Chen Lian.** 2023*a*. "Dampening General Equilibrium: Incomplete Information and Bounded Rationality." In *Handbook of Economic Expectations*. 613–645.

**Angeletos, George-Marios, and Chen Lian.** 2023*b*. "Determinacy without the Taylor Principle." *Journal of Political Economy*, 131(8): 2125–2164.

**Angeletos, George-Marios, and Jennifer La'O.** 2009. "Incomplete Information, Higher-Order Beliefs and Price Inertia." *Journal of Monetary Economics*, 56(SUPPL.): S19–S37.

**Angeletos, George-Marios, and Karthik A. Sastry.** 2021. "Managing Expectations: Instruments Versus Targets." *The Quarterly Journal of Economics*, 136(4): 2467–2532.

**Angeletos, George-Marios, and Zhen Huo.** 2021. "Myopia and Anchoring." *American Economic Review*, 111(4): 1166–1200.

**Arad, Ayala, and Ariel Rubinstein.** 2012. "The 11–20 Money Request Game: A Level-*k* Reasoning Study." *American Economic Review*, 102(7): 3561–3573.

**Auclert, Adrien, Bence Bardóczy, Matthew Rognlie, and Ludwig Straub.** 2021. "Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models." *Econometrica*, 89(5): 2375–2408.

**Auclert, Adrien, Matthew Rognlie, and Ludwig Straub.** 2020. "Micro Jumps, Macro Humps: Monetary Policy and Business Cycles in an Estimated HANK Model." *NBER Working Paper*, w26647.

**Auclert, Adrien, Matthew Rognlie, and Ludwig Straub.** 2024. "The Intertemporal Keynesian Cross." *Journal of Political Economy*, 132(12): 4068–4121.

**Azeredo da Silveira, Rava, Yeji Sung, and Michael Woodford.** 2024. "Optimally Imprecise Memory and Biased Forecasts." *American Economic Review*, 114(10): 3075–3118.

**Barberis, Nicholas.** 2018. "Psychology-Based Models of Asset Prices and Trading Volume." In *Handbook of Behavioral Economics*. Vol. 1, 79–175. Elsevier B.V.

**Barberis, Nicholas, Andrei Shleifer, and Robert W. Vishny.** 1998. "A Model of Investor Sentiment." *Journal of Financial Economics*, 49(3): 307–343.

**Bastianello, Francesca, and Paul Fontanier.** 2025*a*. "Expectations and Learning from Prices." *Review of Economic Studies*, 92(3): 1341–1374.

**Bastianello, Francesca, and Paul Fontanier.** 2025*b*. "Partial Equilibrium Thinking, Extrapolation, and Bubbles." *Review of Financial Studies*, , (Forthcoming).

**Bastianello, Francesca, Paul H. Décaire, and Marius Guenzel.** 2025. "Mental Models and Financial Forecasts." *Working Paper*.

**Bauer, Michael D., Carolin E. Pflueger, and Adi Sunderam.** 2024. "Changing Perceptions and Post-Pandemic Monetary Policy." *Proceedings - Economic Policy Symposium - Jackson Hole*.

**Ben Zeev, Nadav, Valerie A. Ramey, and Sarah Zubairy.** 2023. "Do Government Spending Multipliers Depend on the Sign of the Shock?" *AEA Papers and Proceedings*, 113: 382–387.

**Bianchi, Francesco, Martin Lettau, and Sydney C. Ludvigson.** 2022. "Monetary Policy and Asset Valuation." *The Journal of Finance*, 77(2): 967–1017.

**Bianchi, Francesco, Sydney Ludvigson, and Sai Ma.** 2025. "What Hundreds of Economic News Events Say About Belief Overreaction in the Stock Market." *NBER Working Paper*, w32301.

**Bianchi-Vimercati, Riccardo, Martin Eichenbaum, and Joao Guerreiro.** 2024. "Fiscal Stimulus with Imperfect Expectations: Spending vs. Tax Policy." *Journal of Economic Theory*, 217: 105814.

**Blanchard, Olivier, and Roberto Perotti.** 2002. "An Empirical Characterization of the Dynamic Effects of Changes in Government Spending and Taxes on Output." *The Quarterly Journal of Economics*, 117(4): 1329–1368.

**Blanchard, Olivier J., and Jordi Galí.** 2010. "The Macroeconomic Effects of Oil Price Shocks: Why Are the 2000s so Different from the 1970s?" In *International Dimensions of Monetary Policy*. , ed. Jordi Galí and Mark Gertler, 373–421. Chicago: University of Chicago Press / (National Bureau of Economic Research Conference Report).

**Boivin, Jean, Michael T. Kiley, and Frederic S. Mishkin.** 2010. "How Has the Monetary Transmission Mechanism Evolved Over Time?" In *Handbook of Monetary Economics*. Vol. 3, 369–422. Elsevier Ltd.

**Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer.** 2018. "Diagnostic Expectations and Credit Cycles." *The Journal of Finance*, 73(1): 199–227.

**Bordalo, Pedro, Nicola Gennaioli, Yueran Ma, and Andrei Shleifer.** 2020. "Overreaction in Macroeconomic Expectations." *American Economic Review*, 110(9): 2748–2782.

**Candia, Bernardo, Olivier Coibion, and Yuriy Gorodnichenko.** 2024. "The Inflation Expectations of U.S. Firms: Evidence from a New Survey." *Journal of Monetary Economics*, 145: 103569.

**Cattaneo, Matias D., Richard K. Crump, Max H. Farrell, and Yingjie Feng.** 2024. “On Binscatter.” *American Economic Review*, 114(5): 1488–1514.

**Cieslak, Anna.** 2018. “Short-Rate Expectations and Unexpected Returns in Treasury Bonds.” *The Review of Financial Studies*, 31(9): 3265–3306.

**Cieslak, Anna, and Pavol Povala.** 2015. “Expected Returns in Treasury Bonds.” *Review of Financial Studies*, 28(10): 2859–2901.

**Clarida, Richard, Jordi Galí, and Mark Gertler.** 1999. “The Science of Monetary Policy: A New Keynesian Perspective.” *Journal of Economic Literature*, 37(4): 1661–1707.

**Cloyne, James, Ezgi Kurt, and Paolo Surico.** 2023. “Who Gains from Corporate Tax Cuts?” *NBER Working Paper*, w31278.

**Cloyne, James, Joseba Martinez, Haroon Mumtaz, and Paolo Surico.** 2023. “Do Tax Increases Tame Inflation?” *AEA Papers and Proceedings*, 113: 377–381.

**Cloyne, James, Joseba Martinez, Haroon Mumtaz, and Paolo Surico.** 2024. “Taxes, Innovation and Productivity.” *Working Paper*.

**Cochrane, John H., and Monika Piazzesi.** 2002. “The Fed and Interest Rates—A High-Frequency Identification.” *American Economic Review*, 92(2): 90–95.

**Coibion, Olivier, and Yuriy Gorodnichenko.** 2015. “Information Rigidity and the Expectations Formation Process: A Simple Framework and New Facts.” *American Economic Review*, 105(8): 2644–2678.

**Coibion, Olivier, Dimitris Georgarakos, Yuriy Gorodnichenko, and Maarten Van Rooij.** 2023*a*. “How Does Consumption Respond to News about Inflation? Field Evidence from a Randomized Control Trial.” *American Economic Journal: Macroeconomics*, 15(3): 109–152.

**Coibion, Olivier, Yuriy Gorodnichenko, and Saten Kumar.** 2018. “How Do Firms Form Their Expectations? New Survey Evidence.” *American Economic Review*, 108(9): 2671–2713.

**Coibion, Olivier, Yuriy Gorodnichenko, Edward S. Knotek, and Raphael Schoenle.** 2023*b*. “Average Inflation Targeting and Household Expectations.” *Journal of Political Economy Macroeconomics*, 1(2): 403–446.

**Coibion, Olivier, Yuriy Gorodnichenko, Saten Kumar, and Jane Ryngaert.** 2021. “Do You Know That I Know That You Know. . . ? Higher-Order Beliefs in Survey Data.” *The Quarterly Journal of Economics*, 136(3): 1387–1446.

**Cooper, Ilan, and Richard Priestley.** 2009. “Time-Varying Risk Premiums and the Output Gap.” *Review of Financial Studies*, 22(7): 2801–2833.

**Corsetti, Giancarlo, André Meier, and Gernot J. Müller.** 2012. “Fiscal Stimulus with Spending Reversals.” *Review of Economics and Statistics*, 94(4): 878–895.

**Dal Bó, Ernesto, Pedro Dal Bó, and Erik Eyster.** 2018. “The Demand for Bad Policy When Voters Underappreciate Equilibrium Effects.” *The Review of Economic Studies*, 85(2): 964–998.

**Daniel, Kent, David Hirshleifer, and Avanidhar Subrahmanyam.** 1998. “Investor Psychology and Security Market Under- and Overreactions.” *The Journal of Finance*, 53(6): 1839–1885.

**De Grauwe, Paul.** 2011. "Animal Spirits and Monetary Policy." *Economic Theory*, 47(2-3): 423–457.

**Dogra, Keshav.** 2024. "Paradoxes and Problems in the Causal Interpretation of Equilibrium Economics." *Working Paper*.

**Egger, Dennis, Johannes Haushofer, Edward Miguel, Paul Niehaus, and Michael Walker.** 2022. "General Equilibrium Effects of Cash Transfers: Experimental Evidence From Kenya." *Econometrica*, 90(6): 2603–2643.

**Eusepi, Stefano, and Bruce Preston.** 2018. "The Science of Monetary Policy: An Imperfect Knowledge Perspective." *Journal of Economic Literature*, 56(1): 3–59.

**Evans, George W., and Seppo Honkapohja.** 2001. *Learning and Expectations in Macroeconomics. Frontiers of Economic Research*, Princeton:Princeton University Press.

**Farhi, Emmanuel, and Iván Werning.** 2019. "Monetary Policy, Bounded Rationality, and Incomplete Markets." *American Economic Review*, 109(11): 3887–3928.

**Farmer, Leland E., Emi Nakamura, and Jón Steinsson.** 2024. "Learning about the Long Run." *Journal of Political Economy*, 132(10): 3334–3377.

**Gabaix, Xavier.** 2014. "A Sparsity-Based Model of Bounded Rationality." *The Quarterly Journal of Economics*, 129(4): 1661–1710.

**Gabaix, Xavier.** 2020. "A Behavioral New Keynesian Model." *American Economic Review*, 110(8): 2271–2327.

**Gabaix, Xavier, and David I. Laibson.** 2001. "The 6D Bias and the Equity-Premium Puzzle." *NBER Macroeconomics Annual*, 16: 257–312.

**Galí, Jordi.** 2015. *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications - Second Edition.* Princeton University Press.

**Galí, Jordi, J. David López-Salido, and Javier Vallés.** 2007. "Understanding the Effects of Government Spending on Consumption." *Journal of the European Economic Association*, 5(1): 227–270.

**García-Schmidt, Mariana, and Michael Woodford.** 2019. "Are Low Interest Rates Deflationary? A Paradox of Perfect-Foresight Analysis." *American Economic Review*, 109(1): 86–120.

**Georganas, Sotiris, Paul J. Healy, and Roberto A. Weber.** 2015. "On the Persistence of Strategic Sophistication." *Journal of Economic Theory*, 159: 369–400.

**Greenwood, Robin, and Samuel G. Hanson.** 2015. "Waves in Ship Prices and Investment." *The Quarterly Journal of Economics*, 130(1): 55–109.

**Gürkaynak, Refet S., Brian Sack, and Eric Swanson.** 2005. "The Sensitivity of Long-Term Interest Rates to Economic News: Evidence and Implications for Macroeconomic Models." *American Economic Review*, 95(1): 425–436.

**Haaland, Ingar, Christopher Roth, and Johannes Wohlfart.** 2023. "Designing Information Provision Experiments." *Journal of Economic Literature*, 61(1): 3–40.

**Haavelmo, Trygve.** 1943. "The Statistical Implications of a System of Simultaneous Equations." *Econometrica*, 11(1): 1.

**Hansen, Lars Peter, and Thomas J. Sargent.** 2007. *Robustness.* Princeton, N.J:Princeton University Press.

**Hanson, Samuel G., and Jeremy C. Stein.** 2015. "Monetary Policy and Long-Term Real Rates." *Journal of Financial Economics*, 115(3): 429–448.

**Hanson, Samuel G., David O. Lucca, and Jonathan H. Wright.** 2021. "Rate-Amplifying Demand and the Excess Sensitivity of Long-Term Rates." *Quarterly Journal of Economics*, 136(3): 1719–1781.

**Hayashi, Fumio.** 1982. "Tobin's Marginal q and Average q: A Neoclassical Interpretation." *Econometrica*, 50(1): 213.

**Heckman, James, and Rodrigo Pinto.** 2015. "Causal Analysis after Haavelmo." *Econometric Theory*, 31(1): 115–151.

**Imbens, Guido W.** 2020. "Potential Outcome and Directed Acyclic Graph Approaches to Causality: Relevance for Empirical Practice in Economics." *Journal of Economic Literature*, 58(4): 1129–1179.

**Iovino, Luigi, and Dmitriy Sergeyev.** 2023. "Central Bank Balance Sheet Policies Without Rational Expectations." *Review of Economic Studies*, 90(6): 3119–3152.

**Joslin, Scott, Marcel Priebsch, and Kenneth J. Singleton.** 2014. "Risk Premiums in Dynamic Term Structure Models with Unspanned Macro Risks." *The Journal of Finance*, 69(3): 1197–1233.

**Känzig, Diego R.** 2021. "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements." *American Economic Review*, 111(4): 1092–1125.

**Kilian, Lutz, and Robert J. Vigfusson.** 2011. "Are the Responses of the U.S. Economy Asymmetric in Energy Price Increases and Decreases?: Are Responses of the U.S. Economy Asymmetric?" *Quantitative Economics*, 2(3): 419–453.

**Kindleberger, Charles P.** 1978. *Manias, Panics, and Crashes: A History of Financial Crises.* New York: Basic Books.

**King, Robert G., and Sergio T. Rebelo.** 1999. "Chapter 14 Resuscitating Real Business Cycles." In *Handbook of Macroeconomics*. Vol. 1, 927–1007.

**Kwon, Spencer Yongwook, and Johnny Tang.** 2024. "Extreme Categories and Overreaction to News." *SSRN Electronic Journal*.

**Lucas, Jr., Robert E.** 1972. "Expectations and the Neutrality of Money." *Journal of Economic Theory*, 4(2): 103–124.

**Ludvigson, Sydney C., and Serena Ng.** 2009. "Macro Factors in Bond Risk Premia." *Review of Financial Studies*, 22(12): 5027–5067.

**Lusardi, Annamaria, and Olivia S. Mitchell.** 2011. "Financial Literacy and Retirement Planning in the United States." *Journal of Pension Economics and Finance*, 10(4): 509–525.

**Maćkowiak, Bartosz, and Mirko Wiederholt.** 2009. "Optimal Sticky Prices under Rational Inattention." *American Economic Review*, 99(3): 769–803.

**Malmendier, Ulrike, and Stefan Nagel.** 2016. "Learning from Inflation Experiences." *The Quarterly Journal of Economics*, 131(1): 53–87.

**Mankiw, N. Gregory, and Ricardo Reis.** 2002. "Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve." *The Quarterly Journal of Economics*, 117(4): 1295–1328.

**Mendes, Arthur, Wataru Miyamoto, Thuy Lan Nguyen, Steven Pennings, and Leo Feler.** 2024. "The Macroeconomic Effects of Cash Transfers: Evidence from Brazil." *Working Paper*.

**Mertens, Karel, and Morten O. Ravn.** 2013. "The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States." *American Economic Review*, 103(4): 1212–1247.

**Miranda-Agrippino, Silvia, and Giovanni Ricco.** 2021. "The Transmission of Monetary Policy Shocks." *American Economic Journal: Macroeconomics*, 13(3): 74–107.

**Molavi, Pooya.** 2025. "Simple Models and Biased Forecasts." *Working Paper*.

**Molavi, Pooya, Alireza Tahbaz-Salehi, and Andrea Vedolin.** 2024. "Model Complexity, Expectations, and Asset Prices." *Review of Economic Studies*, 91(4): 2462–2507.

**Moll, Benjamin.** 2025. "The Trouble with Rational Expectations in Heterogeneous Agent Models: A Challenge for Macroeconomics." *Working Paper*.

**Nagel, Rosemarie.** 1995. "Unraveling in Guessing Games: An Experimental Study." *The American Economic Review*, 85(5): 1313–1326.

**Nimark, Kristoffer.** 2008. "Dynamic Pricing and Imperfect Common Knowledge." *Journal of Monetary Economics*, 55(2): 365–382.

**Olivei, Giovanni, and Silvana Tenreyro.** 2007. "The Timing of Monetary Policy Shocks." *American Economic Review*, 97(3): 636–663.

**Pearl, Judea.** 2009. *Causality: Models, Reasoning, and Inference.* . 2 ed., Cambridge University Press.

**Pennings, Steven.** 2021. "Cross-Region Transfer Multipliers in a Monetary Union: Evidence from Social Security and Stimulus Payments." *American Economic Review*, 111(5): 1689–1719.

**Rabin, Matthew.** 2002. "Inference by Believers in the Law of Small Numbers." *The Quarterly Journal of Economics*, 117(3): 775–816.

**Ramey, Valerie A.** 2011. "Identifying Government Spending Shocks: It's All in the Timing." *The Quarterly Journal of Economics*, 126(1): 1–50.

**Ramey, Valerie A.** 2013. "Government Spending and Private Activity." In *Fiscal Policy after the Financial Crisis*. , ed. Alberto Alesina and Francesco Giavazzi, 19–55. University of Chicago Press.

**Ramey, Valerie A.** 2016. "Macroeconomic Shocks and Their Propagation." In *Handbook of Macroeconomics*. Vol. 2, 71–162. Elsevier B.V.

**Rottman, Benjamin Margolin, and Reid Hastie.** 2014. "Reasoning about Causal Relationships: Inferences on Causal Networks." *Psychological Bulletin*, 140(1): 109–139.

**Rozenblit, Leonid, and Frank Keil.** 2002. "The Misunderstood Limits of Folk Science: An Illusion of Explanatory Depth." *Cognitive Science*, 26(5): 521–562.

**Sims, Christopher A.** 2003. "Implications of Rational Inattention." *Journal of Monetary Economics*, 50(3): 665–690.

**Spiegler, Ran.** 2016. "Bayesian Networks and Boundedly Rational Expectations." *The Quarterly Journal of Economics*, 131(3): 1243–1290.

**Spiegler, Ran.** 2020. "Behavioral Implications of Causal Misperceptions." *Annual Review of Economics*, 12(1): 81–106.

**Stantcheva, Stefanie.** 2021. "Understanding Tax Policy: How Do People Reason?" *The Quarterly Journal of Economics*, 136(4): 2309–2369.

**Stantcheva, Stefanie.** 2023*a*. "How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible." *Annual Review of Economics*, 15(1): 205–234.

**Stantcheva, Stefanie.** 2023*b*. "Understanding of Trade." *NBER Working Paper*.

**Tinbergen, Jan.** 1930. "Determination and Interpretation of Supply Curves: An Example [Bestimmung Und Deutung von Angebotskurven. Ein Beispiel]." *Zeitschrift fur Nationalokonomie*, 1(5): 669–679.

**Wachter, Jessica A, and Michael Jacob Kahana.** 2024. "A Retrieved-Context Theory of Financial Decisions." *The Quarterly Journal of Economics*, 139(2): 1095–1147.

**Waldmann, Michael R.,** ed. 2017. *The Oxford Handbook of Causal Reasoning.* Vol. 1, Oxford University Press.

**Wilson, Robert A., and Frank Keil.** 1998. "The Shadows and Shallows of Explanation." *Minds and Machines*, 8(1): 137–159.

**Woodford, Michael.** 2003*a*. "Imperfect Common Knowledge and the Effects of Monetary Policy." In *Knowledge, Information, and Expectations in Modern Macroeconomics: In Honor of Edmund S. Phelps*. , ed. Philippe Aghion, Roman Frydman, Joseph E. Stiglitz and Michael Woodford, 25–58. Princeton:Princeton University Press.

**Woodford, Michael.** 2003*b*. *Interest and Prices: Foundations of a Theory of Monetary Policy.* Princeton University Press.

**Wu, Lingxuan.** 2023. "Mental Macro-Finance Models: Evidence and Theory." *SSRN Electronic Journal*.

# Online Appendix for
# "Thinking about the Economy, Deep or Shallow?"

Pierfrancesco Mei Lingxuan Wu

December 17, 2025

## A Survey Details

### A.1 Survey Design and Sample

**Detailed survey design.** We design our survey to elicit the general public's directional beliefs about changes in a host of macroeconomic variables (such as prices, labor hours, and interest rates) in response to a set of hypothetical macroeconomic shocks (such as oil supply shocks and monetary policy shocks).

Our survey builds on Andre et al. (2022), which ask respondents to forecast changes (in levels) of inflation and the unemployment rate in response to hypothetical macroeconomic shocks.[29] Grounded in our theory, two innovative features of our design are to inquire about a number of major macroeconomic variables and to only elicit beliefs about the directional responses. Inquiring about a host of macroeconomic variables traces out the path of shock propagation and tests if perception fails sequentially as suggested by our theory. For example, one of our questions concerns labor hours and asks respondents whether the average worker will work for more, fewer, or the same number of hours during a typical week in response to shocks. Comparing people's beliefs about labor hours to those about aggregate demand sheds light on their understanding of firms' input choice. Eliciting directional assessments instead of level forecasts lowers the cognitive strain and lets us focus on the qualitative aspect of people's mental models. For instance, their responses about directional price change reveal whether participants understand the Phillips curve, rather than their potentially different perceptions about its slope.[30]

[29] Haaland, Roth and Wohlfart (2023) discuss the potential and limitations of hypothetical vignettes.

[30] Andre et al. (2022) further ask participants to select the relevant ones from a list of potential channels and show that the selected channels predict their forecasts. For example, one channel in the oil supply shock vignette is "due to lower incomes or job loss, households cut back on their spending." We simply ask for directional assessments for a set of variables, without showing them any directional statement that is objectively true or false, which may influence their responses.

Respondents are randomized into three groups with probability one half, one quarter and one quarter, each receiving two shock scenarios in random order. Groups 1, 2, and 3 receive the following pairs of shocks respectively: oil supply shock and monetary policy shock, government spending shock and personal income tax shock, corporate income tax shock and transfer payment shock. In each scenario, we introduce a shock that realizes now and persists for 12 months (except for the transfer payment shock), and then elicit respondents' beliefs about changes in the economy over the next 12 months. We stress the exogeneity of the shocks and state clearly that the shocks are publicly announced or broadcasted and are common knowledge to everyone in the US. Appendix A.2 shows the phrasing of these hypothetical shocks.

Our questions cover a range of macro variables, divided into four blocks presented in random order, each on a separate page. The four blocks correspond to choices and decision-relevant variables of firms, households, the central bank, and the federal fiscal policy. We ask people's opinions about the average US business and household, to avoid any potential peculiarity of their own situations. Each block contains variables that the block either responds to or decides on, summing up to 12 to 16 distinct variables for each shock scenario, listed in Table B1. Several key variables, such as price and total labor hours, are included in more than one block (as one agent's choice and as other agents' decision-relevant variables), resulting in a total of about 22 questions for each shock scenario. For each question, respondents select among "up," "down," "unchanged," or "I don't know" to indicate their perceived directional changes of the specific variable in response to the shocks. The directional responses of most variables we elicit to these shocks are well-established in the empirical literature, as surveyed in Table B2.

In addition, to contrast our depth of thinking against level-$k$ thinking in the game theoretical literature, we play the popular game of "guess 2/3 of the average" à la Nagel (1995) to measure respondents' game theoretical sophistication. Each respondent selects a number between 0 and 100, and they are informed that the number closest to the 2/3 of the average wins the game.[31] Based on each respondent's answer $g_i$, we compute their level-$k$ as $k_i \equiv \log\left(\frac{g_i}{50}\right)/\log\left(\frac{2}{3}\right)$, assuming that a level-0 player randomly selects a number.

[31]Our baseline design runs this game without incentives for two reasons. First, offering a prize requires collecting respondents' email addresses, which some may be averse to, leading to selection bias. Second, since we cannot monitor whether respondents use search engines or other sources (though we ask them not to), we choose not to link compensation to performance and inform them of this. Nonetheless, we conducted an incentivized version with a subsample of 300 respondents, where we confirm our additional finding that depth of thinking correlates across shock scenarios but does not correlate with level-$k$ thinking.

We also assess respondents' financial literacy using questions from Lusardi and Mitchell (2011) and collect other information, including gender, age, race, ZIP code, household composition, education, main occupation and additional employment, political affiliation, household income, assets, and debts. Figure A1 illustrates the flow of our survey.

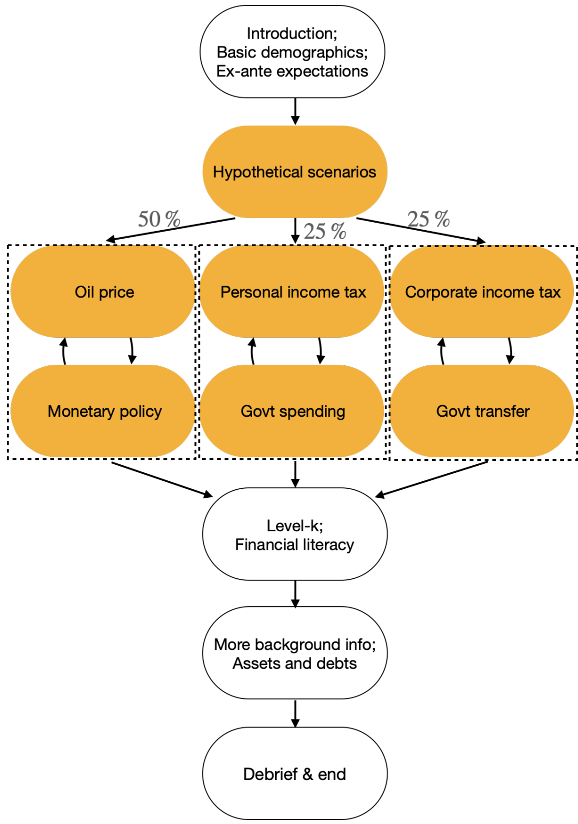



Figure A1: Survey structure

**Sample.** We conducted an online survey of 1,000 US households in June and October 2024. The survey was distributed through LUCID Marketplace, a platform that is widely used for research and is made up of hundreds of suppliers with a diverse set of recruitment methodologies, ensuring that the sample does not overweigh any particular segment of the population. The vast majority are double opt-in suppliers. Suppliers incentivize their respondents with loyalty reward points or gift cards or cash payments. The median completion time for our survey is 16 minutes.

We focus on US residents who are in the labor force at the time of the interview, and are aged between 25 and 65. Conditional on these characteristics, the survey is constrained through quotas to be broadly representative of the US population along the dimensions of gender, age, total gross household income, and race, based on the Annual Social and Economic Supplement data of the Current Population Survey in 2022. Table A1 notes that our sample is largely representative of the US population.[32]

We drop people who fail the attention or sanity checks placed in our survey or spend too much or too little time completing the survey to form our main sample. Our results are robust to various sample selection criteria.

Table A1: Sample statistics

| | US population | Survey sample |
|---|---|---|
| ***Gender*** | | |
| Male | .53 | .48 |
| ***Age*** | | |
| 25-29 years old | .13 | .13 |
| 30-39 years old | .28 | .29 |
| 40-49 years old | .25 | .29 |
| 50-59 years old | .24 | .21 |
| 60-65 years old | .10 | .07 |
| ***Household income*** | | |
| $0-$19,999 | .04 | .11 |
| $20,000-$39,999 | .11 | .17 |
| $40,000-$69,999 | .20 | .24 |
| $70,000-$124,999 | .29 | .34 |
| $125,000+ | .36 | .15 |
| ***Race*** | | |
| White | .61 | .72 |
| Black/African-American | .12 | .13 |
| Hispanic/Latino | .18 | .08 |
| Asian/Asian-American | .07 | .03 |
| Other | .02 | .03 |
| ***Employment status*** | | |
| Full time employed | .78 | .72 |
| Part time employed | .09 | .12 |
| Self-employed | .10 | .09 |
| Unemployed | .03 | .07 |

*Notes*: Shares may not exactly add up to 1 due to rounding errors.

## A.2 Hypothetical Scenarios and Other Questions

### Group 1.

**Oil supply shock (oil).** *"Since the beginning of 2024, the average price for one barrel of WTI crude oil, which is a major benchmark for oil prices in the US, has been around $80.*

*Now, imagine that the price of crude oil unexpectedly increases due to production problems in*

[32] As known in the literature (Stantcheva, 2023*a*), online samples are hard to reach the tails of income distribution and tend to skew towards white and non-Hispanic respondents. Further, we show in Appendix B.3 household income does not correlate with depth of thinking.

*the Middle East. For the next 12 months, the price for one barrel of crude oil will be, on average, \$20 higher than its current level.*

*This price increase is publicly broadcasted by major news outlets and is common knowledge to everyone in the US.*

*We will now ask you a few short questions to understand how you think the US economy would be affected by such an increase in oil price."*

**Monetary policy shock (MP).** *"The Federal Reserve, often referred to simply as "the Fed," is the central bank of the United States that conducts the nation's monetary policy to help regulate the economy. It sets a key interest rate known as the Federal Funds Rate. This is the rate at which banks lend to each other, and it affects the economy in many ways. The Federal Funds Rate influences the interest rates for savings accounts, credit card balances, mortgages, loans, and others. As of now, the Federal Funds Rate set by the Federal Reserve is 4.75%.*

*Now, imagine that the Federal Reserve unexpectedly raises the Federal Funds Rate by 0.5 percentage points, changing it to 5.25%, and announces that it will maintain this rate for the next 12 months.*

*This interest rate raise is publicly announced and is common knowledge to everyone in the US. The Federal Reserve clarifies that this decision is made with no changes in their assessment of economic conditions.*

*We will now ask you a few short questions to understand how you think the US economy would be affected by this raise in the interest rate."*

**Group 2.**

**Government spending shock (G).** *"Since 2000, US federal government spending has averaged about 25% of the US Gross Domestic Product (GDP), which is the total value of all goods and services produced in the country.*

*Now, imagine that the US federal government unexpectedly announces a new defense program, leading to an increase in federal government spending over the next 12 months. And the additional spending will be directed domestically. Specifically, federal government spending relative to US GDP will increase by about 2% over the next 12 months.*

*This increase is publicly announced and is common knowledge to everyone in the US. The government clarifies that this change is temporary and occurs without any alterations in its assessment of national security or economic conditions.*

*We will now ask you a few short questions to understand how you think the US economy would be affected by this increase in federal government spending."*

**Personal income tax shock (PIT).** *"In 2023, a typical household earning the median income is subject to a 12% federal personal income tax rate. Collectively, all households paid about 2.2 trillion dollars in federal personal income taxes in 2023, which is about 8% of the US Gross Domestic Product (GDP), the total value of all goods and services produced in the country.*

*Now, imagine that the US federal government unexpectedly announces a 2% increase in the federal personal income tax rate over the next 12 months.*

*This increase is publicly announced and is common knowledge to everyone in the US. The government clarifies that the change is temporary and occurs without any changes in its assessment of economic conditions.*

*We will now ask you a few short questions to understand how you think the US economy would be affected by this increase in the federal personal income tax rate."*

**Group 3.**

**Transfer payment shock (TP).** *"Imagine that today the US federal government unexpectedly announces that each taxpayer will receive a one-time transfer payment worth, on average, $1,200. This one-time payment, which will not be taxed, will be available in bank accounts or as a check in mailboxes within three months.*

*Taking into account that around 200 million US taxpayers will receive the payment, the total payments disbursed will be approximately 240 billion dollars, which is about 1% of the US Gross Domestic Product (GDP), the total value of all goods and services produced in the country.*

*The payment is publicly announced and is common knowledge to everyone in the US. The government clarifies that the payment is a one-time event and occurs without any changes in its assessment of economic conditions.*

*We will now ask you a few short questions to understand how you think the US economy would be affected by this transfer payment by the federal government"*

**Corporate income tax shock (CIT).** *"Since 2017, there has been a 21% federal corporate income tax rate in place. The taxable income is a business's revenue minus expenses. In 2023, all corporations together paid about 420 billion dollars, which is about 1.5% of the US Gross Domestic Product (GDP), the total value of all goods and services produced in the country.*

*Now, imagine that the US federal government unexpectedly announces a 2% increase in the federal corporate income tax rate over the next 12 months.*

*The increase is publicly announced and is common knowledge to everyone in the US. The government clarifies that the change is temporary and occurs without any changes in its assessment of economic conditions.*

*We will now ask you a few short questions to understand how you think the US economy would be affected by the increase in federal corporate income tax rate."*

**Financial literacy.** Three questions are asked for each respondent.

1. *"Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?: more than today; less than today; exactly the same; don't know."*

2. *"Do you think that the following statement is 'true' or 'false'? Buying a company stock usually provides a safer return than a stock mutual fund.: true; false; don't know."*

3. *"Suppose you had $100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?: more than $102; less than $102; exactly the same; don't know."*

**Level-*k* thinking ("guess 2/3 of the average" game).** *"Imagine you are playing a game with about 300 other people chosen randomly from across the United States.*

*Please choose a number between 0 and 100, inclusive.*

*We will take your number, as well as the numbers chosen by other participants, to calculate the average number. The winning number will be the number that is closest to two-thirds (2/3) of the average number. Specifically, we sum the chosen numbers by everyone and divide by the number of participants. Multiply the result by 2/3. The winning number is the one closest to the last result.*

*The winner will get an electronic gift card at any popular merchant worth $30, which will be split when there are multiple winners. The winner will be contacted in a few days at the conclusion of this study, using the email provided below."*

# B Appendix to Survey Findings

## B.1 Correct Directional Responses and Causal Distance

Table B1: Variables elicited in forecast part of our survey

| Variable | Abbrev. | Group 1 Oil ↑ | Group 1 MP ↑ | Group 2 G ↑ | Group 2 PIT ↑ | Group 3 CIT ↑ | Group 3 TP ↑ |
|---|---|---|---|---|---|---|---|
| **Firms-related** | **bus** | | | | | | |
| Nominal marginal cost | mc | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Demand | Y | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Interest rate | i | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Corporate income tax rate | CIT | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Prices | p | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Intermediate inputs | x | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Investment | I | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Total hours | N | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Unemployment rate | u | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Dividends/post-tax profits | div | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| **Households-related** | **hh** | | | | | | |
| Interest rate | i | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Prices | p | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Hours | N | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Personal income tax rate | PIT | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Pre-tax nominal wage | W | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Durable consumption | D | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Non-durable consumption | ND | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| **Central-bank-related** | **fed** | | | | | | |
| Unemployment rate | u | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Inflation | p | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Interest rate | i | ✓ | × | ✓ | ✓ | ✓ | ✓ |
| **Government-related** | **gov** | | | | | | |
| Borrowing/repayment | B | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Tax revenue | TR | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

## Table B2: Literature review of directional impulse responses

| | | Oil price ↑ | Monetary policy (MP) ↑ | Transfer payment (TP)‡ ↑ |
|---|---|---|---|---|
| Output | Y | Down (Känzig) | Down (Ramey) | Up (Pennings) |
| Interest rate | i | Up (Känzig) | Up (Ramey, as shock itself) | |
| Price | p | Up (Känzig) | Down (MAR), insignificant or up (Ramey, classic price puzzle) | |
| Unemployment | u | Up (Känzig) | Up (Ramey) | Down (MMNPF) |
| Labor hours | N | Down (BG) | Down (MAR) | |
| Nonresidential investment | I | Down (Känzig) | Down (BKM) | |
| Durable consumption | D | Down (Känzig) | Down (BKM) | Up (EHMNW) |
| Nondurables & services | ND | Down (Känzig) | Down (BKM) | Up (EHMNW) |
| Nominal wage | W | Up (BG) | Down (OT, as real wage and price both down), insignificantly down (MAR) | Up (EHMNW) |
| Dividend/post-tax profits | div | | | Up (EHMNW) |
| **References** | | Känzig (2021, figs 8-10/A.7)<br>Blanchard and Galí (2010, fig 7.6.A) | Ramey (2016, figs 2-3)<br>Miranda-Agrippino and Ricco (2021, fig 7)<br>Boivin, Kiley and Mishkin (2010, fig 4, post-84)<br>Olivei and Tenreyro (2007, figs 10-14) | Pennings (2021, tab 1)<br>Mendes et al. (2024, tab 5)<br>Egger et al. (2022, tabs 1, 3) |

| | | Government spending (G) ↑ | Personal income tax (PIT) ↑ | Corporate income tax (CIT) ↑ |
|---|---|---|---|---|
| Output | Y | Up (Ramey16) | Down (MR) | Down (MR) |
| Interest rate | i | Insignificant (Ramey11) | Insignificant (MR) | Insignificant (MR) |
| Price | p | Up (Ramey16 unreported result) | Down (CMMS23), insignificantly down (MR) | Insignificantly up (CMMS23, MR) |
| Unemployment | u | Down (Ramey13*) | Up (MR) | Up (CKS), insignificantly up (MR) |
| Labor hours | N | Up (Ramey11*) | Down (CMMS24, MR) | Insignificantly down (CMMS24), insignificant (MR) |
| Nonresidential investment | I | Down (Ramey11) | Down (MR) | Down (MR) |
| Durable consumption | D | Down (Ramey11**) | Down (MR) | Down (CMMS24†), insignificantly up (MR) |
| Nondurables & services | ND | Down (Ramey11**) | Down (CMMS24†), insignificantly down (MR) | Down (CMMS24†), insignificantly up (MR) |
| Nominal wage | W | Up (Ramey11) | Down (CMMS24, as real wage and price both down) | Down (CMMS24, as real wage down and price insignificant) |
| Dividend/post-tax profits | div | | | Down (CKS) |
| Personal income tax rate | PIT | Up (Ramey11) | Up (MR, as shock itself) | Insignificantly down (MR) |
| Corporate income tax rate | CIT | Up (Ramey16***) | Insignificantly down (MR) | Up (MR, as shock itself) |
| Tax revenue | TR | | Up (CMMS24‡, MR) | Up (CMMS24††), insignificant (MR) |
| Government spending | G | Up (Ramey16, as shock itself) | Insignificant (MR) | Insignificant (MR) |
| Government debt | B | Up (CMM) | Down (CMMS24, MR) | Insignificant (CMMS24, MR) |
| **References** | | Ramey (2016, fig 5)<br>Ramey (2011, fig X)<br>Ramey (2013, figs 1.11-17)<br>Corsetti, Meier and Müller (2012, fig 1) | Mertens and Ravn (2013, figs 2-4/9/10)<br>Cloyne et al. (2023, fig 1)<br>Cloyne et al. (2024, figs 1-2/B.1/H.8)<br>Cloyne, Kurt and Surico (2023, figs 2/3B) | |

*Notes:* This table lists directional impulse responses at about 1-year horizon across variables to shocks. Shocks considered are oil supply shocks, contractionary monetary policy (MP) shocks, transfer payment (TP) shocks, government spending (G) shocks, and positive shocks in personal income tax (PIT) rate and corporate income tax (CIT) rate. As we are only interested in the directions rather than the magnitudes of responses, we only impose a weak assumption that each variable responds to positive and negative shocks with opposite signs. This is weaker than assuming that the multipliers are the same for positive and negative shocks. Nonetheless, it is worth noting that while some earlier papers advocate for asymmetric multipliers, more recent papers argue that the evidence is weak (e.g., Kilian and Vigfusson (2011) for oil supply shocks and Ben Zeev, Ramey and Zubairy (2023) for government spending shocks). The abbreviation in parentheses indicates the main reference, usually the most recent or most cited paper. All references are listed at the bottom of each column.

‡ For the transfer payment shock, Pennings (2021) and Mendes et al. (2024) provide cross-sectional estimates instead of aggregate ones, in the US and Brazil respectively. Egger et al. (2022) study transfers in Kenya, but these transfers are funded from outside the economy. We drop this shock in a robustness version in Table B3.

* Ramey (2013) shows that the unemployment rate falls in response to government spending shocks. It is mainly driven by government employment, with the response of private employment either insignificant or negative in different specifications. We drop this variable in a robustness version in Table B3, since we elicit participants' opinions about private businesses.

** Ramey (2011, 2016) discusses extensively potential issues with previous work (e.g., Blanchard and Perotti, 2002; Galí, López-Salido and Vallés, 2007) that find a positive consumption response to government spending shocks. We drop this variable in a robustness version in Table B3.

*** Ramey (2016, fig 5) suggests that the average tax rate goes up in response to government spending shocks, calculated as federal current receipts divided by nominal GDP.

† Cloyne et al. (2024, fig 2) show positive consumption responses to decreases in PIT and CIT, but do not split into durables vs. nondurables.

†† Cloyne et al. (2024, fig H.8) show negative primary surplus response (tax revenue minus government spending) to decreases in PIT and CIT.

Table B3: Correct directional responses with robustness versions

| | Oil ↑ | | | MP ↑ | | | G ↑ | | | PIT ↑ | | | CIT ↑ | | | TP ↑ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Version | Va | Vb | Vc | Va | Vb | Vc | Va | Vb | Vc | Va | Vb | Vc | Va | Vb | Vc | Va | Vb | Vc |
| Y | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↑ | ↑ | ↑ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↑ | ↑ | ↑ |
| i | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | | | ↑ | | | ↓ | | | ↓ | | | ↑ |
| p | ↑ | ↑ | ↑ | ↓ | | ↓ | ↑ | | ↑ | ↓ | | ↓ | | | ↓ | | | ↑ |
| u | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↓ | | ↓ | ↑ | ↑ | ↑ | ↑ | | ↑ | ↓ | ↓ | ↓ |
| N | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↑ | ↑ | ↑ | ↓ | ↓ | ↓ | ↓ | | ↓ | ↑ | ↑ | ↑ |
| I | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | | | |
| D | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | | ↓ | ↓ | ↓ | ↓ | ↓ | | ↓ | ↑ | | ↑ |
| ND | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | | ↓ | ↓ | | ↓ | ↓ | | ↓ | ↑ | | ↑ |
| W | ↑ | ↑ | ↑ | ↓ | ↓ | ↓ | ↑ | ↑ | ↑ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↑ | | ↑ |
| div | | | ↓ | | | ↓ | | | ↑ | | | ↓ | ↓ | ↓ | ↓ | ↑ | | ↑ |
| mc | | | ↑ | | | ↓ | | | ↑ | | | ↓ | | | ↓ | | | ↑ |
| PIT | | | | | | | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | | | | | | ↑ |
| CIT | | | | | | | ↑ | | ↑ | | | | ↑ | ↑ | ↑ | | | ↑ |
| TR | | | | | | | | | ↑ | ↑ | ↑ | ↑ | ↑ | | ↑ | | | ↑ |
| B | | | | | | | ↑ | ↑ | ↑ | ↓ | ↓ | ↓ | ↓ | | ↓ | | | ↑ |

*Notes:* Three versions are constructed as follows. Version a is the baseline specification in Table 1.

1. Version a is most closely based on most up-to-date empirical estimates with clear directions.
2. Version b drops from Version a variables for which estimates are noisy or controversy exists. A few noteworthy choices are price response to monetary policy shocks (classic price puzzle), unemployment and consumption responses to government spending shocks (as discussed in the notes of Table B2).
3. Version c makes additional predictions based on theoretical predictions, relative to Version a.

Table B4: Model-implied causal distance with robustness versions

| | Oil | | | MP | | | G | | | PIT | | | CIT | | | TP | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | M1 | M2 | M3 | M1 | M2 | M3 | M1 | M2 | M3 | M1 | M2 | M3 | M1 | M2 | M3 | M1 | M2 | M3 |
| Y | 3 | 3 | 3 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 2 | 2 |
| i | 2 | 2 | 2 | 1 | 1 | 1 | 5 | 3 | 3 | 5 | 3 | 4 | 6 | 4 | 3 | 5 | 4 | 3 |
| p | 1 | 1 | 1 | 4 | 3 | 4 | 4 | 2 | 2 | 4 | 3 | 3 | 5 | 4 | 2 | 4 | 3 | 2 |
| u | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 2 | 2 | 3 | 2 |
| N | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 2 | 2 | 3 | 2 |
| I | | | 3 | | | 2 | | | 2 | | | 3 | | | 2 | | | 2 |
| D | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 2 | 2 |
| ND | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 2 | 2 |
| W | | | 2 | | | 3 | | | 3 | | | 3 | | | 3 | | | 3 |
| div | 4 | 4 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 |
| mc | 1 | 1 | 1 | 3 | 3 | 3 | 3 | 2 | 2 | 3 | 3 | 3 | 4 | 4 | 2 | 3 | 3 | 2 |
| PIT | | | | | | | 1 | 1 | 1 | 1 | 1 | 1 | | | | 1 | 1 | 1 |
| CIT | | | | | | | 1 | 1 | 1 | | | | 1 | 1 | 1 | 1 | 1 | 1 |
| B | | | | | | | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

*Notes:* Three sets of causal distance $D_{vs}$ are constructed as follows, with the oil supply shock interpreted as a cost-push shock, as is standard in the literature. Model 2 is the baseline specification in Table 1.

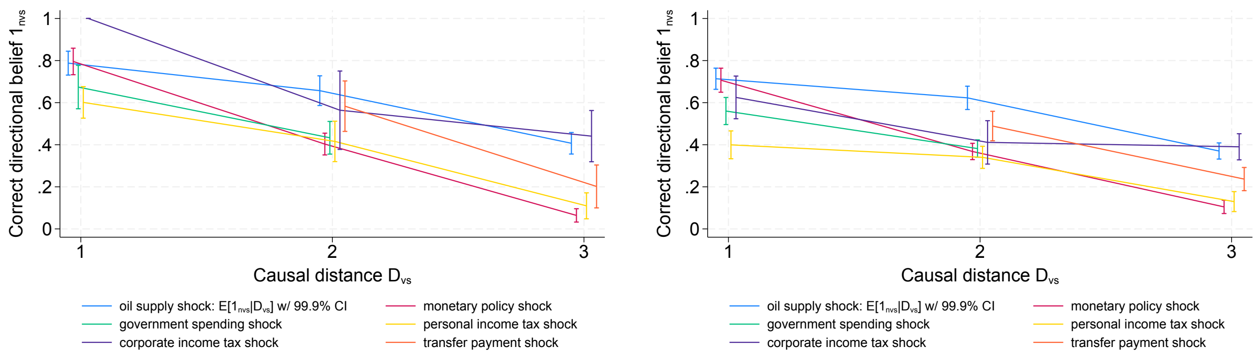
1. Model 1 is the textbook New Keynesian model in the main text of this paper.

2. Model 2 extends Model 1 to feature decreasing-returns production (so that the marginal cost is increasing in quantity) and a Taylor rule of monetary policy that depends on both inflation and unemployment.

3. Model 3 extends Model 1 with capital investment by firms, price and wage rigidity (via a labor union instead of a competitive labor market).

Our directed graph representation of mental models (Figure 1) only includes labor demand and supply shifts $\hat{n}_1^d, \hat{n}_1^s$ and assumes that the real wage $w_1$ requires one additional step beyond demand and supply shifts. For labor hours, we assume that people perceive the change as soon as they perceive a change in either $\hat{n}_1^d$ or $\hat{n}_1^s$. Alternatively, one could assume that labor hours require an additional step beyond these shifts to be recognized as the equilibrium outcome. This difference is absorbed by variable fixed effects in our regression of correct directional belief on causal distance, and is irrelevant when applying shallow thinking in macroeconomic models, since only beliefs about the real wage are decision-relevant, while those about labor hours are not.

## B.2 Additional Results of Predicting Correct Directional Belief




(a) Main sample

(b) Full sample

Figure B1: Expectation of correct directional belief $1_{nvs}$ conditional on causal distance $D_{vs}$

*Notes:* This figure shows the expectation of correct directional belief $1_{nvs}$, conditional on causal distance $D_{vs}$, together with the 99.9% confidence interval, by each shock. Panel (a) shows the main sample of our analysis that filters attention and completion time and panel (b) shows the full sample.

Table B5: Predicting correct directional belief $1_{nvs}$ on subsets of variables and shocks

| | Main | | Oil & MP | | Decision-relevant | | Decision-irrelevant | |
|---|---|---|---|---|---|---|---|---|
| Correct directional belief $1_{nvs}$ | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Causal distance $D_{vs}$ | -0.24*** | -0.27*** | -0.23*** | -0.25*** | -0.37*** | -0.33*** | -0.12*** | -0.32*** |
| | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.07) |
| Constant | 0.99*** | 1.04*** | 0.99*** | 1.03*** | 1.24*** | 1.18*** | 0.71*** | 1.17*** |
| | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) | (0.03) | (0.16) |
| Observations | 10920 | 10763 | 7775 | 7775 | 1560 | 1560 | 4137 | 3902 |
| $R^2$ | 0.24 | 0.63 | 0.26 | 0.66 | 0.56 | 0.85 | 0.24 | 0.63 |
| Individual FE | Yes | Absorbed | Yes | Absorbed | Yes | Absorbed | Yes | Absorbed |
| Individual-variable FE | | Yes | | Yes | | Yes | | Yes |
| Individual-shock FE | | Yes | | Yes | | Yes | | Yes |

Standard errors in parentheses
Standard errors clustered at individual level
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$

*Notes:* Columns (1, 2) present the specification for the main sample that filters attention and completion time. Columns (3, 4) focus on respondents who receive the oil and monetary policy (MP) shocks. Columns (5, 6) include only responses to the household block in the survey, which covers four decision-relevant variables: price, wage, interest rate, and personal income tax. Columns (7, 8) include only responses outside the household block and exclude the four aforementioned decision-relevant variables.

Table B6: Predicting correct directional belief $1_{nvs}$ with robustness versions

| Correct directional belief $1_{nvs}$ | Main (1) | Full (2) | $D_{vs}^{M1}$ (3) | $D_{vs}^{M3}$ (4) | $1_{nvs}^{Vb}$ (5) | $1_{nvs}^{Vc}$ (6) |
|---|---|---|---|---|---|---|
| Causal distance $D_{vs}$ | -0.27*** | -0.23*** | -0.19*** | -0.19*** | -0.20*** | -0.26*** |
| | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) |
| Constant | 1.04*** | 0.91*** | 0.89*** | 0.87*** | 0.93*** | 1.06*** |
| | (0.02) | (0.02) | (0.01) | (0.01) | (0.02) | (0.02) |
| Observations | 10763 | 22023 | 10763 | 12479 | 8335 | 14357 |
| $R^2$ | 0.63 | 0.58 | 0.61 | 0.62 | 0.25 | 0.60 |
| Individual FE | Absorbed | Absorbed | Absorbed | Absorbed | Yes | Absorbed |
| Individual-variable FE | Yes | Yes | Yes | Yes | | Yes |
| Individual-shock FE | Yes | Yes | Yes | Yes | | Yes |

Standard errors in parentheses
Standard errors clustered at individual level
$^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$

*Notes:* Column (1) presents the main specification with individual-variable and individual-shock fixed effects on the main sample that filters attention and completion time. Column (2) examines the full sample, i.e., all respondents who completed the survey. Columns (3, 4) use the distance implied by Model 1 and Model 3 in Table B4, instead of Model 2. Columns (5, 6) use the shock-variable combinations from Version b and Version c in Table B3, instead of Version a. In the case of Version b, which includes fewer shock-variable combinations, there is not enough variation to apply individual-variable and individual-shock fixed effects.

## B.3 Depth of Thinking is Individual Characteristic and Domain-Specific

We further show that the ability to understand shock propagation is indeed an individual characteristic and does not correlate with a classic measure of level-$k$ thinking.

The previous finding that individual fixed effects matter for correct directional beliefs $1_{nvs}$ already suggests that some people get more variables correct than others. To further investigate this, we measure individual $n$'s overall understanding of shock $s$ by a *total depth score (TDS)* as

$$TDS_{ns} \equiv \sum_{v} D_{vs} \cdot 1_{nvs} \tag{B1}$$

To receive a higher TDS, a respondent needs to correctly forecast directional changes in more variables and especially more distant variables.[33]

To the extent that depth of thinking is an individual characteristic as we postulate, we expect each respondent's TDSs to correlate strongly across shocks. To test this, we rank the TDSs from the lowest to the highest for each shock, and correlate the two TDS rank measures across individuals in Table B7. Column (1) confirms this prediction.

[33] This TDS is a more robust measure to noise than the maximum distance of variables whose changes are understood correctly, as we have several variables for each step and respondents may coincidentally get some correct.

Table B7: Total depth score is individual characteristic and domain-specific

| | | | Main | | | Incentivized | Full |
|---|---|---|---|---|---|---|---|
| TDS of 2nd shock (rank) | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| TDS of 1st shock (rank) | 0.31*** | | | | 0.24*** | 0.30** | 0.20*** |
| | (0.05) | | | | (0.05) | (0.11) | (0.04) |
| Level $k$ (rank) | | -0.08 | | | -0.04 | 0.05 | -0.02 |
| | | (0.05) | | | (0.05) | (0.11) | (0.03) |
| Financial literacy (rank) | | | 0.28*** | | 0.23*** | 0.11 | 0.20*** |
| | | | (0.05) | | (0.05) | (0.11) | (0.03) |
| Education (rank) | | | 0.02 | | -0.02 | -0.04 | -0.04 |
| | | | (0.05) | | (0.06) | (0.12) | (0.04) |
| Male | | | -0.03 | | -0.03 | -0.10 | -0.01 |
| | | | (0.03) | | (0.03) | (0.06) | (0.02) |
| Net asset (rank) | | | | 0.05 | 0.05 | 0.04 | 0.08* |
| | | | | (0.06) | (0.06) | (0.11) | (0.04) |
| Income (rank) | | | | 0.06 | 0.06 | 0.11 | 0.03 |
| | | | | (0.06) | (0.07) | (0.13) | (0.05) |
| Constant | 0.38*** | 0.59*** | 0.37*** | 0.49*** | 0.25*** | 0.29* | 0.26*** |
| | (0.03) | (0.03) | (0.05) | (0.04) | (0.06) | (0.14) | (0.04) |
| Observations | 383 | 383 | 383 | 383 | 383 | 118 | 828 |
| $R^2$ | 0.09 | 0.01 | 0.09 | 0.02 | 0.15 | 0.17 | 0.12 |
| Shock group FE | | | | | Yes | Yes | Yes |
| Age group FE | | | | | Yes | Yes | Yes |

Standard errors in parentheses

Standard errors are heteroscedasticity-consistent

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

*Notes:* Columns (1-5) use the main sample that filters attention and completion time. Column (6) focuses on a subsample of the main sample, where the "guess 2/3 of the average" game is incentivized. Column (7) studies the full sample, i.e., all respondents who completed the survey.

In contrast, column (2) suggests that TDS does not correlate with a classic measure of strategic sophistication (level $k$), via a "guess 2/3 of the average" game we play with survey respondents. This connects to findings in the macroeconomic literature that the measured level $k$ does not predict differential consumption response to inflation news by Dutch households (Coibion et al., 2023*a*) or first- and higher-order inflation expectations of New Zealand firm managers (Coibion et al., 2021).[34] We remark that shallow thinking

[34]In the experimental literature, Dal Bó, Dal Bó and Eyster (2018) show that voters prefer policy changes that bring in direct benefits but induce larger indirect costs, but their voting behavior is not correlated with level $k$. Georganas, Healy and Weber (2015) study stability of level $k$ using two families of games: beauty contest games à la Nagel (1995) and undercutting games similar to Arad and Rubinstein (2012). They find that the participants' levels are consistent within the beauty contest family, but do not correlate within the

likely reflects people's limited knowledge about the macroeconomy, a distinct aspect of bounded rationality from limited strategic sophistication. After all, a chess master who could anticipate opponents well may not know macroeconomics, and vice versa.

Column (3) suggests that a measure of financial literacy, based on Lusardi and Mitchell (2011), but not general education, correlates with TDS. This supports the idea that shallow thinking reflects individuals' economic knowledge, for which general education may be too noisy a proxy.

Column (4) shows that households' net asset positions or income do not significantly correlate with TDS. The two significant predictors, TDS and financial literacy, remain significant when all variables are pooled together, as indicated in column (5).

Columns (6) and (7) demonstrate that our findings hold true over a subsample where the "guess 2/3 of the average" game is incentivized, as well as in the full sample, i.e., all respondents who completed the survey.

# C Shallow Thinking in an RBC Economy

Here, we consider an RBC economy, and show that shallow thinking amplifies its responses to persistent productivity shocks and produces a stock market boom and crash.

## C.1 Shallow Thinking in an RBC Economy

We outline the model and shallow thinking thereof, with details in Appendix E.

The RBC economy consists of two types of agents (firms and households) and two competitive markets (the goods market and the labor market). We study a first-order approximation around the steady state. Differing from the convention in the New Keynesian model, here we use capital letters for variables in levels, so that different components of GDP are in the same units.

As in the prior analysis of the New Keynesian economy, we assume that agents observe the sequence of shocks at time $0^-$ and form their expectations $\mathbb{E}^d[\cdot]$ once and for all at time $0^-$, i.e., $\mathbb{E}^d_t[v_s] \equiv \mathbb{E}^d[v_s]$ for $s > t$.

undercutting game family or across two families.

**Firms.** There is a continuum of firms that produce using capital $K_t$ and labor $N_t^d$ with a production function $Y_t = Z_t \left(\frac{K_t}{\alpha}\right)^{\alpha} \left(\frac{N_t^d}{1-\gamma}\right)^{1-\gamma}$. Firms own capital and make investment $I_t$ to increase the capital stock in the next period, $K_{t+1} = (1-\delta) K_t + I_t$. In addition, firms are subject to capital adjustment cost $\Psi(I_t, K_t) = \frac{\psi}{2}\left(\frac{I_t}{K_t} - \delta\right)^2 K_t$, which gives rise to an investment-$q$ relation as in Hayashi (1982), and pay dividends $DIV_t = Y_t - W_t N_t^d - I_t - \Psi(I_t, K_t)$. Firms maximize their values, i.e., the sum of dividends discounted by the gross interest rate. We use $1 + r_t$ to denote the gross interest rate from period $t$ to $t+1$ and assume that it is known in period $t$, as the return of a 1-period bond in zero supply.

In each period, firms choose investment $I_t$, output $Y_t$, and labor demand $N_t^d$, responding to current productivity $Z_t$, the prevailing wage $W_t$, and interest rate $r_t$, as well as future productivities $\{Z_{t+k}\}_{k\geq 1}$ and their expectations $\left\{\overline{\mathbb{E}}\left[W_{t+k}\right], \overline{\mathbb{E}}\left[r_{t+k}\right]\right\}_{k\geq 1}$.

**Households.** There is a continuum of households who live infinitely and maximize their lifetime utility, discounted by $\beta$, which is separable in consumption and labor supply. In each period, households choose consumption $C_t$ and labor supply $N_t^s$, taking as given the prevailing wage $W_t$, interest rate $r_t$ and dividend $DIV_t$, as well as their expectations of the future values of these variables $\left\{\overline{\mathbb{E}}\left[W_{t+k}\right], \overline{\mathbb{E}}\left[r_{t+k}\right], \overline{\mathbb{E}}\left[DIV_{t+k}\right]\right\}_{k\geq 1}$. The household side of the RBC economy is the same as that of the New Keynesian economy, except that they invest in a real bond as opposed to a nominal bond.

**Goods and labor markets.** The interest rate $\{r_t\}_{t\geq 0}$ and wage $\{w_t\}_{t\geq 0}$ arise to clear the goods and labor markets, $Y_t = I_t + \Psi(I_t, K_t) + C_t, N_t^d = N_t^s$.

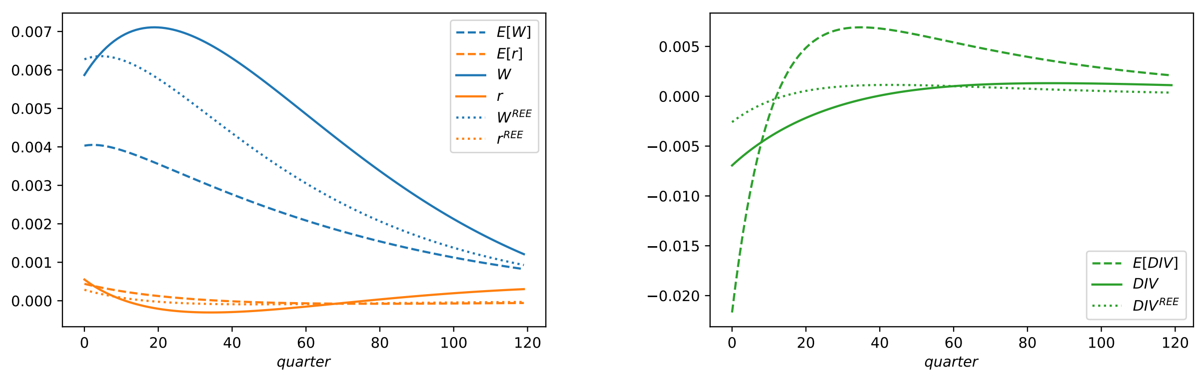
**Shallow thinking.** In terms of belief formation, the system of causal relations consists of firms and households' best responses, as well as the determination of the interest rate and wage by two fictitious auctioneers for the goods and labor markets, respectively. The interest rate depends on the shifts in firms' output and investment and in households' consumption ($\hat{Y}, \hat{I}$ and $\hat{C}$), each of which is a function of decision-relevant variables for firms and households other than the interest rate. Similarly, the wage is determined by the shifts in firms' labor demand and households' labor supply ($\hat{N}^d, \hat{N}^s$), both of which depend on decision-relevant variables other than the wage. Figure E1 illustrates these causal relations, which pin down the average expectations $\overline{\mathbb{E}}[\cdot]$.

**Equilibrium.** Given the average expectations, the equilibrium is determined by agents' best responses and market clearing conditions, period by period.

We adopt a quarterly calibration of the RBC economy, with all parameters listed in Table E1. We assume slight decreasing returns to scale in production ($\alpha < \gamma$) and the existence of a small fringe of rational agents, which we discuss in greater detail in Appendix E.

## C.2 Amplified and Prolonged Output Response and Stock Market Boom-Crash

We study the effects of a persistent productivity shock, with $\rho = 0.979$ following King and Rebelo (1999), on the macroeconomy and asset prices. In particular, we show that shallow thinking amplifies the economy's responses and leads to a stock market boom and crash.



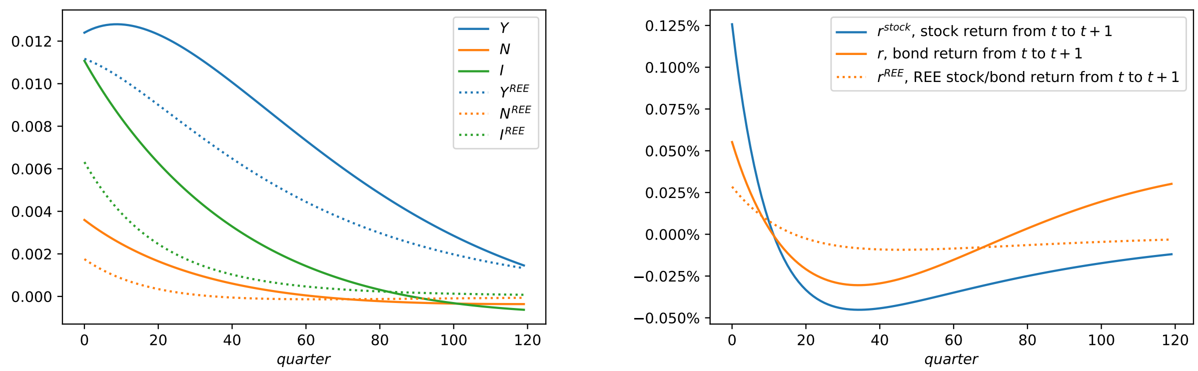

(a) Wage $W$ and interest rate $r$

(b) Dividend $DIV$

Figure C1: Decision-relevant beliefs in response to a productivity shock $Z_t$

*Notes:* Two panels plot the shallow thinking beliefs (in dashed lines), the shallow thinking equilibrium outcomes (in solid lines) and the rational expectations equilibrium outcomes (in dotted lines) of wage $W$, interest rate $r$, and dividend $DIV$, in response to a persistent productivity shock $Z$.

Figure C1 illustrates in dashed lines beliefs about the interest rate $r$, wage $W$, and dividend $DIV$, which are relevant for firms' and households' decisions. In comparison, the solid lines indicate their equilibrium values, and the dotted lines indicate the rational expectations equilibrium (REE). Panel C1a suggests that shallow agents underappreciate the wage response. That occurs because agents believe that the firms will produce more by hiring and investing more in response to a productivity shock, but fail to recognize that firms' behavior will push up wages in the economy. Panel C1b shows that shallow agents perceive a much more volatile stream of dividends compared to the equilibrium values or the REE. This is because they believe firms will invest more early on by cutting dividends and pay more dividends later. However, in equilibrium, facing elevated wages and changes in interest rates, firms will not invest as much as shallow agents believe.




(a) Output $Y$, labor hours $N$ and investment $I$

(b) Stock return $r^{stock}$ and bond return $r$

Figure C2: Equilibrium outcomes in response to productivity shock $Z_t$

*Notes:* Two panels plot the shallow thinking equilibrium outcomes (in solid lines) and the rational expectations equilibrium outcomes (in dotted lines) of output $Y$, labor hours $N$, investment $I$, stock return $r^{stock}$ and bond return $r$ (which is the interest rate), in response to a persistent productivity shock. In the rational expectations equilibrium, the stock and bond returns coincide.

Figure C2 shows the responses of macroeconomic variables and asset returns in solid lines, compared to the rational expectations equilibrium in dotted lines. Panel C2a demonstrates that the output response (in blue) under shallow thinking is hump-shaped and more persistent. The responses of investment and labor hours (in green and orange) are almost twice as large.

Upon observing the productivity shock, the stock market experiences a positive revaluation at time $0^{-}$. Panel C2b suggests that, after the revaluation, the stock excess return relative to bond from $t$ to $t+1$, $xr_t^{stock} = r_t^{stock} - r_t$, is initially positive but turns negative after a few quarters. This occurs because agents underestimate dividends in the short term but overestimate them in the long term, as shown in Figure C1b. This pattern of positive stock excess returns turning negative is consistent with the classic Kindleberger (1978) narrative of crises and many real-world episodes, in which technological or financial innovations lead to booms and crashes in asset markets.

Table C1 summarizes the key moments of the impulse responses to a persistent productivity shock, with the rational expectations equilibrium in column (1) and shallow

Table C1: Moments of RBC impulse responses

| | (1) REE | (2) ST | (3) ST firms | (4) ST households |
|---|---|---|---|---|
| ***Shock (same for all columns)*** | | | | |
| Half life of productivity shock $Z_t$ | 33 | 33 | 33 | 33 |
| Shock size $\mathrm{d}Z_0/\overline{Z}$ | 1.0% | 1.0% | 1.0% | 1.0% |
| ***Macroeconomy*** | | | | |
| Half life of $Y_t$ | 49 | 69 | 71 | 66 |
| Quarter of peak output $Y_t$ (hump shape) | 0 | 9 | 3 | 11 |
| Peak output $\mathrm{d}Y_t/\overline{Y}$ | 1.12% | 1.28% | 1.22% | 1.98% |
| Peak hours $\mathrm{d}N_t/\overline{N}$ | 0.18% | 0.36% | 0.32% | 1.15% |
| Peak investment $\mathrm{d}I_t/\overline{I}$ | 3.54% | 6.2% | 5.66% | 17.68% |
| Peak consumption $\mathrm{d}C_t/\overline{C}$ | 0.8% | 0.83% | 0.79% | 1.33% |
| ***Asset prices*** | | | | |
| Peak stock price $\mathrm{d}P_t/\overline{P}$ | 0.85% | 2.55% | 2.59% | 2.59% |
| Quarter of excess return $xr_t^{stock}$ turning negative | $xr_t^{stock} = 0$ always | 12 | 9 | $xr_t^{stock} < 0$ always |

*Notes:* Four columns correspond to the rational expectations equilibrium (REE), the shallow thinking equilibrium (ST), a model in which firms are shallow but households are rational, and a model in which households are shallow but firms are rational. In the latter two cases which feature belief disagreement, to illustrate the stock market dynamics, we simply assume that the stock is priced by the shallow agents.

thinking equilibrium in (2). Quantitatively, shallow thinking leads to amplified and more persistent responses in labor hours, investment and the stock price relative to the rational expectations equilibrium. Qualitatively, it produces hump-shaped responses in consumption and output, as well as a stock market boom and crash, which are absent under rational expectations. Columns (3) and (4) further examine cases where only firms or households exhibit shallow thinking, suggesting that both parties' beliefs are important.

# D Appendix to the New Keynesian Model

## D.1 The Infinite-Horizon New Keynesian Model

**Firms.** There is a continuum of firms indexed by $j \in [0, 1]$ in this economy subject to Calvo price rigidity. Each firm chooses labor demand $N_{jt}^s$, pays dividend $DIV_{jt}$, and sets price $P_{jt}^*$ when it can, taking as given the aggregate inflation rate $\pi_t$, the real wage $W_t$, and the aggregate demand $C_t$. They agree on the steady state of the economy but may have heterogeneous beliefs about the economy's response to shocks.

Each firm produces a differentiated good, using the same constant-returns production technology using labor hours $Y_{jt} = N_{jt}^d$, which together forms a bundle with constant elasticity $\varepsilon$ of substitution (CES) that the households consume. At the steady state, they each charge a markup $\mu \equiv \frac{\varepsilon}{\varepsilon-1}$. The log-linearized real dividend and aggregate labor demand are

$$div_t = c_t - \frac{1}{\mu - 1} w_t \tag{D1}$$

$$n_t^d = c_t \tag{D2}$$

since the price dispersion only introduces second-order changes as in Galí (2015).

Each firm resets its price with independent probability $1-\theta$ in any period and fulfills its demand period by period. When considering its reset price $P_{jt}^*$, each firm maximizes its discounted sum of profits

$$\max_{P_{jt}^*} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_{jt} \left[ \beta^k \frac{C_{t+k}^{-\sigma}}{C_t^{-\sigma}} \frac{P_{jt}^* - W_{t+k} P_{t+k}}{P_{t+k}} Y_{j,t+k|t} \right]$$

where $\beta^k \frac{C_{t+k}^{-\sigma}}{C_t^{-\sigma}}$ is the discount factor and $W_{t+k}P_{t+k}$ is the nominal marginal cost, subject to the sequence of demand constraints

$$Y_{j,t+k|t} = \left( \frac{P_{jt}^*}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k}$$

The first-order condition is

$$0 = \sum_{k=0}^{\infty} \theta^k \mathbb{E}_{jt} \left[ \beta^k \frac{C_{t+k}^{-\sigma}}{C_t^{-\sigma}} \frac{(1-\varepsilon) P_{jt}^* + \varepsilon W_{t+k} P_{t+k}}{P_{t+k}} Y_{j,t+k|t} \right]$$

and thus

$$p_{jt}^* = p_{t-1} + (1-\beta\theta) \sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_{jt} \left[ \sum_{l=0}^{k} \pi_{t+l} + w_{t+k} \right]$$

Aggregate inflation emerges from the pricing behavior of the $(1-\theta)$ share of resetting firms as $\pi_t = (1-\theta)\left(p_{jt}^* - p_{t-1}\right)$. Following the tradition, we consider a cost-push shock $\epsilon_t^\pi$ for inflation

$$\pi_t = \theta\kappa w_t + (1-\theta)\pi_t + \theta\kappa \sum_{k=1}^{\infty} (\beta\theta)^k \overline{\mathbb{E}}_t [w_{t+k}] + (1-\theta) \sum_{k=1}^{\infty} (\beta\theta)^k \overline{\mathbb{E}}_t [\pi_{t+k}] + \epsilon_t^\pi \tag{D3}$$

with $\kappa \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}$ capturing the slope of the Phillips curve and $\overline{\mathbb{E}}_t[\cdot]$ being the average expectations. Importantly, we do not move $\pi_t$ on the right-hand side to the left. We intentionally preserve the dependence of $\pi_t$ on itself, which encapsulates the within-period complementarity in individual price setting as each firm takes aggregate inflation as given.

**Households.** There is a continuum of households who live infinitely indexed by $h \in [0,1]$. Each household chooses consumption $C_{ht}$ and labor supply $N^s_{ht}$, taking as given the gross nominal interest rates $R_{t-1}$, the inflation rate $\pi_t \equiv P_t/P_{t-1} - 1$, the real wage $W_t$, and the real dividend $DIV_t$. They agree on the steady state of the economy but may have heterogeneous beliefs about the economy's response to shocks.

They each maximize

$$\max_{\{C_{ht}, N^s_{ht}\}_{t\geq 0}} \mathbb{E}_{h,t=0} \sum_{t=0}^{\infty} \beta^t \left( \frac{C_{ht}^{1-\sigma} - 1}{1-\sigma} - \frac{\left(N^s_{ht}\right)^{1+\nu}}{1+\nu} \right)$$

subject to the budget constraint

$$C_{ht} + A_{ht} = \frac{R_{t-1}}{1+\pi_t} A_{h,t-1} + W_t N^s_{ht} + DIV_t$$

where $C_{ht}$ is a CES bundle of goods in the economy, $A_{ht}$ is the period-$t$ saving.

We derive the log-linearized aggregate consumption and labor supply functions as follows. The log-linearized life-time budget constraint is

$$\mathbb{E}_{ht} \sum_{k=0}^{\infty} \beta^k \left( c_{h,t+k} - \underbrace{\frac{\overline{WN}}{\overline{C}}}_{\mu^{-1}} \left( w_{t+k} + n^s_{h,t+k} \right) - \underbrace{\frac{\overline{DIV}}{\overline{C}}}_{1-\mu^{-1}} div_{t+k} \right) - \beta^{-1} a_{h,t-1} = 0$$

where the lower-case variables denote the log deviation from the corresponding steady-state values, and $\mu = \frac{\overline{C}}{\overline{WN}}$ denotes the ratio of consumption to labor income at the steady state (which equals firms' steady state markup $\mu$).

Log-linearizing the consumption-labor FOC $W_t C_{ht}^{-\sigma} = \left(N_{ht}^s\right)^\nu$ yields

$$n_{ht}^s = \frac{1}{\nu}\left(w_t - \sigma c_{ht}\right)$$

Plugging this into the budget constraint gives

$$\mathbb{E}_{ht}\sum_{k=0}^{\infty}\beta^k\left(\left(1+\mu^{-1}\sigma\nu^{-1}\right)c_{h,t+k} - \mu^{-1}\left(1+\nu^{-1}\right)w_{t+k} - \left(1-\mu^{-1}\right)div_{t+k}\right) - \beta^{-1}a_{h,t-1} = 0$$

$$\mathbb{E}_{ht}\sum_{k=0}^{\infty}\beta^k\left(c_{h,t+k} - \frac{\mu^{-1}(1+\nu)}{\mu^{-1}\sigma+\nu}w_{t+k} - \frac{\left(1-\mu^{-1}\right)\nu}{\mu^{-1}\sigma+\nu}div_{t+k}\right) - \frac{\beta^{-1}\nu}{\mu^{-1}\sigma+\nu}a_{h,t-1} = 0$$

Log-linearizing the Euler condition $C_{ht}^{-\sigma} = \mathbb{E}_{ht}\left[\frac{\beta R_t}{1+\pi_{t+1}}C_{h,t+1}^{-\sigma}\right]$ gives

$$c_{ht} = \mathbb{E}_{ht}\left(c_{h,t+1} - \sigma^{-1}\left(i_t - \pi_{t+1}\right)\right)$$

Combining this with the budget constraint to substitute $c_{h,t+k}$ gives rise to

$$\mathbb{E}_{ht}\sum_{k=0}^{\infty}\beta^k\left(c_{ht} + \sum_{l=0}^{k-1}\sigma^{-1}\left(i_{t+l} - \pi_{t+l+1}\right) - \frac{\mu^{-1}(1+\nu)}{\mu^{-1}\sigma+\nu}w_{t+k} - \frac{\left(1-\mu^{-1}\right)\nu}{\mu^{-1}\sigma+\nu}div_{t+k}\right) - \frac{\beta^{-1}\nu}{\mu^{-1}\sigma+\nu}a_{h,t-1} = 0$$

$$\frac{1}{1-\beta}c_{ht} - \sum_{k=0}^{\infty}\beta^k\mathbb{E}_{ht}\left(\frac{(1+\nu)}{\sigma+\mu\nu}w_{t+k} + \frac{(\mu-1)\nu}{\sigma+\mu\nu}div_{t+k}\right) + \frac{\sigma^{-1}\beta}{1-\beta}\sum_{k=0}^{\infty}\beta^k\mathbb{E}_{ht}\left(i_{t+k} - \pi_{t+k+1}\right) - \frac{\beta^{-1}\mu\nu}{\sigma+\mu\nu}a_{h,t-1} = 0$$

which leads to the aggregate consumption function, once aggregated across all households

$$c_t = -\sigma^{-1}\beta i_t - \sigma^{-1}\beta\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}_t\left[i_{t+k}\right] + \sigma^{-1}\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}_t\left[\pi_{t+k}\right] + (1-\beta)\left[\frac{(\mu-1)\nu}{\sigma+\mu\nu}div_t + \frac{(1+\nu)}{\sigma+\mu\nu}w_t\right]$$
$$+ (1-\beta)\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}_t\Big[\frac{(\mu-1)\nu}{\sigma+\mu\nu}div_{t+k} + \frac{(1+\nu)}{\sigma+\mu\nu}w_{t+k}\Big] \tag{D4}$$

Using the consumption-labor FOC $n_{ht}^s = \frac{1}{\nu}\left(w_t - \sigma c_{ht}\right)$ again, we get the aggregate labor supply function

$$n_t^s = \nu^{-1}\beta i_t + \nu^{-1}\beta\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}_t\left[i_{t+k}\right] - \nu^{-1}\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}_t\left[\pi_{t+k}\right] - (1-\beta)\frac{(\mu-1)\sigma}{\sigma+\mu\nu}div_t + \nu^{-1}\left(1 - \sigma\frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu}\right)w_t$$

$$- (1-\beta) \sum_{k=1}^{\infty} \beta^k \overline{\mathbb{E}}_t \Big[ \frac{(\mu - 1)\,\sigma}{\sigma + \mu\nu} div_{t+k} + \frac{(1+\nu)\,\sigma\nu^{-1}}{\sigma + \mu\nu} w_{t+k} \Big] \tag{D5}$$

where $\overline{\mathbb{E}}_t\,[\cdot]$ denotes the average expectations.

**Central bank.** The central bank follows a Taylor rule with a monetary policy shock $\epsilon_t^i$,

$$i_t = \phi \pi_t + \epsilon_t^i \tag{D6}$$

**Labor market.** Last, to close the model, the wage arises by equilibrating labor supply and demand

$$n_t^s = n_t^d \tag{D7}$$

**Equilibrium.** We study a *temporary equilibrium* in which agents maximize their utilities, taking as given the average expectations $\overline{\mathbb{E}}_t\,[\cdot]$, and markets clear. In this New Keynesian model, equations (D1-D7) characterizes the equilibrium given the average expectations.

## D.2 3-Variable Representation

A popular way of representing the textbook New Keynesian model is to use three variables—the interest rate $i_1$, inflation $\pi_1$, and consumption $c_1$ (i.e., output). To derive this representation from (1-7), we first solve for the other variables $V_1^{other} = \left(div_1, \hat{n}_1^d, \hat{n}_1^s, w_1\right)$ in (8) in terms of $(i_1, \pi_1, c_1)$. Substituting $V_1^{other}$ into the consumption function (4) yields $c_1 = -\sigma i_1$. Then, substituting $V_1^{other}$ into the Phillips curve (3) and using $c_1 = -\sigma i_1$ to eliminate $i_1$, we obtain $\pi_1 = \kappa\,(\sigma + \nu)\,c_1 + \theta^{-1}\epsilon_1^{\pi}$, which expresses inflation solely as a function of output $c_1$. The Taylor rule (6) is kept unchanged.

$$\begin{pmatrix} i_1 \\ \pi_1 \\ c_1 \end{pmatrix} = \begin{pmatrix} 0 & \phi & 0 \\ 0 & 0 & \kappa\,(\sigma+\nu) \\ -\sigma & 0 & 0 \end{pmatrix} \begin{pmatrix} i_1 \\ \pi_1 \\ c_1 \end{pmatrix} + \begin{pmatrix} \epsilon_1^i \\ \theta^{-1}\epsilon_1^{\pi} \\ 0 \end{pmatrix} \tag{D8}$$

This representation expresses inflation $\pi_1$ as a function of consumption $c_1$, rather than in the best response form (3), by canceling the pricing complementarity term $\pi_1$ on the right-hand side with the left-hand side in (3) and implicitly substituting the equilibrium value of the wage, thereby failing to reflect firms' best responses. While (D8) shares the same fixed point in terms of $(i_1, \pi_1, c_1)$ as our representation (8), it is not a system of best responses and involves a different iterative process to arrive at the fixed point. Furthermore, this 3-variable representation is still incomplete in that it does not take a stance on how other variables, e.g., wage $w_1$ and dividend $div_1$, depend on these three variables.

## D.3 Belief Misreaction and Strength of GE Feedback

We offer a synthesis of belief misreaction under shallow thinking. To start with, consider a simple model with a single general equilibrium (GE) loop. Beliefs about a variable underreact (or overreact) to shocks that directly affect itself if the single GE loop amplifies (or offsets) the direct effect of these shocks, as shallow agents underappreciate the GE loop, in line with existing theories reviewed by Angeletos and Lian (2023*a*). Additionally, our theory suggests that beliefs about a variable underreact to shocks that directly affect other variables (e.g., inflation in response to cost-push shocks), because shallow agents underperceive shock propagation. This latter prediction extends to more complex models with multiple GE loops.

In models with multiple GE loops, such as our workhorse macroeconomic models, whether beliefs of variables over- or underreact to shocks that directly affect themselves depends jointly on the strength and length of these GE loops. For example, consider the inflation response to a cost-push shock $\epsilon_1^\pi$. Under our calibration, shallow agents perceive the cost-push shock $\epsilon_1^\pi$ as amplified, even though it is actually offset, i.e., $\frac{\partial \overline{\mathbb{E}}_0[\pi_1]}{\partial \epsilon_1^\pi} > 1 > \frac{\partial \pi_1}{\partial \epsilon_1^\pi}$. Shallow thinking flips the sign of the perceived net GE effect, because shallow agents underappreciate the long, strong offset loop involving the monetary policy reaction. The order of operations is key because GE feedback loops of varying length are dampened differently, due to the Jensen's inequality. The implication could differ if one ignored length by first collapsing multiple GE loops in the model into a single net effect (which, in this case, is offset) and then applying dampening.

To illustrate, we study inflation expectations, varying the strength of the offset loop by adjusting the Taylor rule coefficient $\phi$, as depicted in Figure D1. If, instead of $\phi = 1.5$, $\phi$ takes on an intermediate value, e.g., $\phi = 1$, the long offset loop is too weak to turn the true net effect into offset. In that case, the true net effect is amplification, but shallow agents perceive even *more* amplification, i.e., $\frac{\partial \overline{\mathbb{E}}_0[\pi_1]}{\partial \epsilon_1^\pi} > \frac{\partial \pi_1}{\partial \epsilon_1^\pi} > 1$, as they overweigh the short amplification loop relative to the long offset loop. Again, if one naively collapsed multiple GE loops into a net amplification effect and dampened it, the perceived amplification would be less rather than more.

With an even lower Taylor rule coefficient $\phi$, e.g., $\phi = 0$ in a case where the policy rate is constrained at the zero lower bound (ZLB) in response to a deflationary shock,[35] the long offset loop is weak or non-existent, leaving the short amplification loop as the dominant force. That leads us towards a simple case with only one GE loop that amplifies the direct effect, which we began this discussion with. As a result, shallow agents perceive less amplification on net than the true effect, i.e., $\frac{\partial \pi_1}{\partial \epsilon_1^\pi} > \frac{\partial \overline{\mathbb{E}}_0[\pi_1]}{\partial \epsilon_1^\pi} > 1$.

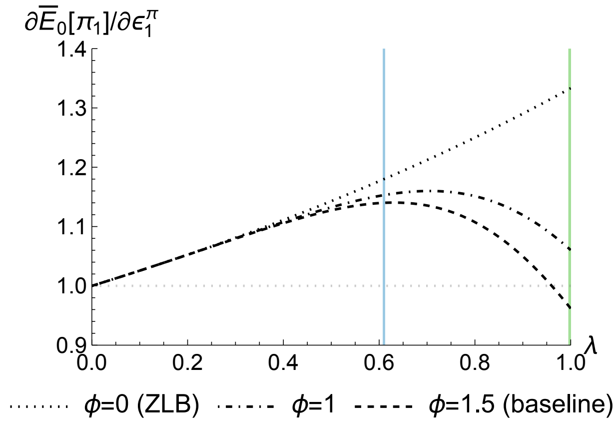


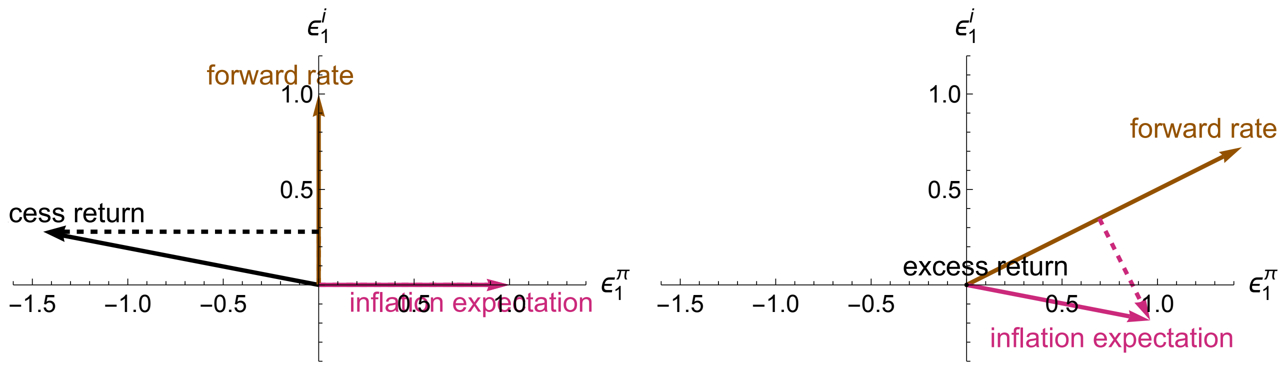

Figure D1: Inflation expectation $\overline{\mathbb{E}}_0[\pi_1]$ in response to cost-push news shocks $\epsilon_1^\pi$, varying Taylor rule coefficient $\phi$

*Notes:* This figure plots the average interest expectation $\overline{\mathbb{E}}_0[i_1]$ (relative to the size of the shock) as a function of the shallow thinking parameter $\lambda$, in response to news about a cost-push shock $\epsilon_1^\pi$, under different values of the Taylor rule coefficient $\phi$. The blue line indicates our calibration of $\lambda$, while the green line represents rational expectations ($\lambda = 1$).

## D.4 Additional Results of Bond Returns

**Bond excess returns with transitory news shocks.** Figure D2 shows the loadings of bond excess returns and predictors on transitory news shocks, under $\lambda = 0, 1$, complementing Figure 7b.

[35] The Taylor principle requires $\phi > 1$ for determinacy of the infinite-horizon New Keynesian model under rational expectations. In our 2-period setting, determinacy is not a concern. With persistent shocks, shallow thinking beliefs are uniquely defined by a formula similar to (10), which helps select an equilibrium. For further discussion on determinacy, which is not our focus, see Farhi and Werning (2019), Gabaix (2020) and Angeletos and Lian (2023*b*).




(a) Loadings on news shocks ($\lambda = 0$)

(b) Loadings on news shocks ($\lambda = 1$)

Figure D2: Predictability of bond excess returns, with transitory news shocks (cont'd)

*Notes:* These panels plot the loadings of bond excess returns and predictors on transitory news shocks, under $\lambda = 0, 1$, complementing Figure 7b. In the case of $\lambda = 0$, the forward rate and inflation expectation point along the y- and x-axes, respectively. In the case of $\lambda = 1$, the excess return is 0.

**Sensitivity of long-term rates to short-term rates with persistent shocks.** We consider an economy impacted by persistent monetary policy shocks $\epsilon_t^i = \rho^t \epsilon^i$ and cost-push shocks $\epsilon_t^\pi = \rho^t \epsilon^\pi$. Suppose that in each period, the two shocks have variances $\sigma_i^2$ and $\sigma_\pi^2$, and are independently distributed.

In this environment, the yield of a $T$-period bond, $y_t^{(T)}$, is determined by the expectations hypothesis as

$$y_t^{(T)} = \frac{i_t + \sum_{k=1}^{T-1} \overline{\mathbb{E}}\left[i_{t+k}\right]}{T} \tag{D9}$$

We consider a univariate regression of the long-term yield $y_t^{(T)}$ on the short-term interest rate $i_t$ à la Hanson, Lucca and Wright (2021)

$$y_t^{(T)} = \beta^{OLS} i_t + \alpha + \epsilon_0 \tag{D10}$$

Since the economy is affected by two different shocks, the univariate regression coefficient $\beta^{OLS}$ depends on the mix of these shocks. To shed light on this, we express the responses of interest rate expectation $\overline{\mathbb{E}}\left[i_{t+k}\right]$ and equilibrium interest rate $i_t$ to shocks as

$$\begin{aligned}\overline{\mathbb{E}}\left[i_{t+k}\right] &= \rho^k \gamma_i^{exp} \epsilon_t^i + \rho^k \gamma_\pi^{exp} \epsilon_t^\pi, \ \forall k \geq 1 \\ i_t &= \gamma_i^{ST} \epsilon_t^i + \gamma_\pi^{ST} \epsilon_t^\pi\end{aligned}$$

where the $\gamma$ coefficients are implicit functions of model parameters including $\lambda$, and $\rho^k$ stems from the fact that beliefs are mean-reverting at the same rate $\rho$. The $\gamma^{exp}$ coefficients agree with $\gamma^{ST}$ coefficients under $\lambda = 1$ (rational expectations). We denote the relative importance of monetary policy shocks in driving the short-term interest rate as $\zeta_i \equiv \frac{(\gamma_i^{ST})^2\sigma_i^2}{(\gamma_i^{ST})^2\sigma_i^2+(\gamma_\pi^{ST})^2\sigma_\pi^2} \in [0, 1]$, and the relative importance of cost-push shocks as $1 - \zeta_i$.

According to (D9), the long-term yield follows

$$y_t^{(T)} = T^{-1}\left(\gamma_i^{ST} + \rho\frac{1-\rho^{T-1}}{1-\rho}\gamma_i^{exp}\right)\epsilon_t^i + T^{-1}\left(\gamma_\pi^{ST} + \rho\frac{1-\rho^{T-1}}{1-\rho}\gamma_\pi^{exp}\right)\epsilon_\pi^i$$

Therefore, the univariate regression coefficient $\beta^{OLS}$ from (D10) is

$$\beta^{OLS} = \frac{\text{Cov}\left(y_t^{(T)}, i_t\right)}{\text{Var}(i_t)} = \underbrace{T^{-1}\left(1 + \rho\frac{1-\rho^{T-1}}{1-\rho}\frac{\gamma_i^{exp}}{\gamma_i^{ST}}\right)}_{\equiv\beta_i^{IV}}\zeta_i + \underbrace{T^{-1}\left(1 + \rho\frac{1-\rho^{T-1}}{1-\rho}\frac{\gamma_\pi^{exp}}{\gamma_\pi^{ST}}\right)}_{\equiv\beta_\pi^{IV}}(1-\zeta_i) \quad \text{(D11)}$$

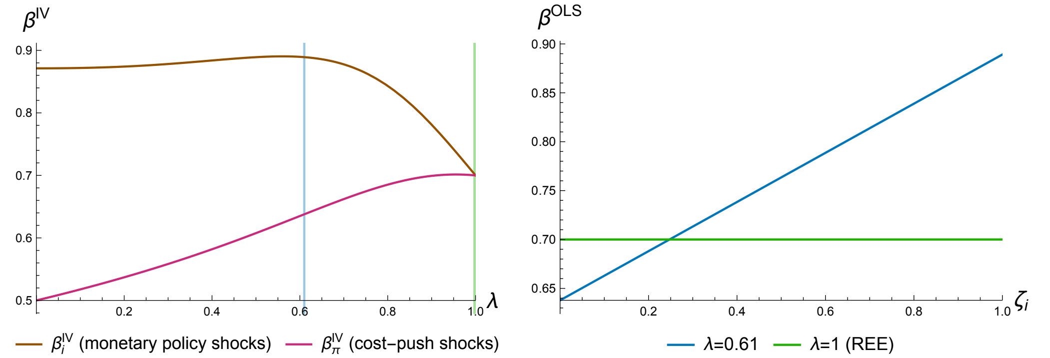
which linearly interpolates between $\beta_i^{IV}$, the conditional sensitivity under monetary shocks, and $\beta_\pi^{IV}$, the conditional sensitivity under cost-push shocks, based on the relative importance $\zeta_i$.

As $\gamma_i^{exp} = \gamma_i^{ST}$ and $\gamma_\pi^{exp} = \gamma_\pi^{ST}$ under rational expectations, we have

$$\beta^{OLS,REE} \equiv T^{-1}\frac{1-\rho^T}{1-\rho} \quad \text{(D12)}$$

which is independent of the shock mix $\zeta_i$.

Figure D3 plots these univariate regression coefficients $\beta_i^{IV}, \beta_\pi^{IV}$ and $\beta^{OLS}$, for a 2-period bond ($T = 2$) under a mild shock persistence $\rho = 0.4$. Panel D3a suggests that the conditional sensitivity to monetary policy shocks $\beta_i^{IV}$ (in brown) is higher than the conditional sensitivity to cost-push shocks $\beta_\pi^{IV}$ (in purple), as long as $\lambda < 1$. Consequently, under our calibrated $\lambda$, the unconditional sensitivity $\beta^{OLS}$ increases in the relative importance of the monetary policy shocks $\zeta_i$, as shown in panel D3b.




(a) Conditional sensitivities $\beta_i^{IV}$ and $\beta_\pi^{IV}$ as functions of shallow thinking parameter $\lambda$

(b) Unconditional sensitivity $\beta^{OLS}$ as a function of relative importance of monetary policy shocks $\zeta_i$

Figure D3: Univariate regression coefficients of long-term rates on short-term rates

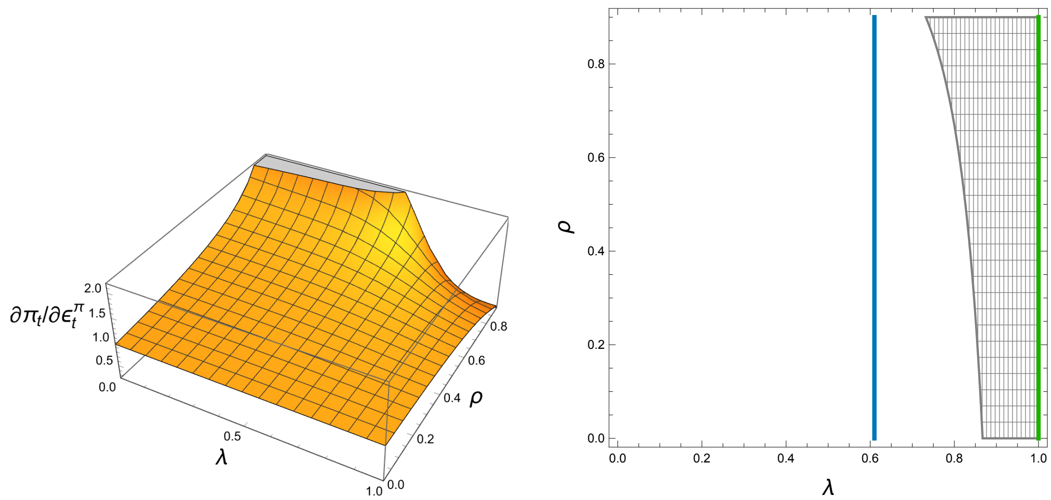
*Notes:* Panel (a) plots the conditional sensitivities $\beta_i^{IV}, \beta_\pi^{IV}$ of the long-term rate with $T = 2$ on the short-term rate, conditional on monetary policy shocks (in brown) and cost-push shocks (in purple) respectively, under $\rho = 0.4$, as functions of the shallow thinking parameter $\lambda$.
Panel (b) plots the unconditional sensitivity $\beta^{OLS}$ of the long-term rate with $T = 2$, under $\rho = 0.4$, as a function of the relative importance of monetary policy shocks $\zeta_i$. The blue line indicates our calibrated $\lambda$, while the green line represents the rational expectations equilibrium (REE) with $\lambda = 1$.

## D.5 Additional Results of Persistent Shocks

**The effect of more persistent cost-push shocks on equilibrium inflation.** Figure 38 suggests that under our calibrated $\lambda$, a more persistent cost-push shock leads to lower inflation, contrary to the rational expectations equilibrium. Figure D4 shows the equilibrium inflation response to a persistent cost-push shock, varying the shallow thinking parameter $\lambda$ and the persistence of the shock $\rho$. Panel D4a suggests that the inflation response increases in $\rho$ when $\lambda$ is low and decreases in $\rho$ when $\lambda$ is high. Panel D4b shows, in the gray meshed area, the combination of $(\lambda, \rho)$ values under which a more persistent cost-push shock leads to lower inflation than a purely transitory shock, i.e., $\frac{\partial \pi_t}{\partial \epsilon_t^\pi} < \frac{\partial \pi_t}{\partial \epsilon_t^\pi}|_{\rho=0}$. In the case of purely transitory shock ($\rho = 0$), the equilibrium is independent of $\lambda$, as agents observe all variables in the same period. The vertical blue line (our calibrated $\lambda$) is completely outside the meshed region, whereas the vertical green line (rational expectations equilibrium) is entirely within it.

**Persistent monetary policy shocks.**

**Proposition D1.** (Persistent monetary policy shock) *The rational-expectations equilibrium*




(a) Equilibrium inflation $\pi_t$ in response to cost-push shock $\epsilon_t^\pi$, varying $\lambda$ and $\rho$

(b) Combination of $(\lambda, \rho)$ values under which $\frac{\partial \pi_t}{\partial \epsilon_t^\pi} < \frac{\partial \pi_t}{\partial \epsilon_t^\pi}|_{\rho=0}$

Figure D4: Inflation response to persistent cost-push shock $\epsilon_t^\pi$

*Notes:* Panel (a) plots the equilibrium inflation $\pi_t$ (relative to the size of the shock) in response to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$, as a function of the shallow thinking parameter $\lambda$ and the shock persistence $\rho$. Panel (b) shows, in the gray meshed area, the $(\lambda, \rho)$ combinations under which a persistent cost-push shock leads to lower inflation than a purely transitory shock, i.e., $\frac{\partial \pi_t}{\partial \epsilon_t^\pi} < \frac{\partial \pi_t}{\partial \epsilon_t^\pi}|_{\rho=0}$. In the case of purely transitory shock ($\rho = 0$), the equilibrium is independent of $\lambda$, as agents observe all variables in the same period. The vertical line in blue indicates our calibration of $\lambda$, which is completely outside the gray area. In contrast, the vertical green line representing the rational expectations equilibrium lies entirely within the area.

*(REE) response to a persistent monetary policy shock $\epsilon_t^i = \rho^t \epsilon^i$ features*

$$i_t^{REE} = \left[ 1 + \frac{\frac{\phi\theta\kappa}{1-\beta\theta\rho} \frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}}{1 - \frac{(1-\theta)+\rho\theta\kappa\frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}}{1-\beta\theta\rho}} \right]^{-1} \epsilon_t^i \tag{D13}$$

$$\pi_t^{REE} = -\frac{\frac{\theta\kappa}{1-\beta\theta\rho} \frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}}{1 - \frac{(1-\theta)+\rho\theta\kappa\frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}}{1-\beta\theta\rho}} i_t^{REE} \tag{D14}$$

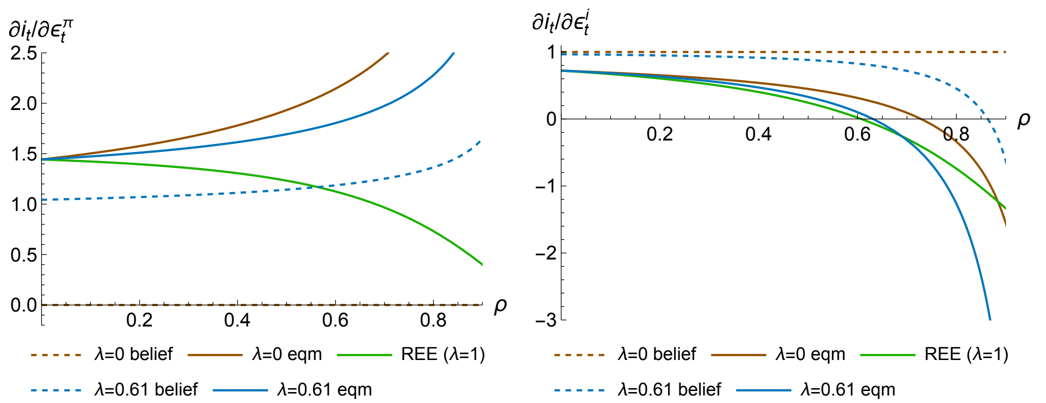
*whereas the average expectations under shallow thinking are*

$$\overline{\mathbb{E}}\left[i_t\right] = \left[ 1 + \frac{\frac{\lambda^4\phi\theta\kappa K(\lambda,\rho)}{1-\beta\theta\rho}}{1 - \frac{\lambda(1-\theta)+\lambda^3\rho\theta\kappa K(\lambda,\rho)}{1-\beta\theta\rho}} \right]^{-1} \epsilon_t^i \tag{D15}$$

$$\overline{\mathbb{E}}\left[\pi_t\right] = -\frac{\frac{\lambda^3\theta\kappa K\left(\lambda,\rho\right)}{1-\beta\theta\rho}}{1-\frac{\lambda(1-\theta)+\lambda^3\rho\theta\kappa K\left(\lambda,\rho\right)}{1-\beta\theta\rho}}\overline{\mathbb{E}}\left[i_t\right] \tag{D16}$$

*with $K(\lambda,\rho)$ increasing in $\lambda$ and $\rho$ under our calibration and $K(1,\rho) = \frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}$.*

This proposition nests Proposition 5 with $\rho = 0$. Figure D5 shows the responses of interest rate expectations and equilibrium interest rate to persistent cost-push shocks and monetary policy shocks.




(a) Persistent cost-push shock $\epsilon_t^\pi = \rho^t\epsilon^\pi$ (b) Persistent monetary policy shock $\epsilon_t^i = \rho^t\epsilon^i$

Figure D5: Interest rate $i_t$ in response to persistent shocks

*Notes:* These panels plot the responses of the interest rate to persistent cost-push shocks $\epsilon_t^\pi$ and monetary policy shocks $\epsilon_t^i$, parallel to Figure 9b. Under our calibration, the average interest rate expectation is higher than its equilibrium value in response to monetary policy shocks, but is lower in response to cost-push shocks, regardless of $\rho$.

### D.6 Proofs

**Proof of Proposition 1.** It follows directly from Section 2.1. We explicitly write down the system as

$$\underbrace{\begin{pmatrix} i_1 \\ \pi_1 \\ div_1 \\ \hat{n}_1^d \\ c_1 \\ \hat{n}_1^s \\ w_1 \end{pmatrix}}_{V_1} = \underbrace{\begin{pmatrix} 0 & \phi & 0 & 0 & 0 & 0 & 0 \\ 0 & 1-\theta & 0 & 0 & 0 & 0 & \theta\kappa \\ 0 & 0 & 0 & 0 & 1 & 0 & -\frac{1}{\mu-1} \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ -\sigma^{-1}\beta & 0 & \frac{(1-\beta)(\mu-1)\nu}{\sigma+\mu\nu} & 0 & 0 & 0 & \frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu} \\ \nu^{-1}\beta & 0 & -\frac{(1-\beta)(\mu-1)\sigma}{\sigma+\mu\nu} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{s-1} & 0 & -e^{s-1} & 0 \end{pmatrix}}_{M} \underbrace{\begin{pmatrix} i_1 \\ \pi_1 \\ div_1 \\ \hat{n}_1^d \\ c_1 \\ \hat{n}_1^s \\ w_1 \end{pmatrix}}_{V_1} + \underbrace{\begin{pmatrix} \epsilon_1^i \\ \epsilon_1^\pi \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}}_{S_1} \tag{D17}$$

with $e^s = \nu^{-1}\left[1 - \sigma\frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu}\right]$.$\square$

**Proof of Proposition 2.** Assumptions 1 and 2 imply that the average beliefs are

$$\begin{aligned} \overline{\mathbb{E}}_0\left[V_1\right] &\equiv \sum_{d=1}^{\infty} \mathbb{P}\left(d=n\right)\mathbb{E}_0^d\left[V_1\right] \\ &= \sum_{d=1}^{\infty} \mathbb{P}\left(d=n\right)\sum_{n=1}^{d} M^{n-1}S_1 = \sum_{n=1}^{\infty} \mathbb{P}\left(d\geq n\right)M^{n-1}S_1 = \sum_{n=1}^{\infty} \lambda^{n-1}M^{n-1}S_1 \end{aligned}$$

which can be recast as (14).$\square$

**Proof of Proposition 3.** (15) follows from Assumptions 2 and 3 and Definition 2. Since the causal distance $D_{vs}$ is bounded above, an ordinary least squares estimation identifies a negative slope $\gamma$. Further, the conditional expectation (15) minimizes the mean squared error of $\mathbb{E}^{population}\left(1_{nvs} - h\left(D_{vs}\right)\right)$ among all possible predictor $h\left(D_{vs}\right)$. Since it is a case of the exponential family $b_1 \cdot b_2^{D_{vs}-1} + b_0$, we can exactly identify $\lambda$ with $b_2$.$\square$

**Proofs of Propositions 4 and 5.** Directly solving (1-7) in response to the cost-push shock $\epsilon_1^\pi$ and the monetary policy shock $\epsilon_1^i$ yields the equilibrium. Solving the fixed point of (1-7) with cross-variable relations dampened by $\lambda$ as in (14) gives the average expectations.

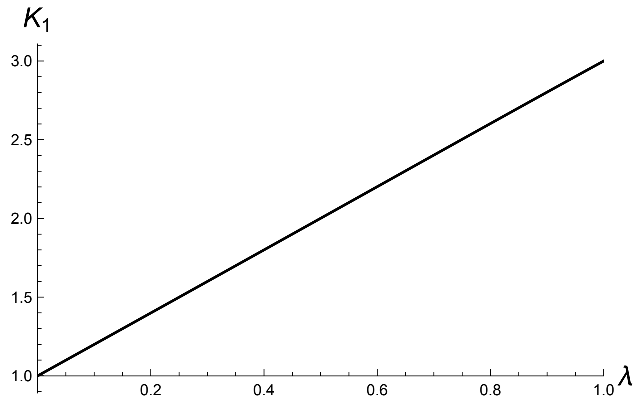



Figure D6: $K_1(\lambda)$ under our calibration

The dampening of Keynesian cross is

$$K_1(\lambda) = \frac{\frac{\beta(\sigma+\lambda\nu)}{\sigma\left(1-\frac{(1-\beta)(1+\nu)\sigma}{(\mu\nu+\sigma)}\right)}}{1-\lambda^2\frac{(1-\beta)(\mu-1)\nu}{(\mu\nu+\sigma)}-\lambda^3\frac{(1-\beta)\nu\left(1+\nu-\sigma-\lambda\nu+\lambda(\mu-1)\frac{(1-\beta)(1+\nu)\sigma}{(\mu\nu+\sigma)}\right)}{(\mu\nu+\sigma)-(1-\beta)(1+\nu)\sigma}} \tag{D18}$$

which depends on $\beta, \mu, \sigma, \nu$ but no other parameters when $\lambda \in [0, 1)$ and obtains $\sigma^{-1}(\nu+\sigma)$ at $\lambda = 1$. Under our calibration, $K_1(\lambda)$ increases in $\lambda$ as illustrated in Figure D6.□

**Proof of Proposition 6.** It follows direction from Section 5.1 and Appendix D.1.□

**Proof of Propositions 7 and D1.** Though the proposition only concerns a cost-push shock $\epsilon_t^\pi$, we also consider a monetary policy shock $\epsilon_t^i$ in the proof, which does not add much complexity. This proof will nest the Proofs of Proposition 4 and 5 by setting $\rho = 0$.

We solve the average expectations, which will nest the rational expectations equilibrium with $\lambda = 1$. We guess and verify that beliefs mean-revert at the same rate $\rho$.

Agents' actions (D1-D6) characterize $\left\{\overline{\mathbb{E}}[div_t], \overline{\mathbb{E}}\left[\hat{n}_t^d\right], \overline{\mathbb{E}}[\pi_t], \overline{\mathbb{E}}[c_t], \overline{\mathbb{E}}\left[\hat{n}_t^s\right], \overline{\mathbb{E}}[i_t]\right\}$ as

$$\overline{\mathbb{E}}[div_t] = \lambda\overline{\mathbb{E}}[c_t] - \lambda\frac{1}{\mu-1}\overline{\mathbb{E}}[w_t] \tag{D19}$$

$$\overline{\mathbb{E}}\left[\hat{n}_t^d\right] = \lambda\overline{\mathbb{E}}[c_t] \tag{D20}$$

$$\overline{\mathbb{E}}[\pi_t] = \lambda\theta\kappa\frac{\overline{\mathbb{E}}[w_t]}{1-\beta\theta\rho} + \lambda(1-\theta)\frac{\overline{\mathbb{E}}[\pi_t]}{1-\beta\theta\rho} + \epsilon_t^\pi \tag{D21}$$

$$\overline{\mathbb{E}}\left[c_t\right] = -\lambda\sigma^{-1}\beta\left(\frac{\overline{\mathbb{E}}\left[i_t\right]}{1-\beta\rho} - \frac{\rho\overline{\mathbb{E}}\left[\pi_t\right]}{1-\beta\rho}\right) + \lambda\left(1-\beta\right)\frac{\left(\mu-1\right)\nu}{\sigma+\mu\nu}\frac{\overline{\mathbb{E}}\left[div_t\right]}{1-\beta\rho} + \lambda\left(1-\beta\right)\frac{\left(1+\nu\right)}{\sigma+\mu\nu}\frac{\overline{\mathbb{E}}\left[w_t\right]}{1-\beta\rho} \tag{D22}$$

$$\overline{\mathbb{E}}\left[\hat{n}_t^s\right] = \lambda\nu^{-1}\beta\left(\frac{\overline{\mathbb{E}}\left[i_t\right]}{1-\beta\rho} - \frac{\rho\overline{\mathbb{E}}\left[\pi_t\right]}{1-\beta\rho}\right) - \lambda\left(1-\beta\right)\frac{\left(\mu-1\right)\sigma}{\sigma+\mu\nu}\frac{\overline{\mathbb{E}}\left[div_t\right]}{1-\beta\rho} \tag{D23}$$

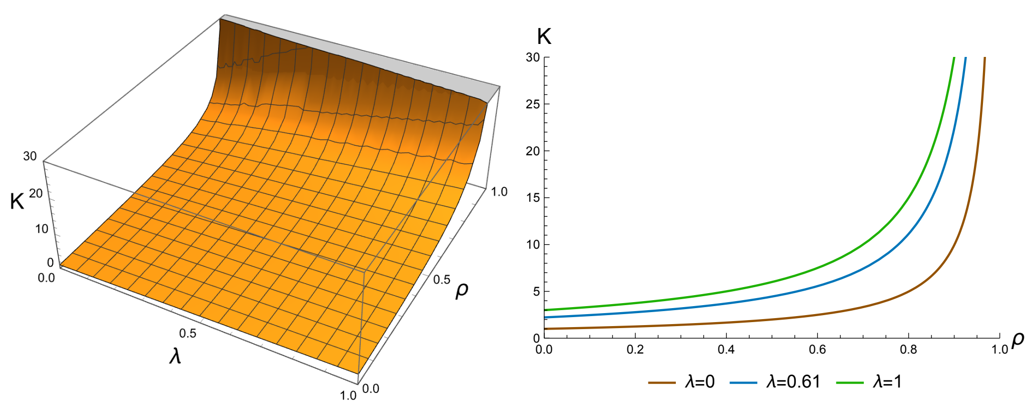
$$\overline{\mathbb{E}}\left[i_t\right] = \lambda\phi\overline{\mathbb{E}}\left[\pi_t\right] + \epsilon_t^i \tag{D24}$$

Note that in constructing $\overline{\mathbb{E}}\left[\hat{n}_t^d\right]$ and $\overline{\mathbb{E}}\left[\hat{n}_t^s\right]$, we take away their dependence on the wage $w$ from $n^d$ and $n^s$ in (D2, D5).

Last, the labor market clearing condition (D7) combined with the dependence of $n^d$ and $n^s$ on $w$ from (D2, D5) determines $\overline{\mathbb{E}}\left[w_t\right]$

$$\nu^{-1}\left(1-\sigma\frac{\left(1-\beta\right)\left(1+\nu\right)}{\left(1-\beta\rho\right)\left(\sigma+\mu\nu\right)}\right)\overline{\mathbb{E}}\left[w_t\right] = \lambda\overline{\mathbb{E}}\left[\hat{n}_t^d\right] - \lambda\overline{\mathbb{E}}\left[\hat{n}_t^s\right] \tag{D25}$$

Solving (D19-D25) gives the average expectations (40) and (41), which nest the rational expectations equilibrium (38) and (39) with $\lambda = 1$.




(a) $K\left(\lambda,\rho\right)$ (b) $K\left(\lambda,\rho\right)$ , varying $\rho$ under fixed $\lambda$

Figure D7: $K\left(\lambda,\rho\right)$ under our calibration

The dampening of Keynesian cross under persistent shocks is

$$K\left(\lambda,\rho\right) = \frac{\frac{\beta\left(\sigma+\lambda\nu\right)}{\sigma\left(1-\beta\rho\right)\left(1-\frac{\left(1-\beta\right)\left(1+\nu\right)\sigma}{\left(1-\beta\rho\right)\left(\mu\nu+\sigma\right)}\right)}}{1-\lambda^2\frac{\left(1-\beta\right)\left(\mu-1\right)\nu}{\left(1-\beta\rho\right)\left(\mu\nu+\sigma\right)} - \lambda^3\frac{\left(1-\beta\right)\nu\left(1+\nu-\sigma-\lambda\nu+\lambda\left(\mu-1\right)\frac{\left(1-\beta\right)\left(1+\nu\right)\sigma}{\left(1-\beta\rho\right)\left(\mu\nu+\sigma\right)}\right)}{\left(1-\beta\rho\right)\left(\mu\nu+\sigma\right)-\left(1-\beta\right)\left(1+\nu\right)\sigma}} \tag{D26}$$

which depends on $\rho, \beta, \mu, \sigma, \nu$ but no other parameters when $\lambda \in [0, 1)$ and obtains $\frac{\sigma^{-1}(\nu+\sigma)}{1-\rho}$ at $\lambda = 1$. Under our calibration, $K(\lambda, \rho)$ increases both in $\lambda$ and $\rho$ as shown in Figure D7.$\square$

# E Appendix to the RBC Model

## E.1 The RBC Model

Under our assumption that agents observe all variables in each period and have expectations formed once and for all after observing the shocks, i.e., $\mathbb{E}_t^d[v_s] \equiv \mathbb{E}^d[v_s]$ for $s > t$, we omit the time subscript on expectations for simplicity.

**Firms.** There is a continuum of firms with identical production technology and capital stock at the steady state but differing beliefs about impulse responses to shocks. To simplify notation, we consider a generic firm without firm-specific indexing, noting that once we aggregate across firms, the average expectations determine aggregate behavior under a first-order approximation around the steady state.

Firms maximize

$$\max_{\{N_t^d, I_t, Y_t\}} \mathbb{E} \sum_{t=0}^{\infty} \prod_{k=0}^{t-1} (1 + r_k)^{-1} DIV_t$$

subject to

$$DIV_t = Y_t - W_t N_t^d - I_t - \frac{\psi}{2}\left(\frac{I_t}{K_t} - \delta\right)^2 K_t \tag{E1}$$

$$Y_t = Z_t \left(\frac{K_t}{\alpha}\right)^{\alpha} \left(\frac{N_t^d}{1-\gamma}\right)^{1-\gamma} \tag{E2}$$

$$K_{t+1} = (1-\delta) K_t + I_t \tag{E3}$$

In each period, firms choose labor demand $N_t^d$ to satisfy

$$Z_t \left(\frac{K_t}{\alpha}\right)^{\alpha} \left(\frac{N_t^d}{1-\gamma}\right)^{-\gamma} = W_t$$

$$N_t^d = (1-\gamma)\left(\frac{Z_t}{W_t}\right)^{\frac{1}{\gamma}} \left(\frac{K_t}{\alpha}\right)^{\frac{\alpha}{\gamma}} \tag{E4}$$

We form the Lagrangian after plugging in the optimal labor demand as

$$\mathbb{E}\sum_{t=0}^{\infty}\prod_{k=0}^{t-1}R_k^{-1}\left[\gamma Z_t^{\frac{1}{\gamma}}W_t^{1-\frac{1}{\gamma}}\left(\frac{K_t}{\alpha}\right)^{\alpha/\gamma}-I_t-\frac{\psi}{2}\left(\frac{I_t}{K_t}-\delta\right)^2K_t\right]+\prod_{k=0}^{t-1}R_k^{-1}q_t\left[(1-\delta)K_t+I_t-K_{t+1}\right]$$

Its FOC w.r.t. $I_t$ is

$$-\left[1+\psi\left(\frac{I_t}{K_t}-\delta\right)\right]+q_t=0$$

which pins down the investment as

$$I_t=\left(\frac{q_t-1}{\psi}+\delta\right)K_t \tag{E5}$$

Its FOC w.r.t. $K_{t+1}$ is

$$\mathbb{E}\left[Z_{t+1}^{\frac{1}{\gamma}}W_{t+1}^{1-\frac{1}{\gamma}}\left(\frac{K_{t+1}}{\alpha}\right)^{\frac{\alpha}{\gamma}-1}-\frac{\psi}{2}\left(\frac{I_{t+1}}{K_{t+1}}-\delta\right)^2+\psi\left(\frac{I_{t+1}}{K_{t+1}}-\delta\right)\frac{I_{t+1}}{K_{t+1}}+q_{t+1}(1-\delta)\right]-(1+r_t)q_t=0$$

and thus

$$q_t=(1+r_t)^{-1}\,\mathbb{E}\left[Z_{t+1}^{\frac{1}{\gamma}}W_{t+1}^{1-\frac{1}{\gamma}}\left(\frac{K_{t+1}}{\alpha}\right)^{\frac{\alpha}{\gamma}-1}-\frac{\psi}{2}\left(\frac{I_{t+1}}{K_{t+1}}-\delta\right)^2+\psi\left(\frac{I_{t+1}}{K_{t+1}}-\delta\right)\frac{I_{t+1}}{K_{t+1}}+(1-\delta)q_{t+1}\right] \tag{E6}$$

(E1-E6) implicitly characterize the dependence of output $Y_t$, investment (including adjustment cost) $I_t+\Psi_t$ and labor demand $N_t^d$ on the past and current interest rates $\{r_s\}_{0\leq s\leq t}$ and wages $\{W_s\}_{0\leq s\leq t}$, as well as expectations of future interest rates $\{\mathbb{E}[r_s]\}_{s>t}$ and wages $\{\mathbb{E}[W_s]\}_{s>t}$. As we adopt a first-order approximation around the steady state, such implicit dependence is linear. As a result, once aggregated across all firms, the aggregate behavior depends on the average expectations of future interest rates $\left\{\overline{\mathbb{E}}[r_s]\right\}_{s>t}$ and wages $\left\{\overline{\mathbb{E}}[W_s]\right\}_{s>t}$.

**Households.** The household side of the RBC economy is the same as that of the New Keynesian economy, except that they invest in a real bond as opposed to a nominal bond. Thus the consumption and labor supply functions follow from (D4, D5) by replacing $i_{t+k}-\pi_{t+k+1}$ with $\frac{r_{t+k}-\bar{r}}{1+\bar{r}}$, where the normalization translates a level change of the real interest rate into a log change. Consumption and labor supply $C_t, N_t^s$ depend on the interest rate, wage and dividend $\{r_t, W_t, DIV_t\}_{t\geq 0}$ as follows

$$\frac{C_t-\overline{C}}{\overline{C}} = -\sigma^{-1}\beta\frac{r_t-\bar{r}}{1+\bar{r}} - \sigma^{-1}\beta\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}\left[\frac{r_{t+k}-\bar{r}}{1+\bar{r}}\right] + (1-\beta)\left[\frac{(\mu-1)\nu}{\sigma+\mu\nu}\frac{DIV_t}{\overline{DIV}} + \frac{(1+\nu)}{\sigma+\mu\nu}\frac{W_t}{\overline{W}}\right]$$
$$+ (1-\beta)\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}\Big[\frac{(\mu-1)\nu}{\sigma+\mu\nu}\frac{DIV_{t+k}}{\overline{DIV}} + \frac{(1+\nu)}{\sigma+\mu\nu}\frac{W_{t+k}}{\overline{W}}\Big] \tag{E7}$$
$$\frac{N_t^s-\overline{N}}{\overline{N}} = \nu^{-1}\beta\frac{r_t-\bar{r}}{1+\bar{r}} + \nu^{-1}\beta\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}\left[\frac{r_{t+k}-\bar{r}}{1+\bar{r}}\right] - (1-\beta)\frac{(\mu-1)\sigma}{\sigma+\mu\nu}\frac{DIV_t}{\overline{DIV}} + \nu^{-1}\left(1-\sigma\frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu}\right)\frac{W_t}{\overline{W}}$$
$$- (1-\beta)\sum_{k=1}^{\infty}\beta^k\overline{\mathbb{E}}\Big[\frac{(\mu-1)\sigma}{\sigma+\mu\nu}\frac{DIV_{t+k}}{\overline{DIV}} + \frac{(1+\nu)\sigma\nu^{-1}}{\sigma+\mu\nu}\frac{W_{t+k}}{\overline{W}}\Big] \tag{E8}$$

**Goods and labor markets.** The interest rates and wages arise from clearing the goods and labor markets,

$$N_t^s = N_t^d \tag{E9}$$

and

$$Y_t = I_t + \Psi_t + C_t \tag{E10}$$

**Equilibrium.** We study a *temporary equilibrium* in which agents maximize their utilities, taking as given the average expectations $\overline{\mathbb{E}}_t[\cdot]$, and markets clear. In this RBC model, equations (E1-E10) characterizes the equilibrium given the average expectations.

We adopt a quarterly calibration of the RBC economy, with all parameters listed in Table E1. We assume slight decreasing returns to scale in production ($\alpha < \gamma$) and the existence of a small fringe of rational agents, discussed in detail after introducing shallow thinking.

## E.2 Shallow Thinking and Modeling Choices

Similar to Section 5.1, we represent the RBC economy by the sequences of eight variables $\left\{r_t, Y_t, I_t, DIV_t, N_t^d, C_t, N_t^s, W_t\right\}_{t\geq 0}$, which we refer to as $\mathcal{V}$.[36] Six of them are agents' actions, which we collect as $\mathcal{V}^{action}$, whereas the other two, $\{r_t, W_t\}_{t\geq 0}$, are prices formed in the competitive goods and labor market.

[36]For brevity, we ignore the adjustment cost $\Psi_t$, which is 0 to the first order around the steady state since the adjustment cost is quadratic.

Table E1: Quarterly calibration of the RBC economy

| Parameter | Description | Value | Estimate/Target |
|---|---|---|---|
| ***Beliefs*** | | | |
| $\lambda$ | Continuation rate of depth of thinking | 0.61 | Our survey evidence |
| $\vartheta$ | Share of rational agents | 0.1 | |
| ***Firms*** | | | |
| $Z$ | Productivity | 0.33 | Steady state $\overline{Y} = 1$ |
| $\rho$ | Persistence of productivity shocks | 0.979 | King and Rebelo (1999) |
| $\alpha$ | Capital share | 0.25 | |
| $1 - \gamma$ | Labor share | 0.67 | |
| $\delta$ | Capital depreciation rate | 0.025 | |
| $\psi$ | Capital adjustment cost | 1 | |
| ***Households*** | | | |
| $\beta$ | Discount factor | 0.99 | Steady state annual $\bar{r} = 4\%$ |
| $\sigma^{-1}$ | Elasticity of intertemporal substitution (EIS) | 1 | |
| $\nu^{-1}$ | Frisch elasticity | 0.5 | |
| $\varphi$ | Labor disutility scale | 0.81 | Steady state $\overline{N} = 1$ |

We characterize the rational expectations equilibrium (REE) and introduce shallow thinking beliefs accordingly. For REE, each action $\mathbf{v} = (\{v_t\}_{t\geq 0})'$ in $\mathcal{V}^{action}$ can be represented in the sequence space as

$$\mathbf{v}^{REE} = \sum_{\mathbf{u}\in\mathcal{V}} \mathbf{J}_{\mathbf{vu}} \mathbf{u}^{REE} + \epsilon^{v}, \ \forall \mathbf{v} \in \mathcal{V}^{action} \tag{E11}$$

where $\mathbf{J}_{\mathbf{vu}}$ encodes the dependence among sequences of variables. Households' consumption $\{C_t\}_{t\geq 0}$ and labor supply $\left\{N_t^s\right\}_{t\geq 0}$ depend on the interest rate $\{r_t\}_{t\geq 0}$, dividend $\{DIV_t\}_{t\geq 0}$, and wage $\{W_t\}_{t\geq 0}$. Firms' output $\{Y_t\}_{t\geq 0}$, investment $\{I_t\}_{t\geq 0}$, dividend $\{DIV_t\}_{t\geq 0}$, and labor demand $\left\{N_t^d\right\}_{t\geq 0}$ are functions of the interest rate $\{r_t\}_{t\geq 0}$ and wage $\{W_t\}_{t\geq 0}$, subject to the productivity shock $\{Z_t\}_{t\geq 0}$. Differing from the textbook New Keynesian model with no state variable, in the RBC economy firms have a stock of capital. Thus firms' actions depend on not only current and future interest rates and wages, but also past ones. Yet (E11) is general enough to allow for it.

Instead of using (E11) together with market clearing conditions to solve for the REE, as done in Auclert et al. (2021), we use it to represent the causal relations and reinterpret the market clearing conditions as such. We pedagogically write down these expressions by associating each price with a competitive market, i.e., wages with the labor market and interest rates with the goods market.

For labor supply and demand $\mathbf{N}^s, \mathbf{N}^d$, by separating their dependence on the wage $\mathbf{W}$

from the rest, we interpret them as supply and demand curves in the sequence space

$$\mathbf{v}^{REE} = \mathbf{J}_{\mathbf{vW}}\mathbf{W}^{REE} + \hat{\mathbf{v}}^{REE},\ \mathbf{v} \in \left\{\mathbf{N}^s, \mathbf{N}^d\right\}$$

with elasticities $\mathbf{J}_{\mathbf{N}^s\mathbf{W}}, \mathbf{J}_{\mathbf{N}^d\mathbf{W}}$ and shifts $\hat{\mathbf{N}}^{s,REE}, \hat{\mathbf{N}}^{d,REE}$ defined as

$$\hat{\mathbf{v}}^{REE} = \sum_{\mathbf{u}\in\mathcal{V}\backslash\{\mathbf{W}\}} \mathbf{J}_{\mathbf{vu}}\mathbf{u}^{REE} + \epsilon^v,\ \mathbf{v} \in \left\{\mathbf{N}^s, \mathbf{N}^d\right\} \tag{E12}$$

In the RBC economy, both supply and demand elasticities $\mathbf{J}_{\mathbf{N}^s\mathbf{W}}, \mathbf{J}_{\mathbf{N}^d\mathbf{W}}$ are non-zero. By equalizing $\mathbf{N}^{s,REE} = \mathbf{N}^{d,REE}$, we can interpret the wage $\mathbf{W}^{REE}$ as resulting from shifts $\hat{\mathbf{N}}^{d,REE}, \hat{\mathbf{N}}^{s,REE}$

$$\mathbf{W}^{REE} = \left(\mathbf{J}_{\mathbf{N}^s\mathbf{W}} - \mathbf{J}_{\mathbf{N}^d\mathbf{W}}\right)^{-1}\left(\hat{\mathbf{N}}^{d,REE} - \hat{\mathbf{N}}^{s,REE}\right) \tag{E13}$$

For output, investment and consumption $\mathbf{Y}, \mathbf{I}, \mathbf{C}$, by separating their dependence on the interest rate $\mathbf{r}$ from the rest, we interpret them as supply and demand curves as well

$$\mathbf{v}^{REE} = \mathbf{J}_{\mathbf{vr}}\mathbf{r}^{REE} + \hat{\mathbf{v}}^{REE},\ \forall \mathbf{v} \in \{\mathbf{Y}, \mathbf{I}, \mathbf{C}\}$$

with elasticities $\mathbf{J}_{\mathbf{Yr}}, \mathbf{J}_{\mathbf{Ir}}, \mathbf{J}_{\mathbf{Cr}}$ and shifts $\hat{\mathbf{Y}}, \hat{\mathbf{I}}, \hat{\mathbf{C}}$ defined as

$$\hat{\mathbf{v}}^{REE} = \sum_{\mathbf{u}\in\mathcal{V}\backslash\{\mathbf{r}\}} \mathbf{J}_{\mathbf{vu}}\mathbf{u}^{REE} + \epsilon^v,\ \mathbf{v} \in \{\mathbf{Y}, \mathbf{I}, \mathbf{C}\} \tag{E14}$$

By equalizing $\mathbf{Y} = \mathbf{I} + \mathbf{C}$, we can interpret the interest rate as dependent on shifts $\hat{\mathbf{I}}^{REE}, \hat{\mathbf{C}}^{REE}, \hat{\mathbf{Y}}^{REE}$

$$\mathbf{r}^{REE} = \left(\mathbf{J}_{\mathbf{Yr}} - \mathbf{J}_{\mathbf{Ir}} - \mathbf{J}_{\mathbf{Cr}}\right)^{-1}\left(\hat{\mathbf{I}}^{REE} + \hat{\mathbf{C}}^{REE} - \hat{\mathbf{Y}}^{REE}\right) \tag{E15}$$

To determine beliefs, we reason with $\hat{\mathbf{v}}$ as in (E12, E14) instead of $\mathbf{v}$ in (E11). Taking stock, (E11) and (E13, E15) characterize the REE. We can similarly express it as a linear system following Proposition 6. Figure E1 illustrates the system of causal relations that represent the RBC economy.

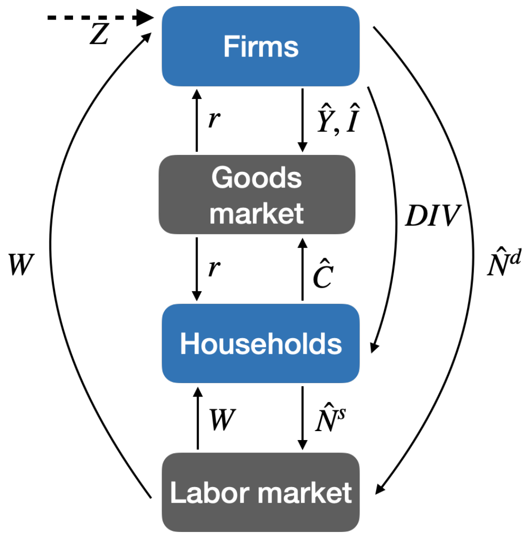



Figure E1: Causal relations of the RBC economy in the sequence space

**Modeling choices.** We make two modeling choices regarding the returns to scale in production and the existence of a fringe of rational agents, to ensure a stationary equilibrium. We leave it for future research to explore other approaches.

Regarding firms' production, we set decreasing returns to scale with $\alpha < \gamma$. The reason is that if firms operate under constant returns, a finite-depth agent who knows the productivity shock $\{Z_t\}_{t\geq 0}$ but does not anticipate changes in wages $\{W_t\}_{t\geq 0}$ or interest rates $\{r_t\}_{t\geq 0}$ would believe that firms expand their capital stock to a different steady level.[37] That introduces nonstationary beliefs, leading to nonstationary equilibrium responses. In such a case, agents' beliefs will be perpetually wrong, unlike in a stationary case where agents correctly understand the steady state but misjudge the temporary impulse responses. For this reason, we consider decreasing returns to production as the generic case, with constant returns approximated by a small degree of decreasing returns.

Regarding the existence of rational agents, we first note that households' forward-looking behavior is necessary for the existence of a stationary equilibrium in the RBC economy. Suppose we study a production economy with labor as a factor in fixed supply and firms like those in our RBC economy. No equilibrium with stationary $\{r_t, W_t\}_{t\geq 0}$ exists to meet the market clearing conditions $N_t^d = \overline{N}$ and $Y_t = I_t$, even if we assume that firms are rational. Hence, it is the forward-looking behavior of households that supports the RBC economy under rational expectations. In modeling shallow thinking of dynamic

[37] If there is no adjustment cost ($\psi = 0$), then the agent expects firms to infinitely expand their capital stock and production. With a positive adjustment cost ($\psi > 0$), the agent expects firms to expand their capital stock to a steady level different from before.

general equilibrium, we nest rational expectations with $\lambda = 1$, but otherwise do not allow beliefs to be directly related to the actual equilibrium for simplicity. That is because the causal relations used in Proposition 6 are those of the rational expectations equilibrium rather than the actual equilibrium. In the RBC economy, if households have beliefs that are entirely disconnected from the actual equilibrium, no stationary equilibrium exists. Thus, we assume that a small $\vartheta$ share of agents (both firms and households) are rational, giving rise to average expectations (for a generic variable $v$) $\overline{\mathbb{E}}^{mix}[v_t] = \vartheta v_t + (1-\vartheta)\overline{\mathbb{E}}[v_t]$ that determine the equilibrium, where $v_t$ is the actual equilibrium outcome and $\overline{\mathbb{E}}[v_t]$ represents the average shallow thinking beliefs. Future work may improve on this ad hoc assumption, using the causal relations of the actual dynamic general equilibrium.

# F Implementing Shallow Thinking with the Sequence-Space Jacobian

We pedagogically write down a 3-step procedure to implement shallow thinking in the RBC economy using the sequence-space Jacobians (SSJ) toolkit developed by Auclert et al. (2021). This procedure can also be applied to other models that can be solved in the sequence space, including models with incomplete markets and idiosyncratic shocks.

**Step 1: characterizing REE.** In the first step, we determine the causal relations as the Jacobians of the rational expectations equilibrium (REE) in terms of agents' best responses, i.e., (E11). We strictly adhere to expressions of agents' choices as functions of decision-relevant variables. For example, household consumption $\{C_t\}_{t\geq 0}$ and labor supply $\left\{N_t^s\right\}_{t\geq 0}$ respond to wages $\{W_t\}_{t\geq 0}$ and dividends $\{DIV_t\}_{t\geq 0}$, instead of aggregate output $\{Y_t\}_{t\geq 0}$. The REE is characterized by these Jacobians together with market clearing conditions.

**Step 2: determining shallow thinking beliefs.** In the second step, we characterize the average shallow thinking expectations $\overline{\mathbb{E}}[\cdot]$ of the sequences of eight variables $\mathcal{V} = \left\{\mathbf{r}, \mathbf{Y}, \mathbf{I}, \mathbf{DIV}, \mathbf{N}^d, \mathbf{C}, \mathbf{N}^s, \mathbf{W}\right\}$ with the REE Jacobians. In order to use the SSJ toolkit to determine shallow thinking beliefs, we construct a *modified model* as a directed *acyclic* graph (DAG) based on the REE Jacobians. We divide the system of equations governing these variables into two groups, one concerning agents' actions and one about prices from competitive markets.

Based on (E11), the first group of equations concerning agents is

$$\overline{\mathbb{E}}[\mathbf{v}] = \lambda \sum_{\mathbf{u}\in\mathcal{V}} \mathbf{J}_{\mathbf{vu}} \overline{\mathbb{E}}[\mathbf{u}] + \epsilon^{v}, \ \forall \mathbf{v} \in \mathcal{V}^{action} \equiv \mathcal{V} \setminus \{\mathbf{r}, \mathbf{W}\} \tag{F1}$$

where $\{\mathbf{J}_{\mathbf{vu}}\}$ are the Jacobians of the REE determined in the first step. To embed this in the SSJ toolkit, we construct the *modified households and firms blocks* as follows. We replicate the households and firms blocks with REE Jacobians and modify all the Jacobians by $\lambda$.[38] Then, we create Jacobians of shifts $\overline{\mathbb{E}}\left[\hat{\mathbf{N}}^d\right], \overline{\mathbb{E}}\left[\hat{\mathbf{N}}^s\right]$ w.r.t. decision-relevant variables by replicating the Jacobians of $\overline{\mathbb{E}}\left[\mathbf{N}^d, \mathbf{N}^s\right]$ and setting their dependence on $\overline{\mathbb{E}}[\mathbf{W}]$ to zero. Similarly, we create Jacobians of shifts $\overline{\mathbb{E}}\left[\hat{\mathbf{Y}}, \hat{\mathbf{I}}, \hat{\mathbf{C}}\right]$ w.r.t. decision-relevant variables by replicating those of $\overline{\mathbb{E}}[\mathbf{Y}, \mathbf{I}, \mathbf{C}]$ and setting their dependence on $\overline{\mathbb{E}}[\mathbf{r}]$ to zero.

According to (E13, E15), the second group about markets consists of

$$\begin{aligned}\overline{\mathbb{E}}[\mathbf{W}] &= \lambda \left(\mathbf{J}_{\mathbf{N}^s\mathbf{W}} - \mathbf{J}_{\mathbf{N}^d\mathbf{W}}\right)^{-1} \left(\overline{\mathbb{E}}\left[\hat{\mathbf{N}}^d\right] - \overline{\mathbb{E}}\left[\hat{\mathbf{N}}^s\right]\right) \\ \overline{\mathbb{E}}[\mathbf{r}] &= \lambda \left(\mathbf{J}_{\mathbf{Yr}} - \mathbf{J}_{\mathbf{Ir}} - \mathbf{J}_{\mathbf{Cr}}\right)^{-1} \left(\overline{\mathbb{E}}\left[\hat{\mathbf{I}}\right] + \overline{\mathbb{E}}\left[\hat{\mathbf{C}}\right] - \overline{\mathbb{E}}\left[\hat{\mathbf{Y}}\right]\right)\end{aligned}$$

which can be rewritten as

$$\lambda \left(\overline{\mathbb{E}}\left[\hat{\mathbf{N}}^d\right] - \overline{\mathbb{E}}\left[\hat{\mathbf{N}}^s\right]\right) - \left(\mathbf{J}_{\mathbf{N}^s\mathbf{W}} - \mathbf{J}_{\mathbf{N}^d\mathbf{W}}\right) \overline{\mathbb{E}}[\mathbf{W}] = \mathbf{0} \tag{F2}$$

$$\lambda \left(\overline{\mathbb{E}}\left[\hat{\mathbf{I}}\right] + \overline{\mathbb{E}}\left[\hat{\mathbf{C}}\right] - \overline{\mathbb{E}}\left[\hat{\mathbf{Y}}\right]\right) - \left(\mathbf{J}_{\mathbf{Yr}} - \mathbf{J}_{\mathbf{Ir}} - \mathbf{J}_{\mathbf{Cr}}\right) \overline{\mathbb{E}}[\mathbf{r}] = \mathbf{0} \tag{F3}$$

We set up *fictitious Walrasian auctioneer blocks* as simple blocks that take $\overline{\mathbb{E}}\left[\hat{\mathbf{N}}^d, \hat{\mathbf{N}}^s, \hat{\mathbf{Y}}, \hat{\mathbf{I}}, \hat{\mathbf{C}}\right]$ and $\overline{\mathbb{E}}[\mathbf{W}, \mathbf{r}]$ as inputs and produce the residuals of (F2, F3) as outputs.

Putting together the modified households and firms blocks and the fictitious Walrasian auctioneer blocks forms a DAG, with $\overline{\mathbb{E}}[\mathbf{W}, \mathbf{r}]$ as unknowns and (F2, F3) as targets. Solving this DAG yields the average shallow thinking expectations $\overline{\mathbb{E}}[\cdot]$ of the eight variables $\mathcal{V} = \left\{\mathbf{r}, \mathbf{Y}, \mathbf{I}, \mathbf{DIV}, \mathbf{N}^d, \mathbf{C}, \mathbf{N}^s, \mathbf{W}\right\}$.

**Step 3: calculating equilibrium given beliefs.** In the last step, we determine the equilibrium given the average shallow thinking expectations. Here we allow for a slight generalization that $\vartheta$ share of agents are rational, hence the average expectations that

[38] In the RBC model, the households and firms blocks have no cyclic dependence. If there were, one could turn a cycle into a target to use the SSJ toolkit.

matter for the equilibrium is

$$\overline{\mathbb{E}}^{mix}[v_t] = \vartheta v_t + (1-\vartheta)\overline{\mathbb{E}}[v_t] \tag{F4}$$

where $v_t$ is the actual equilibrium outcome and $\overline{\mathbb{E}}[v_t]$ represents the average shallow thinking expectations determined above. The REE is nested by either $\vartheta = 1$ or $\lambda = 1$ (hence $\overline{\mathbb{E}}[v_t] = v_t$). Outside the limits, forward-looking agents with beliefs (F4) are surprised in each period $t$ and change their behavior when a decision-relevant variable turns out different from their beliefs ($v_t \neq \overline{\mathbb{E}}^{mix}[v_t]$). They behave as

$$\mathbf{v} = \sum_{\mathbf{u}\in\mathcal{V}} \left(\mathbf{J}_{\mathbf{vu}}\overline{\mathbb{E}}^{mix}[\mathbf{u}] + \check{\mathbf{J}}_{\mathbf{vu}}\left(\mathbf{u} - \overline{\mathbb{E}}^{mix}[\mathbf{u}]\right)\right) + \boldsymbol{\epsilon}^v,\ \forall \mathbf{v}\in\mathcal{V}^{action} \tag{F5}$$

where

$$\left(\check{\mathbf{J}}_{\mathbf{vu}}\right)_{ts} \equiv \begin{cases} (\mathbf{J}_{\mathbf{vu}})_{t-s,0} & s\le t \\ 0 & s>t \end{cases}$$

is the myopic Jacobian that captures responses to variables observed in the past.[39] Rearranging terms and plugging in (F4), we get for any $\mathbf{v}\in\mathcal{V}^{action}$

$$\begin{aligned}\mathbf{v} &= \sum_{\mathbf{u}\in\mathcal{V}}\check{\mathbf{J}}_{\mathbf{vu}}\mathbf{u} + \sum_{\mathbf{u}\in\mathcal{V}}\left(\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}\right)\overline{\mathbb{E}}^{mix}[\mathbf{u}] + \boldsymbol{\epsilon}^v \\ &= \sum_{\mathbf{u}\in\mathcal{V}}\check{\mathbf{J}}_{\mathbf{vu}}\mathbf{u} + \sum_{\mathbf{u}\in\mathcal{V}}\left(\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}\right)\left(\vartheta\mathbf{u}_t + (1-\vartheta)\overline{\mathbb{E}}[\mathbf{u}_t]\right) + \boldsymbol{\epsilon}^v \\ &= \sum_{\mathbf{u}\in\mathcal{V}}\underbrace{\left(\check{\mathbf{J}}_{\mathbf{vu}} + \vartheta\left(\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}\right)\right)}_{\equiv\tilde{\mathbf{J}}_{\mathbf{vu}}}\mathbf{u} + \underbrace{\sum_{\mathbf{u}\in\mathcal{V}}(1-\vartheta)\left(\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}\right)\overline{\mathbb{E}}[\mathbf{u}_t] + \boldsymbol{\epsilon}^v}_{\equiv\tilde{\boldsymbol{\epsilon}}^v}\end{aligned} \tag{F6}$$

where the first part $\tilde{\mathbf{J}}_{\mathbf{vu}}\mathbf{u}$ encodes the dependence among equilibrium outcomes and the second part $\tilde{\boldsymbol{\epsilon}}^v$ is determined independent of the equilibrium. Thus $\tilde{\boldsymbol{\epsilon}}^v$ is equivalent to a shock to forward-looking agents, which we call *pseudo shocks*.

To determine the equilibrium with the SSJ toolkit, we construct a model by replacing the REE Jacobians for forward-looking agents by $\tilde{\mathbf{J}}_{\mathbf{vu}}$ and adding pseudo shocks $\tilde{\boldsymbol{\epsilon}}^v$ in addition to the true shocks $\boldsymbol{\epsilon}^v$. We keep the market clearing conditions as they are. Solving for this model yields the equilibrium given beliefs.

[39]See Auclert et al. (2021) and Auclert, Rognlie and Straub (2020) for the validity of this expression to the first order.